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ME 274: Design Project II

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Course Introduction

- This course, ME 274: SECOND YEAR DESIGN PROJECT is a second year course for BSc Mechanical Engineering.

COURSE OVERVIEW

- You will be taken through the following: **Principles of Mechanical Engineering Design**, **Design of Machine members under static, variable and impact loads.**

COURSE OBJECTIVES

On completion of the course it is expected that you will be able to:

- Study the **Engineering Design process** and apply it in **Design Projects**

Course Outline

The course is divided into four lectures

Lecture 1: Engineering Design Process

Lecture 2: Basic Stress Analysis for Machine Members

Lecture 3: Failure under Static Loading.

Lecture 4: Fatigue Failure under Variable Loading

Assessment

Continuous assessment:			**30%**
	i.	Attendance	05%
	ii.	Project	10%
	iii.	Mid Semester	15%
End of semester examination:			**70%**

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LECTURE 1

ENGINEERING DESIGN PROCESS

Lecture Outline

The Engineering Design Process

- Recognition of Need
- Definition of Functional Requirements
- Planning the Design Project
- Definition of the Problem and Specification
- Conceptual Design
- Introduction to 'Total Design' Philosophy.

Evaluation and Selection of Design Alternatives

- Method A
- Method B
- Implementation of Design
- Desired Attributes of an Engineer

Learning Objectives

After reading this unit you should be able to:

1. Draw and Explain the Engineering Design Flow Chart

2. List the Guidelines for writing a good Project Report.

3. Apply the Design Process Flow Chart to a Design task (Project)

Introduction

- There are many definitions of Design and various ways proposed for modeling or representing the process.

- There is no single way of defining design but, whatever the wording of a definition, design should be seen as encompassing the Total process which starts with **recognition and clarification** of '**a Need**' and continues until a satisfactory **result** is achieved.

- It is therefore **a broad and complex field**, involving many different types of activity, but is essentially concerned with projects. The notion of a project is central to the understanding design.

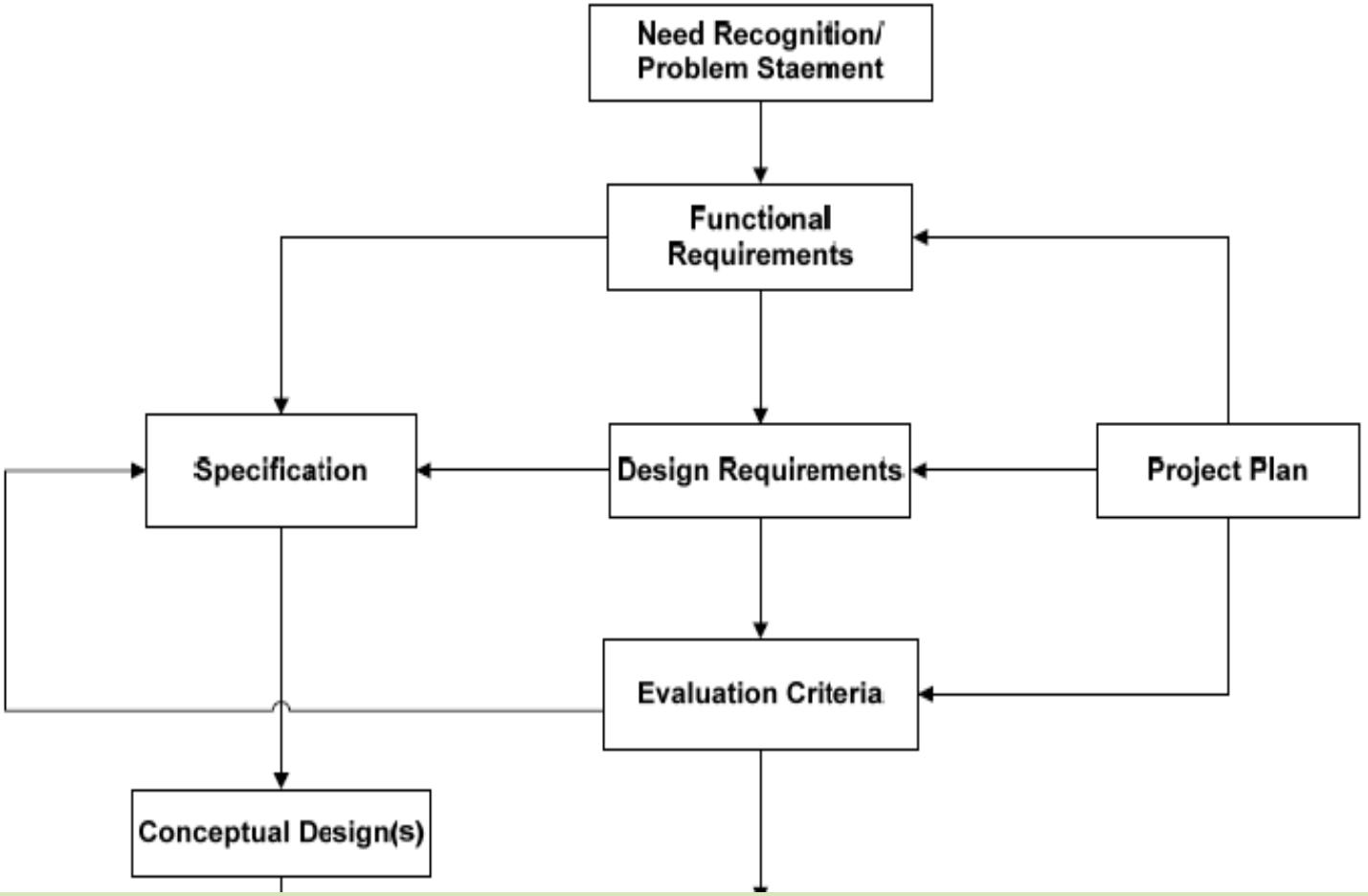
- In this section we shall study the Design process from the **Need recognition stage** to generation of **Conceptual designs**.

Definition

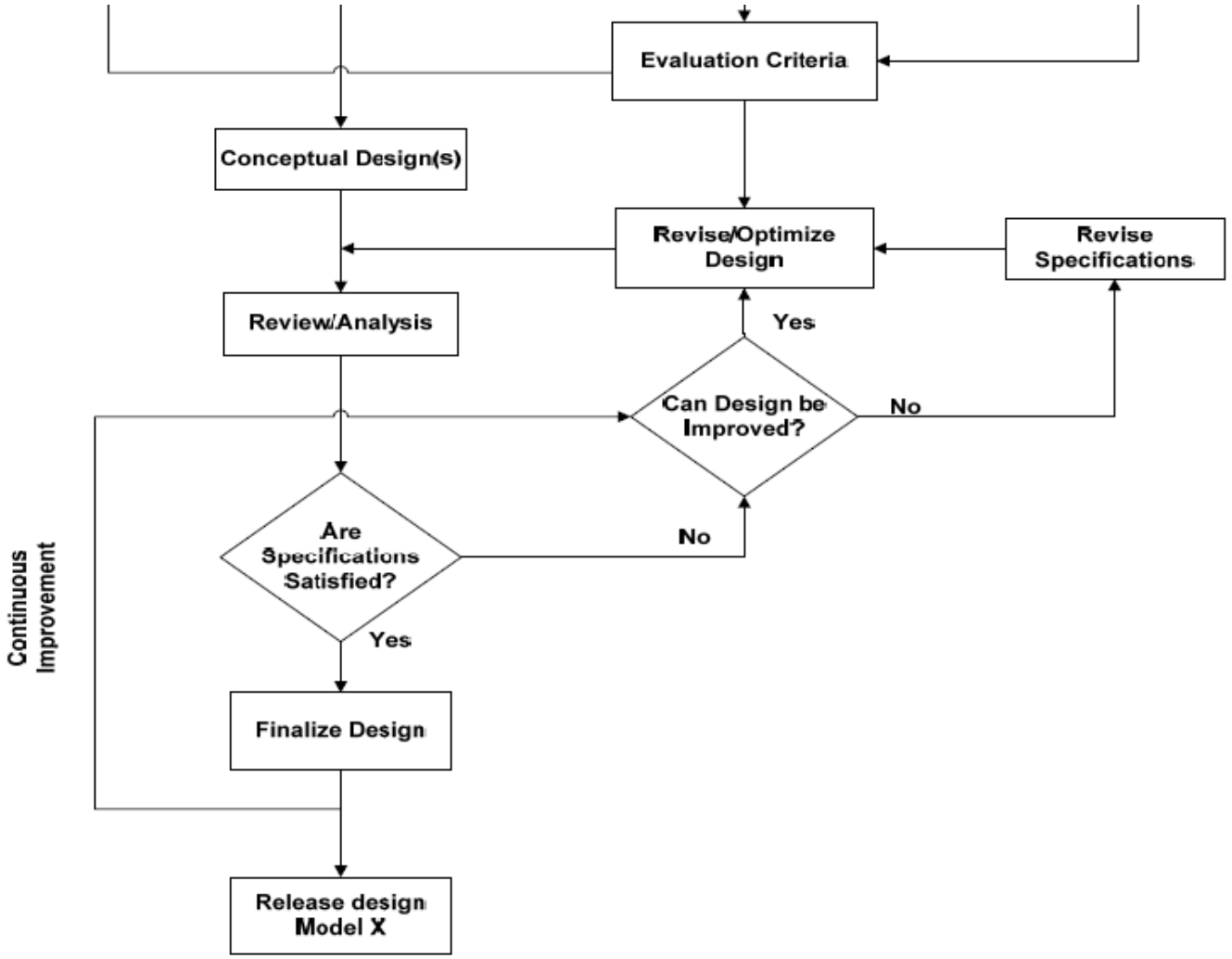
- is the systematic activity necessary from the identification of the market/user need to the selling of the successful product to satisfy that need

- an activity that encompasses product, process, people and organisation.

Design Process



Design Process



Recognition of Need

1. The human being is a wanting animal whose needs depend on what he already has.

2. The needs are arranged in a hierarchy of importance.

3. Once one need is satisfied, another emerges and demands satisfaction.

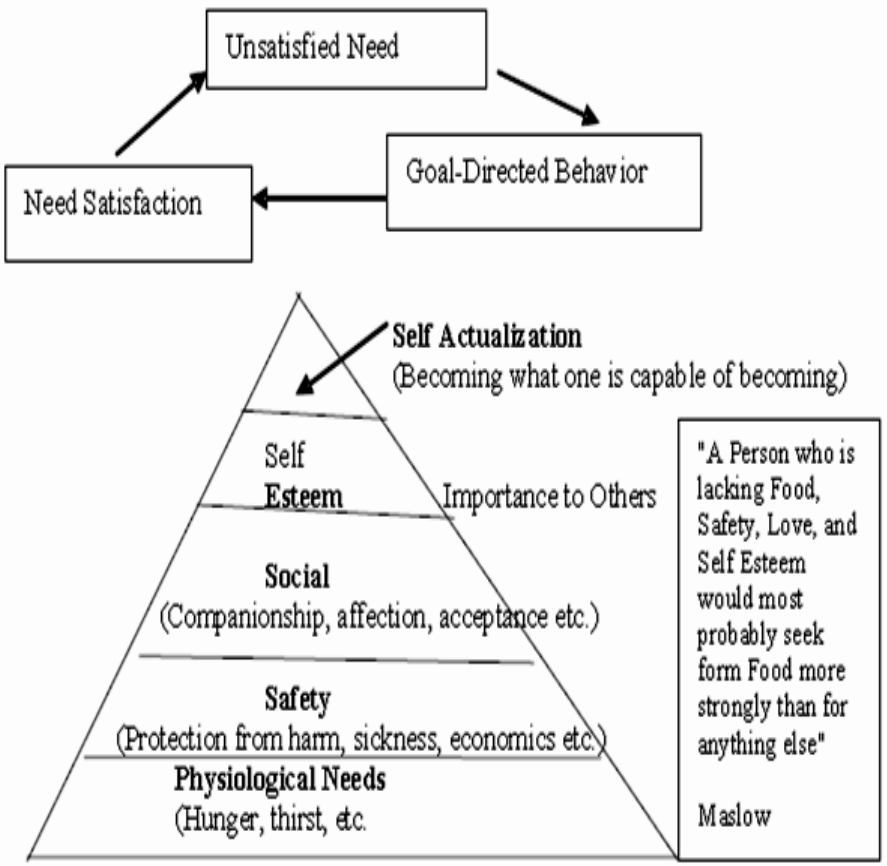


Figure 2:Maslow's Hierarchy of Need

Functional Requirements

- Statements which indicate "**what**" the device is supposed to do.

- They should always employ action phrases such as "**to lift**", "**to push**", "**to support**", "**to transmit**" etc.

- This is the ***Need Statement*** and should be as clear as possible to the intended customer.

Planning the Design Project

1. Identify tasks to be performed

2. Relate tasks to be performed

3. Apply time line to tasks

4. Develop a Gantt chart and/or Activity Path Network and Critical Path.

5. Apply responsibilities to tasks

Definition of Problem and Specification

1. Design Requirements

Detail statements indicating the expected specific and quantitative data with regard to the performance of the device.

They indicate size, power, efficiency, speed, torque, etc.

2. Evaluation Criteria

Functionality, safety, reliability, usability, manufacturability, marketability, and competitiveness (initial cost, operating cost, maintenance cost, aesthetics, ecological and societal considerations).

A sample of the criteria for the sterilizer is presented in the table below

Evaluation Criteria Table

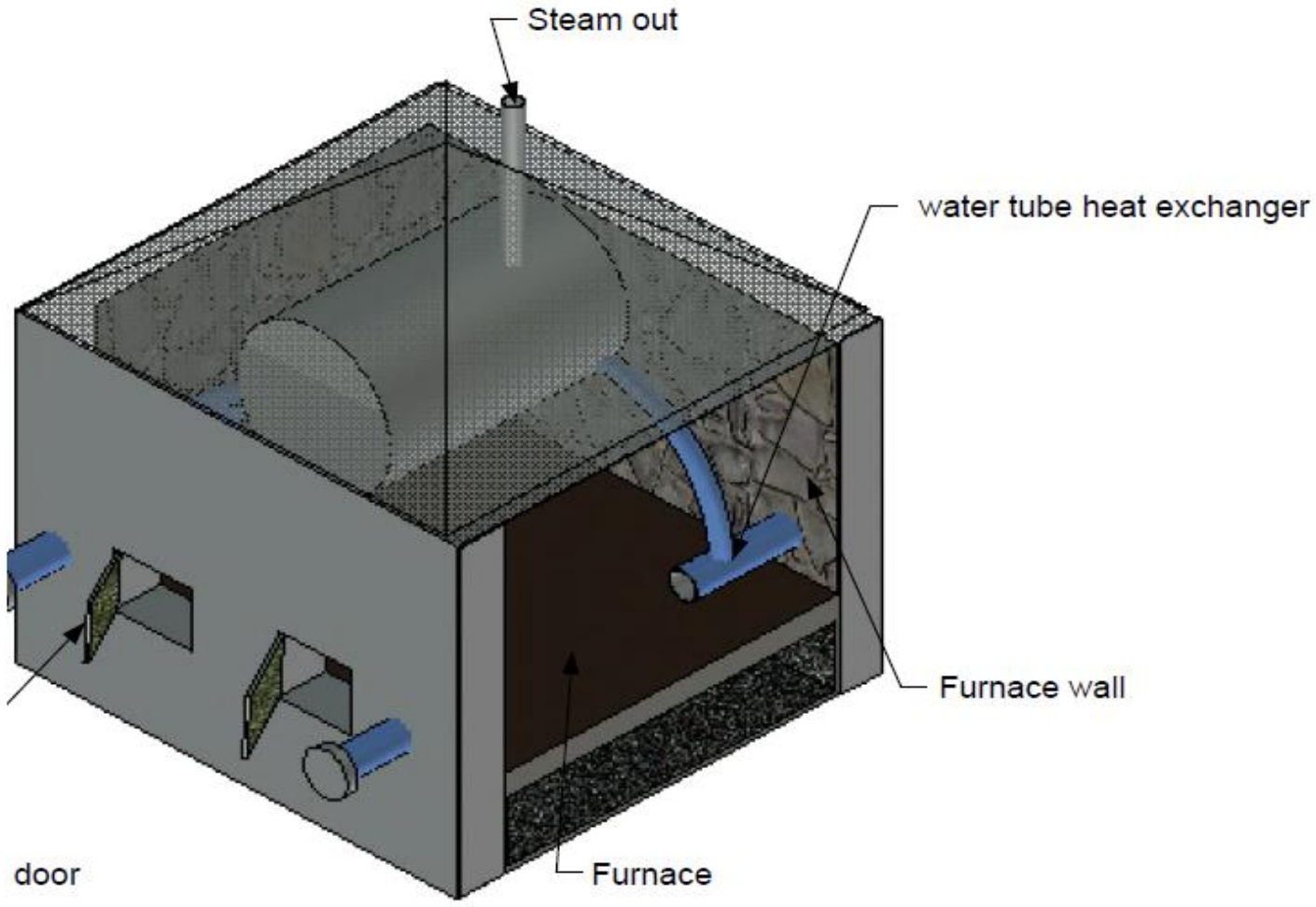
CRITERION	INITIAL COST		EASE OF MANUFACTURE		EASE OF MAITENANCE		ASSEMBLY	
WEIGHTING	0.25		0.25		0.25		0.25	
PARAMETERS	NUMBER OF PARTS		TYPES OF SHAPES		NUMBER OF FASTENERS		EASE OF ASSEMBLY	
	MAG.	Score	MAG.	Score	MAG.	Score	MAG.	Score
	Below 5	5	Very Easy	5	Below 5	5	Very Easy	5
	6 – 10	4	Easy	4	6 – 10	4	Easy	4
	11 – 15	3	Fairly easy	3	11 – 15	3	Fairly easy	3
	16 – 20	2	Difficult	2	16 – 20	2	Difficult	2
	Above 21	1	Very Difficult	1	Above 21	1	Very Difficult	1

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Conceptualize the Design

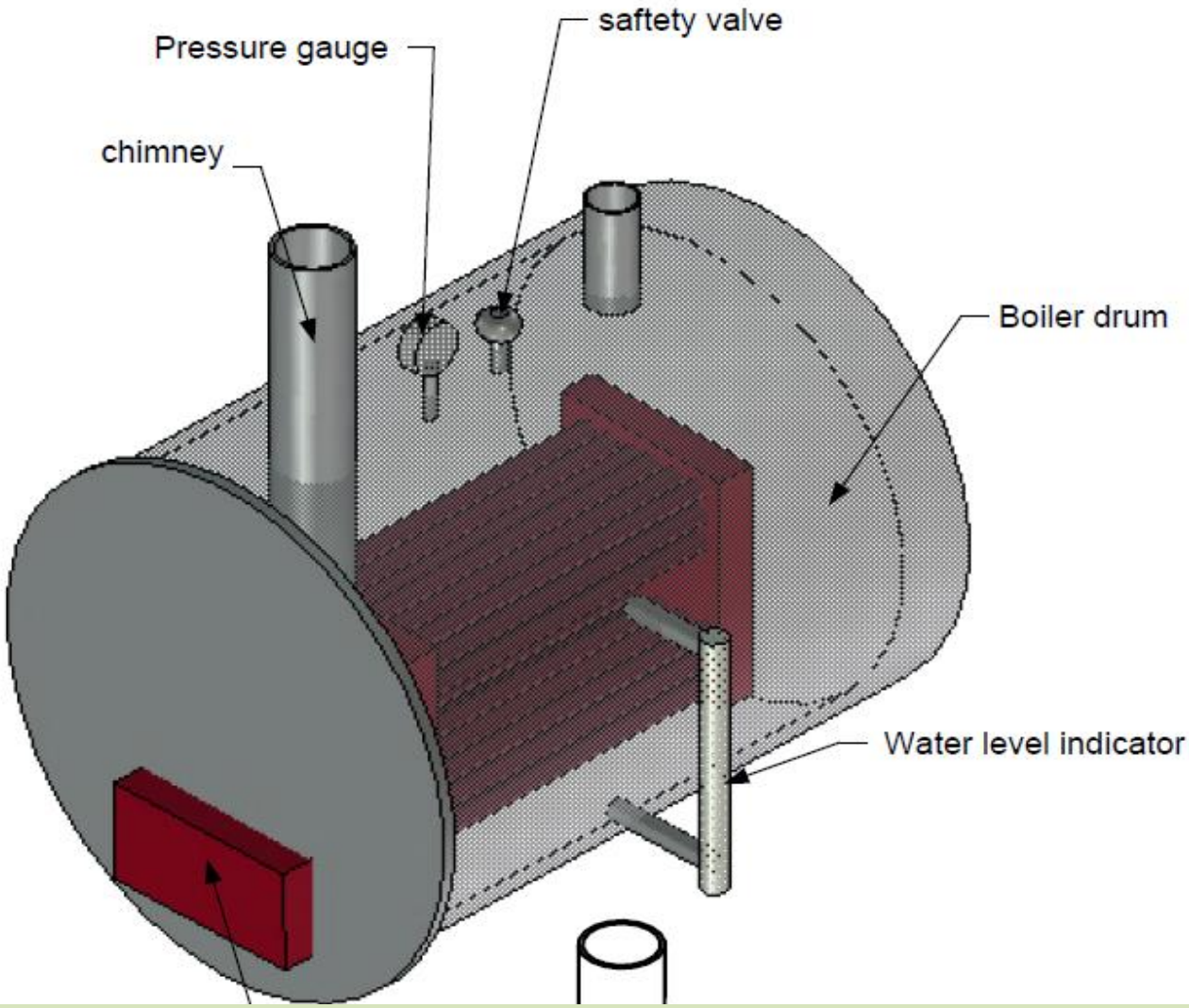
- Research - Gather information about past or similar designs.
- Develop conceptual approaches/techniques.
- Develop conceptual alternate designs
- Perform preliminary evaluation of alternative approaches.
- Select the best alternative.

THE WATER TUBE DESIGN (CONCEPT 1)

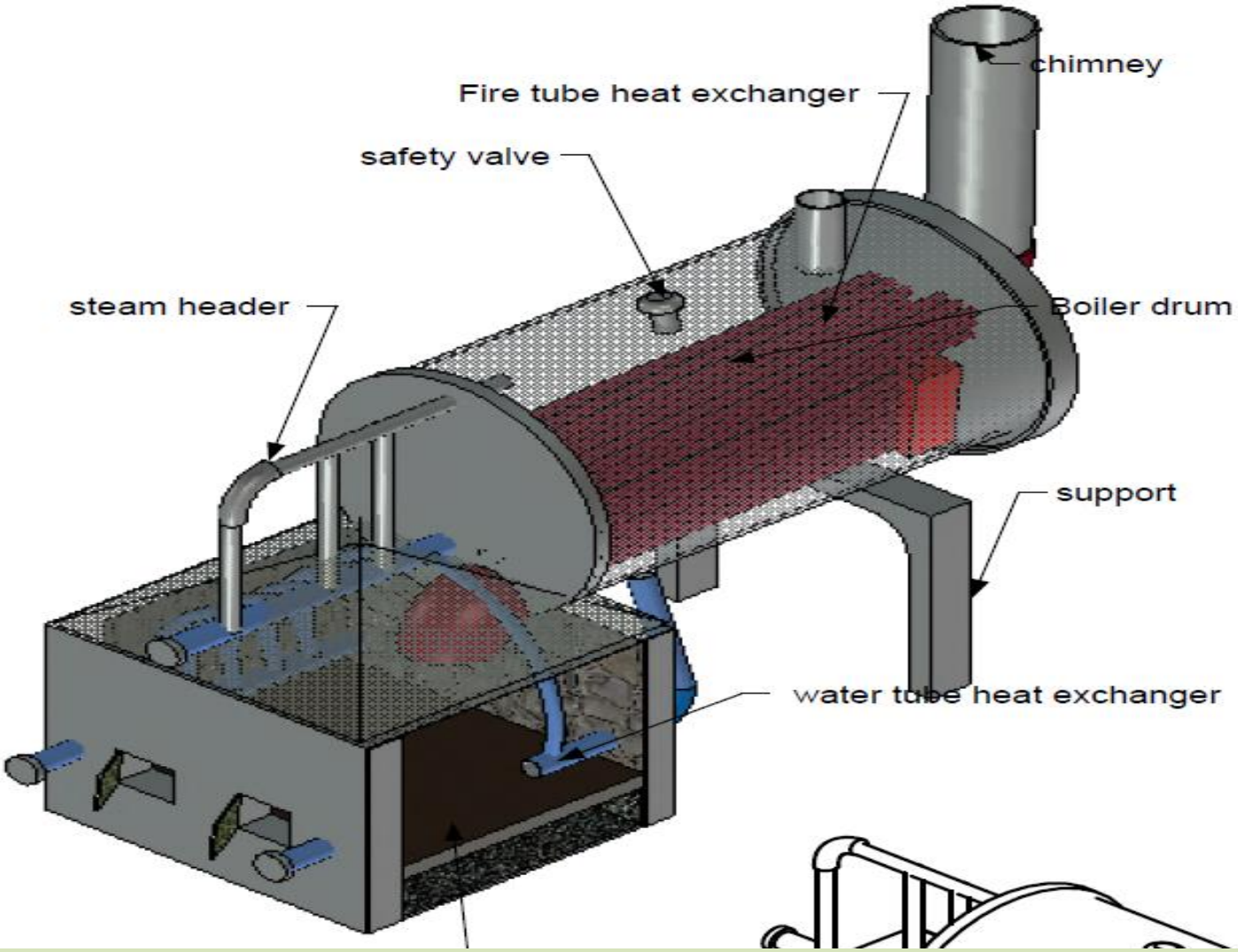


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THE FIRE TUBE DESIGN (CONCEPT 2)



COMBINATION DESIGN (CONCEPT 3)



Evaluation and Selection of Design Alternatives-Method A

1. If each criterion has the same weight as the others (equal weights):
2. Rank each design/process criterion from poor (1) to excellent (5)
3. Sum all rankings for each design/process ($C = \Sigma r_i$)
4. A higher score from step 2 indicates a favourable design/process

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Decision Matrix Table for Sterilizer-Method A

CRITERION	INITIAL COST			EASE OF MANUFACTURE			EASE OF MAINTENANCE			EASE OF ASSEMBLY			TOTAL SCORE
WEIGHT	0.25			0.25			0.25			0.25			
PARAMETERS	NUMBER OF PARTS			TYPES OF SHAPES			NUMBER OF FASTENERS			EASE OF ASSEMBLY			
	MAG.	Score	Value	MAG.	Score	Value	MAG.	Score	Value	MAG.	Score	Value	
CONCEPT 1	**15**	**3.0**	**0.75**	**F. Easy**	**3.0**	**0.75**	**7**	**4.0**	**1.00**	**F. Easy**	**3.0**	**0.75**	**3.25**
CONCEPT 2	**15**	**3.0**	**0.75**	**Easy**	**4.0**	**1.00**	**6**	**4.0**	**1.00**	**Easy**	**4.0**	**1.00**	**3.75**
CONCEPT 3	**20**	**2.0**	**0.50**	**V. Easy**	**5.0**	**1.25**	**12**	**3.0**	**0.75**	**V. Easy**	**5.0**	**1.25**	**3.75**

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Evaluation and Selection of Design Alternatives-Method B

1. If each criterion has a different weight (unequal weights)
2. Assign weights (w_i) to each criterion
3. Rank each design/process criterion from poor (1) to excellent (5)
4. Subtract the average ranking from each ranking (r_i-3)
5. Sum all weighted Score for each design/process, $C_{wi} = \Sigma w_i\ (r_i\text{-}3)$
6. A higher score from step "4" indicates a favorable design/process

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Decision Matrix Table for Sterilizer-Method B

CRITERION	INITIAL COST			EASE OF MANUFACTURE			EASE OF MAINTENANCE			EASE OF ASSEMBLY			TOTAL SCORE
WEIGHT	0.35			0.20			0.30			0.15			
PARAMETERS	NUMBER OF PARTS			TYPES OF SHAPES			NUMBER OF FASTENERS			EASE OF ASSEMBLY			
	MAG.	Score	Value	MAG.	Score	Value	MAG.	Score	Value	MAG.	Score	Value	
CONCEPT 1	15	3.0	1.05	F. Easy	3.0	0.60	7	4.0	1.20	F. Easy	3.0	0.45	3.30
CONCEPT 2	15	3.0	1.05	Easy	4.0	0.80	6	4.0	1.20	Easy	4.0	0.60	3.65
CONCEPT 3	20	2.0	0.70	V. Easy	5.0	1.00	12	3.0	1.05	V. Easy	5.0	0.75	3.50

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Implementation of Design

- Design Analysis (Stress/Sizing, deformation, deflection, manufacturability, etc.)
- Build Prototype (Modern Approach is to build a Computer Model and Simulation)
- Evaluation and Revision

Desired Attributes of an Engineer

- This is a list of basic, durable attributes into which can be mapped discipline specific skills reflecting the diversity of the overall environment in which the engineering profession is practiced.

- This list is a compilation of industry responses and roundtable discussions as distributed by The Boeing Company.

- A good understanding of engineering science, fundamentals Mathematics, Physical and life Science, and Information technology (far more than “Computer Literacy”).

- A good understanding of engineering design and manufacturing process; Engineering is about designing and making things people want.

Basic understanding of the context in which Engineering is Practiced

- Economics and business practices
- The Environment
- Customer and societal needs
- Government regulations

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Good Communication Skills

- Written
- Verbal
- Graphic
- Listening

High Ethical Standards

1. Profession ethics

2. Business and workplace ethics

- Ability to think critically and creatively.

- Profound understanding of the importance of teamwork.

- Flexibility - An ability and self-confidence to adapt to rapid/major technological changes.

- Curiosity and a desire to learn for life.

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Project Topics

1. **Health Equipment**
2. **Mechanical Systems**
3. **Waste Management**
4. **Energy/Thermal Systems**

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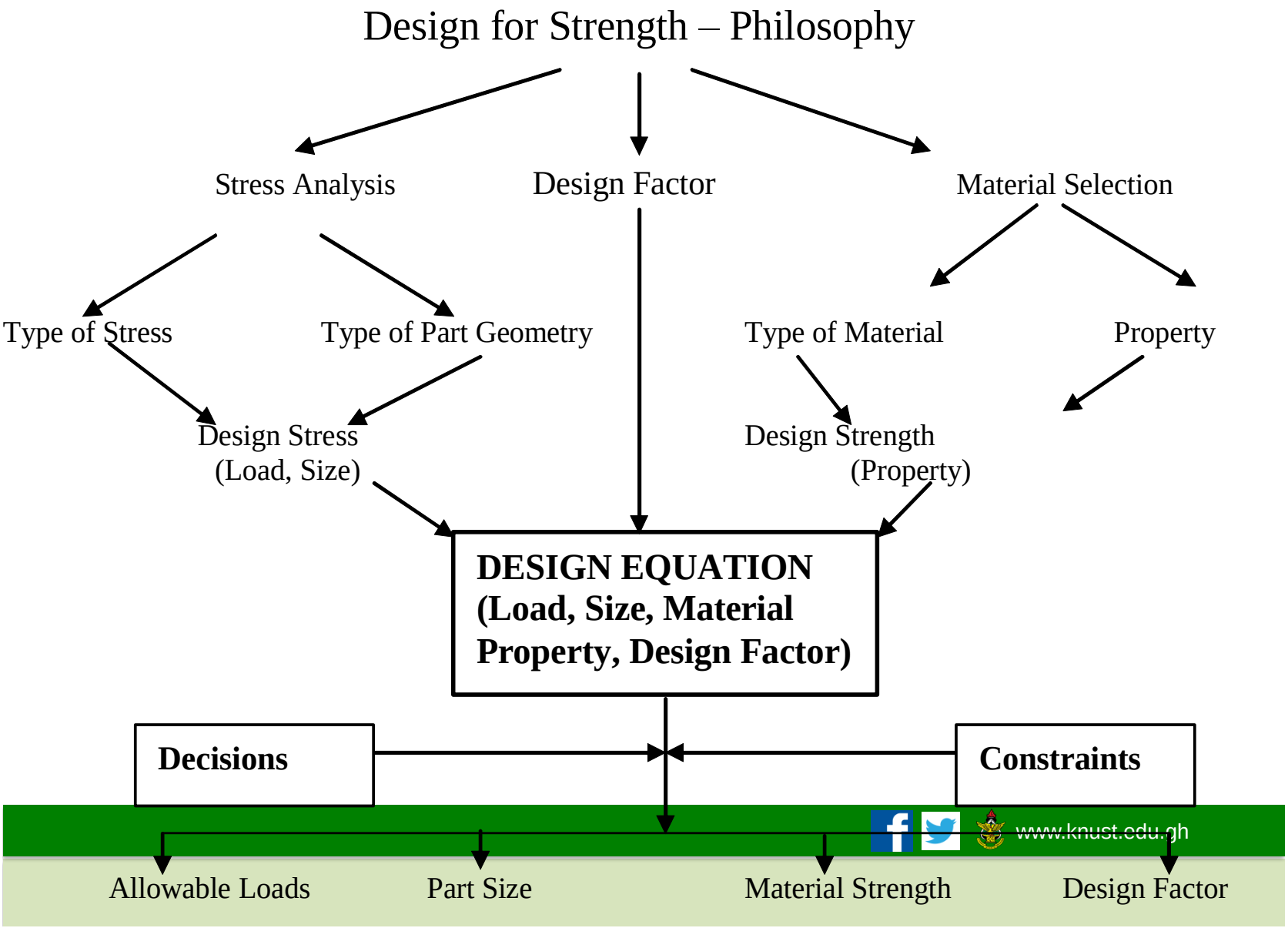
LECTURE 2

LOAD AND STRESS ANALYSIS

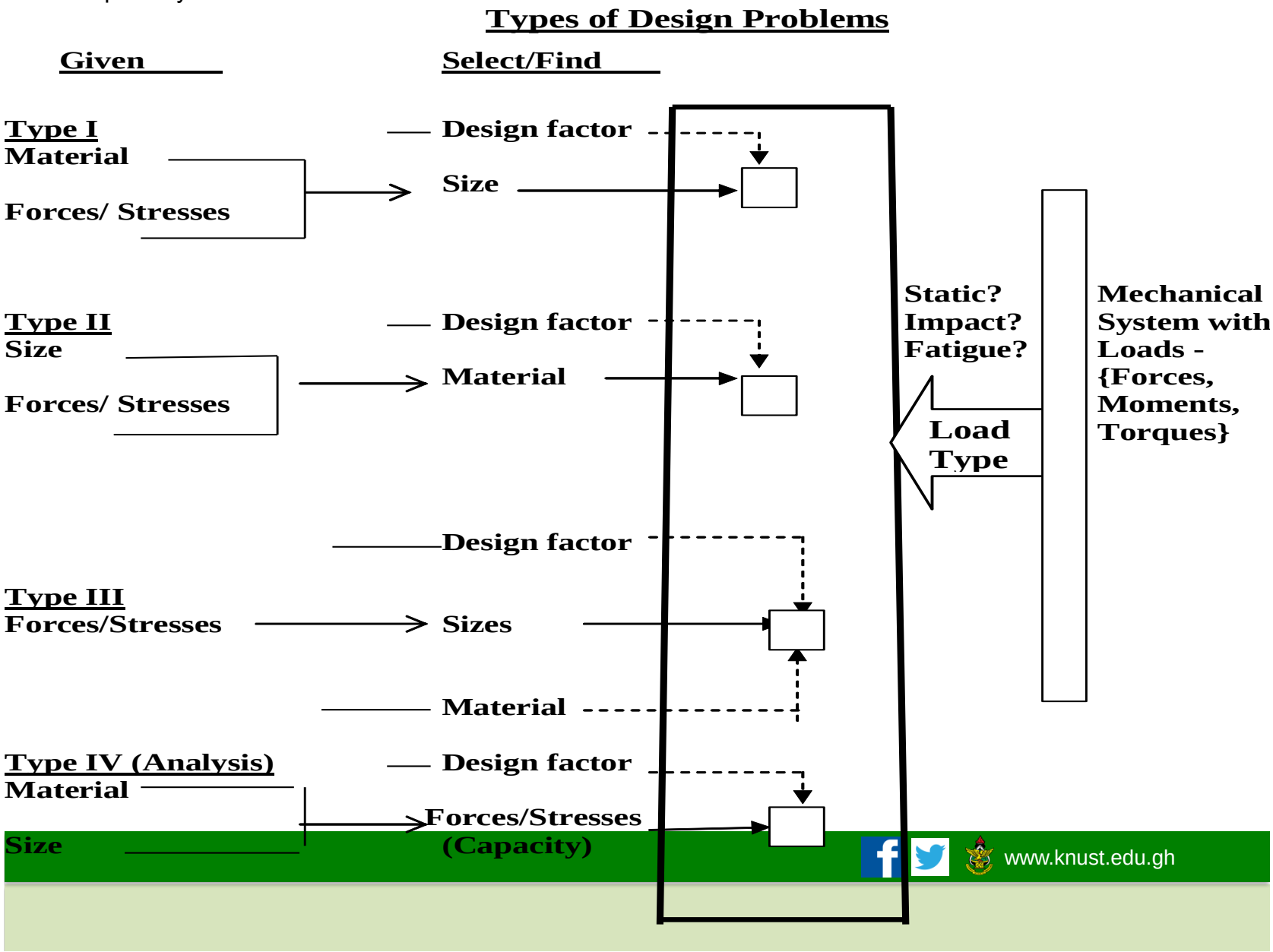
Lecture Outline

- Stress Concentration
- Normal Stress
- Shearing Stress
- Stress Analysis for Pure Loading
- Stress Analysis Combined Loading
- Working Stresses

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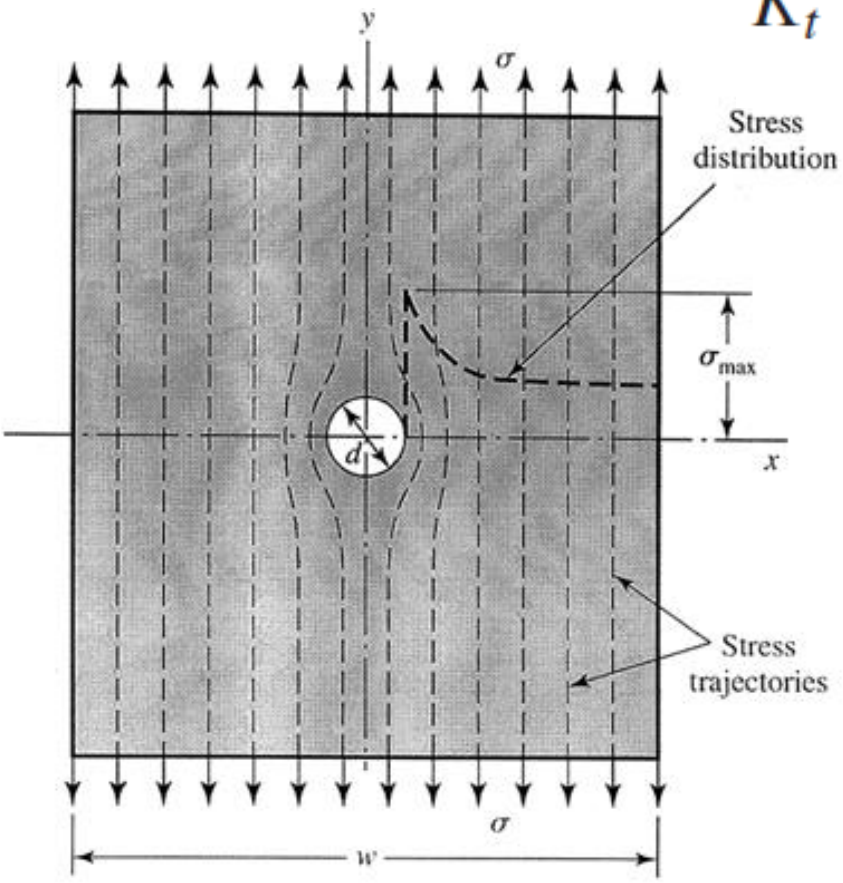
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Stress Concentration

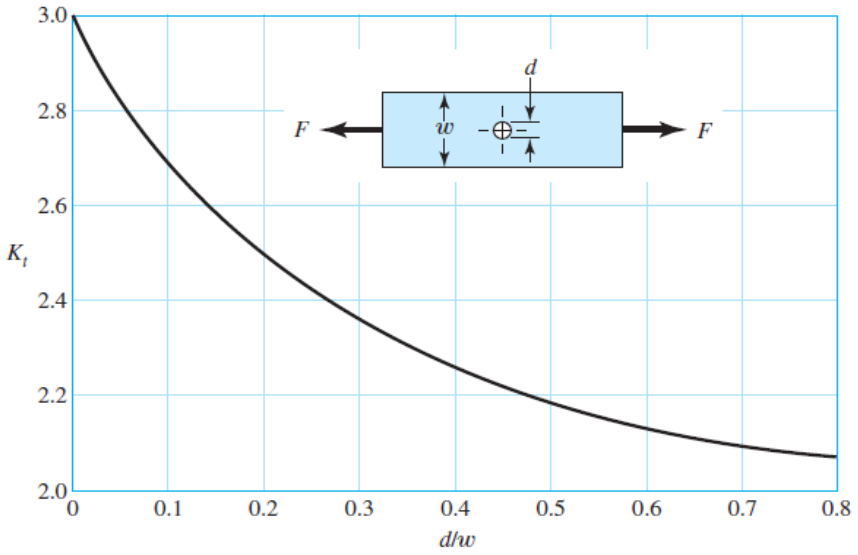
- Localized increase of stress near discontinuities
- K_t is Theoretical (Geometric) Stress Concentration Factor

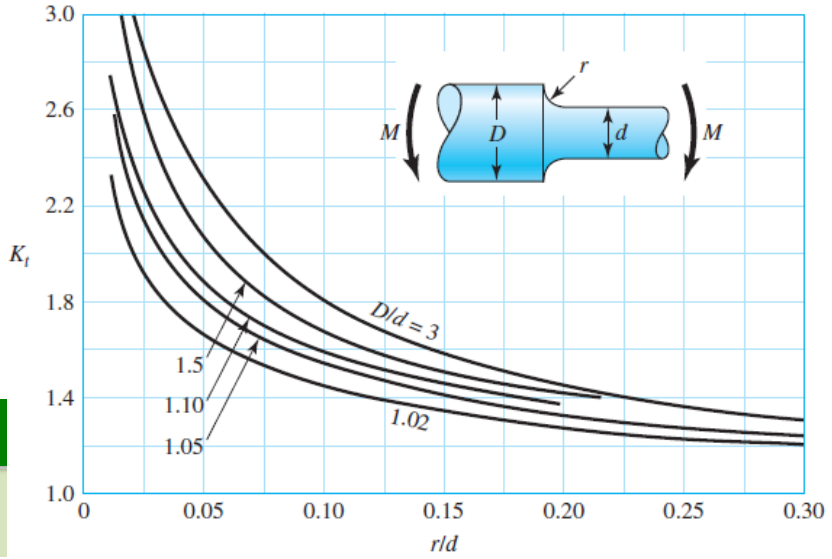
$$K_t = \frac{\sigma_{max}}{\sigma_0} \qquad K_{ts} = \frac{\tau_{max}}{\tau_0}$$



Theoretical Stress Concentration Factor

- Graphs available for standard configurations
- See Appendix A–15 and A–16 for common examples
- Many more in *Peterson's Stress-Concentration Factors*
- Note the trend for higher K_t at sharper discontinuity radius, and at greater disruption





Stress Concentration for Static and Ductile Conditions

- With static loads and ductile materials
  - Highest stressed fibers yield (cold work)
  - Load is shared with next fibers
  - Cold working is localized
  - Overall part does not see damage unless ultimate strength is exceeded
  - Stress concentration effect is commonly ignored for static loads on ductile materials

Techniques to Reduce Stress Concentration

- Increase radius
- Reduce disruption
- Allow “dead zones” to shape flowlines more gradually

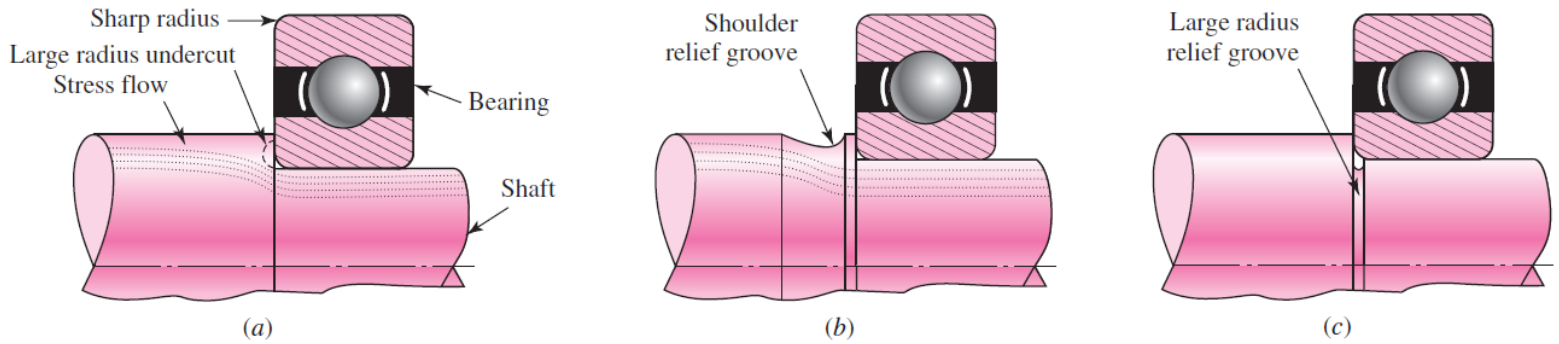
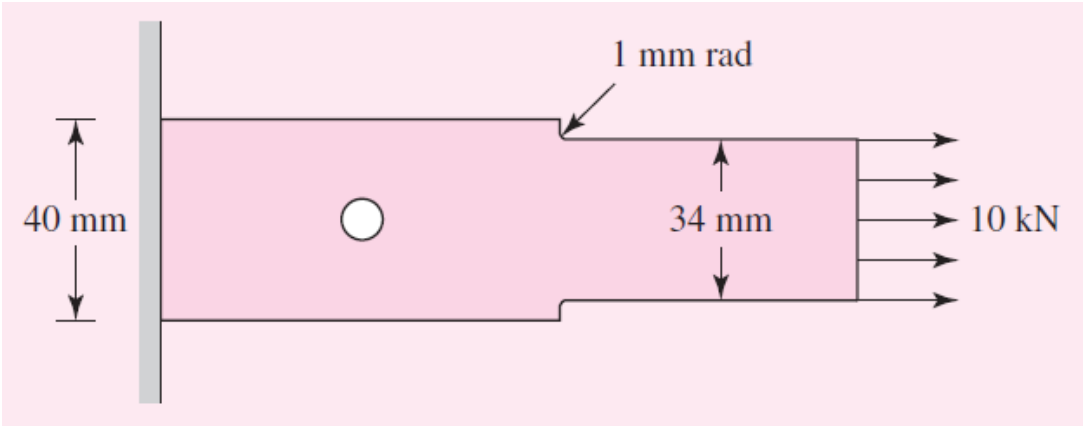


Figure 7–9

Example 2–1

The 2-mm-thick bar shown in Fig. 3–30 is loaded axially with a constant force of 10 kN. The bar material has been heat treated and quenched to raise its strength, but as a consequence it has lost most of its ductility. It is desired to drill a hole through the center of the 40-mm face of the plate to allow a cable to pass through it. A 4-mm hole is sufficient for the cable to fit, but an 8-mm drill is readily available. Will a crack be more likely to initiate at the larger hole, the smaller hole, or at the fillet?



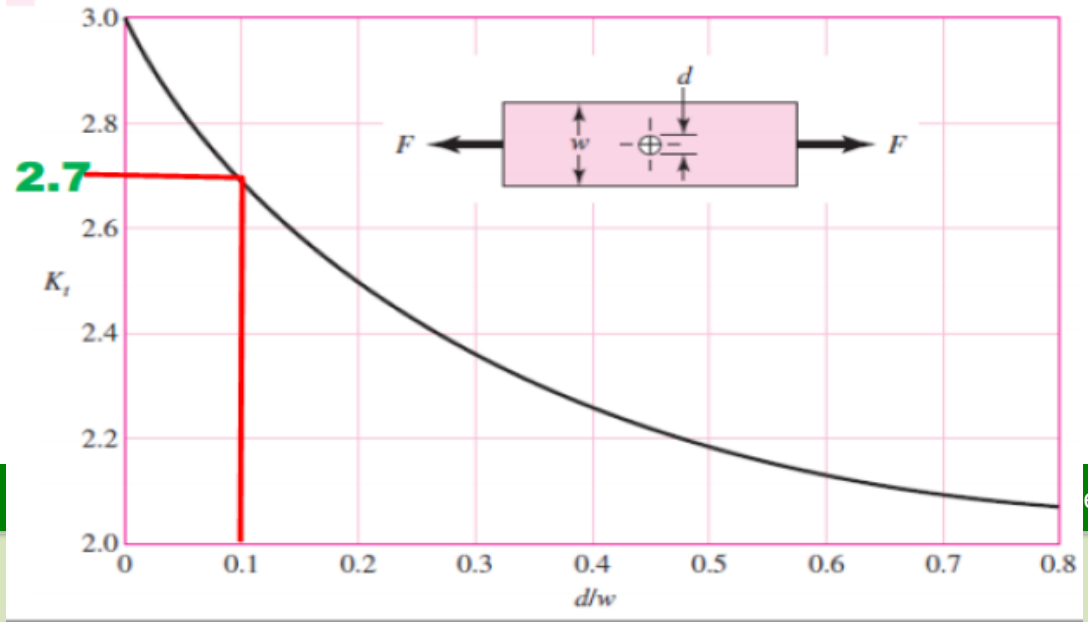
Example 2–1 (continued)

Since the material is brittle, the effect of stress concentrations near the discontinuities must be considered. Dealing with the hole first, for a 4-mm hole, the nominal stress is

$$\sigma_0 = \frac{F}{A} = \frac{F}{(w-d)t} = \frac{10\ 000}{(40-4)2} = 139\ \text{MPa}$$

The theoretical stress concentration factor, from Fig. A–15–1, with $d/w = 4/40 = 0.1$, is $K_t = 2.7$. The maximum stress is

$$\sigma_{\max} = K_t \sigma_0 = 2.7(139) = 380\ \text{MPa}$$



Example 2–1 (continued)

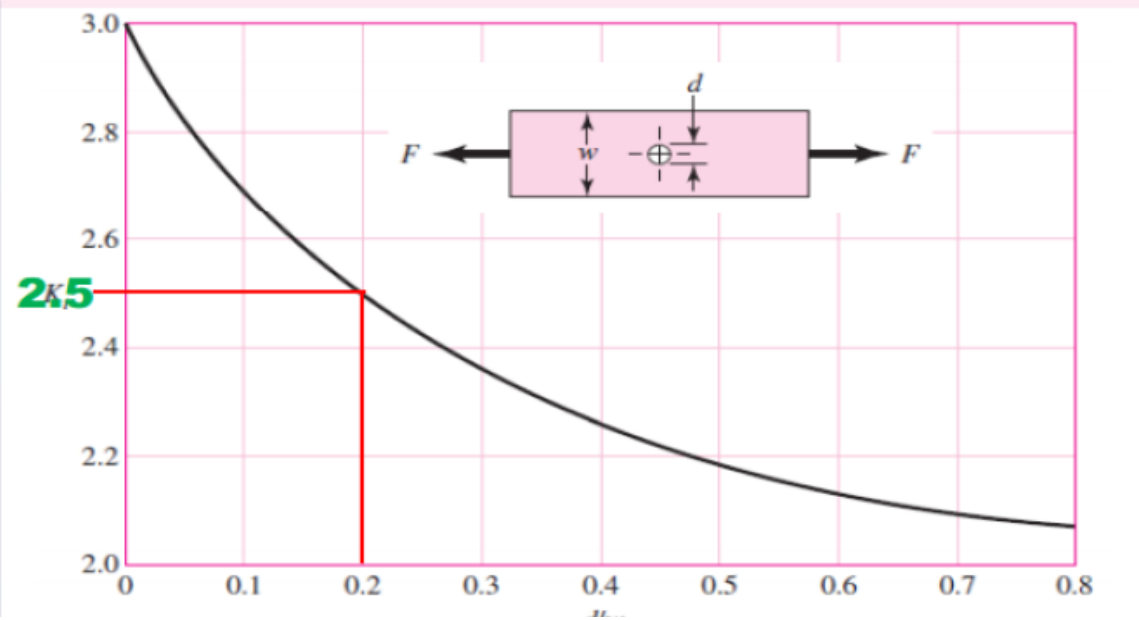
Similarly, for an 8-mm hole,

$$\sigma_0 = \frac{F}{A} = \frac{F}{(w-d)t} = \frac{10\ 000}{(40-8)2} = 156 \text{ MPa}$$

With $d/w = 8/40 = 0.2$, then $K_t = 2.5$, and the maximum stress is

$$\sigma_{\max} = K_t \sigma_0 = 2.5(156) = 390 \text{ MPa}$$

Though the stress concentration is higher with the 4-mm hole, in this case the increased nominal stress with the 8-mm hole has more effect on the maximum stress.



Example 2–1 (continued)

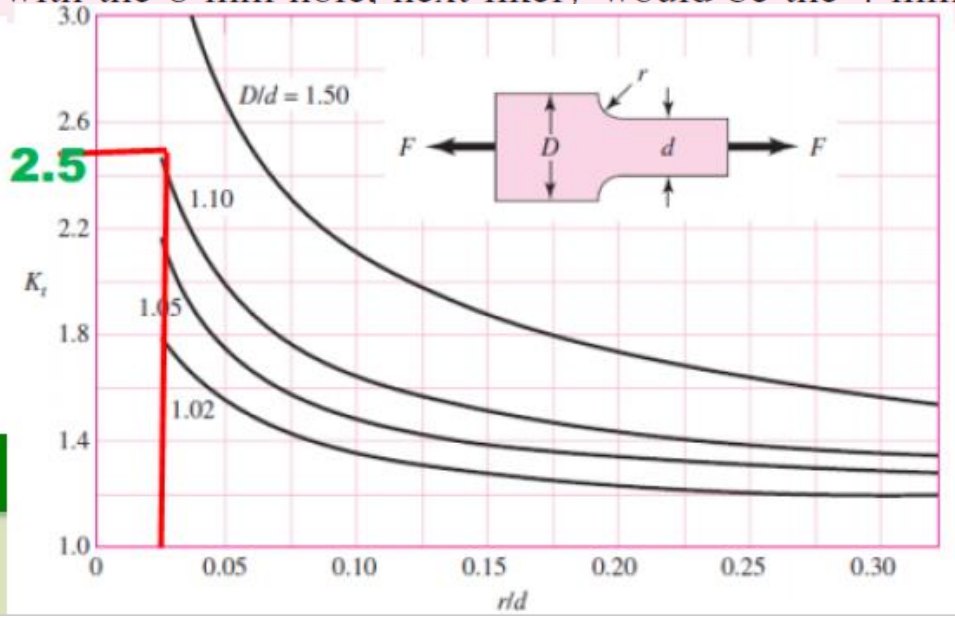
For the fillet,

$$\sigma_0 = \frac{F}{A} = \frac{10\ 000}{(34)2} = 147\ \text{MPa}$$

From Table A–15–5, $D/d = 40/34 = 1.18$, and $r/d = 1/34 = 0.026$. Then $K_t = 2.5$.

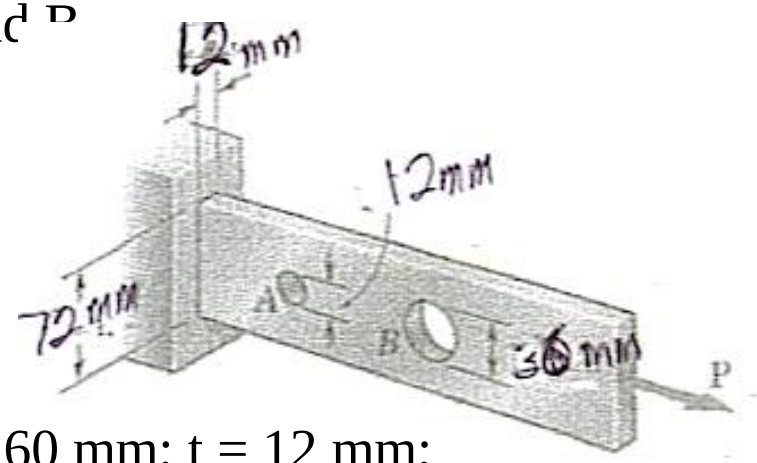
$$\sigma_{\max} = K_t \sigma_0 = 2.5(147) = 368\ \text{MPa}$$

The crack will most likely occur with the 8-mm hole, next likely would be the 4-mm hole, and least likely at the fillet.



Example 2–2

- Two holes have been drilled through a long steel bar that is subjected to a concentric axial load as shown. For P= 8.5 kN, determine the maximum values of the stress at A and B



- **Solution**
- Section Properties

At hole A, r = 6 mm; d = 72 mm – 12 = 60 mm; t = 12 mm;

A_{net}=dt = 7.20 x 10^{-4} m^2

At hole B, r = 18 mm; d = 72 – 36 = 36 mm; t = 12 mm;

A_{net}= 4.32 x 10^{-4} m^2

Example 2–2 (continued)

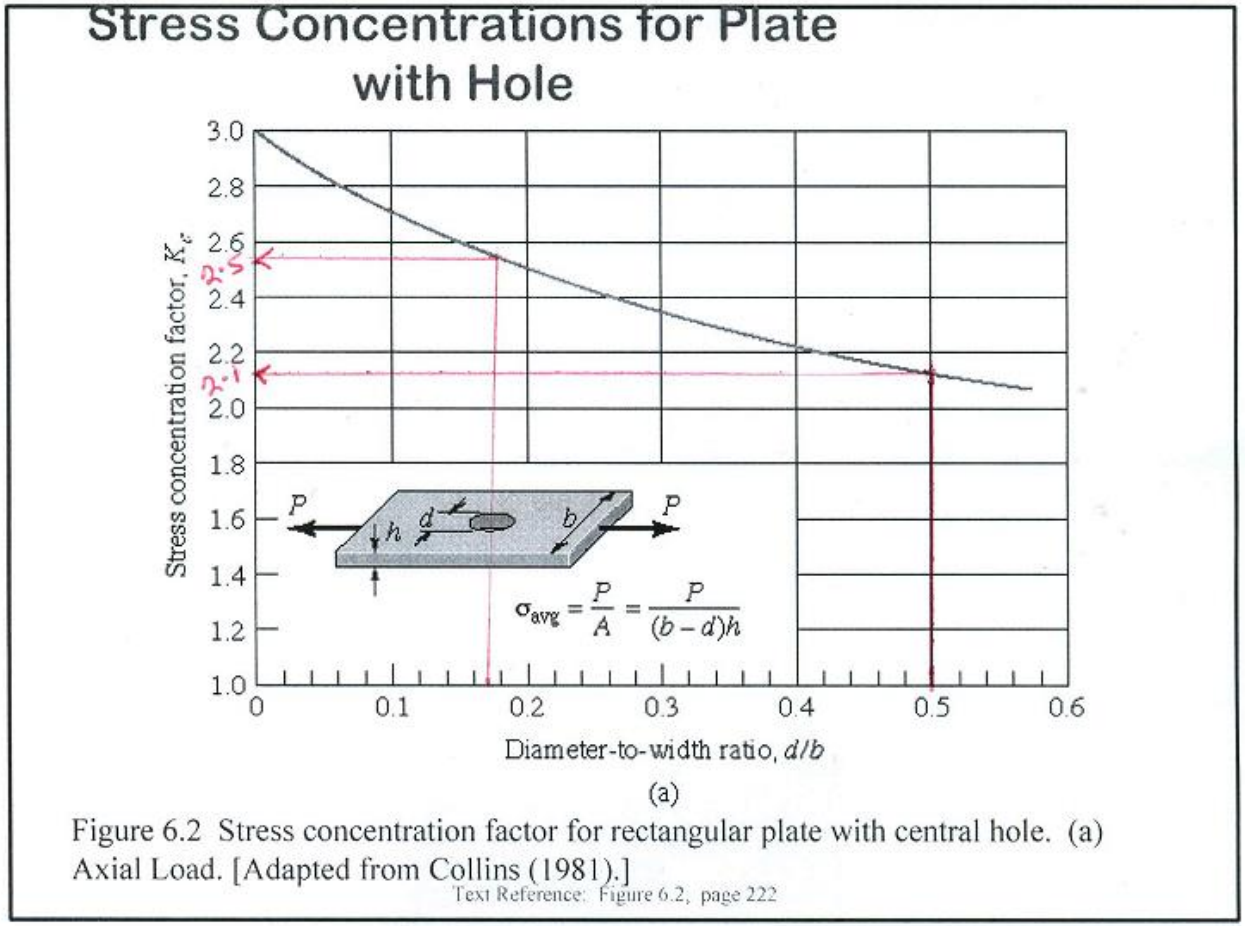


Figure 6.2 Stress concentration factor for rectangular plate with central hole. (a) Axial Load. [Adapted from Collins (1981).]

Text Reference: Figure 6.2, page 222

Example 2–2 (continued)

Force-couple System at A and B

- F = P = 8.5 kN

Axial Stress

at A, $\sigma_{nom} = F/A_{net} = 11.8$ MPa

r/d = 0.10; From Figure, K = 2.50

$$\therefore \sigma_{max} = K\sigma_{nom} = \mathbf{31.86\ MPa}$$

at B, $\sigma_{nom} = F/A_{net} = 19.77$ MPa

r/d = 0.50; From Fig K = 2.10

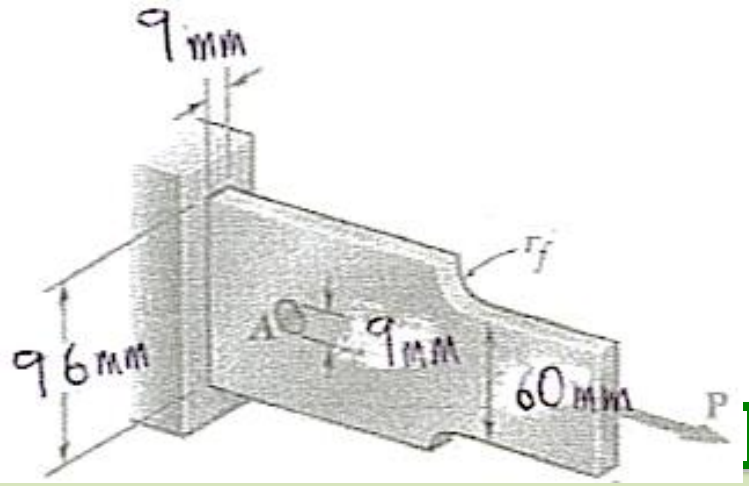
$$\therefore \sigma_{max} = K\sigma_{nom} = \mathbf{41.52\ MPa}$$

Example 3–3

A centric axial load P is applied to the steel bar shown. A hole of 9 mm diameter is drilled through the steel plate. Determine

a. the radius r_f of the fillets for which the same maximum stress occurs at the hole A and at the fillets

b. the corresponding maximum allowable stress if the allowable load **P** is 7.5 kN



Example 2–3 (continued)

Section Properties

At the hole ,r = 4.5 mm; d = 96 – 9 = 87 mm; t = 9 mm;

$A_{net} = dt$ = 7.83 x 10^{-4} m^2

For fillet, D = 96 mm; d = 60 mm;

$A_{net} = dt$ = 5.40 x 10^{-4} m^2

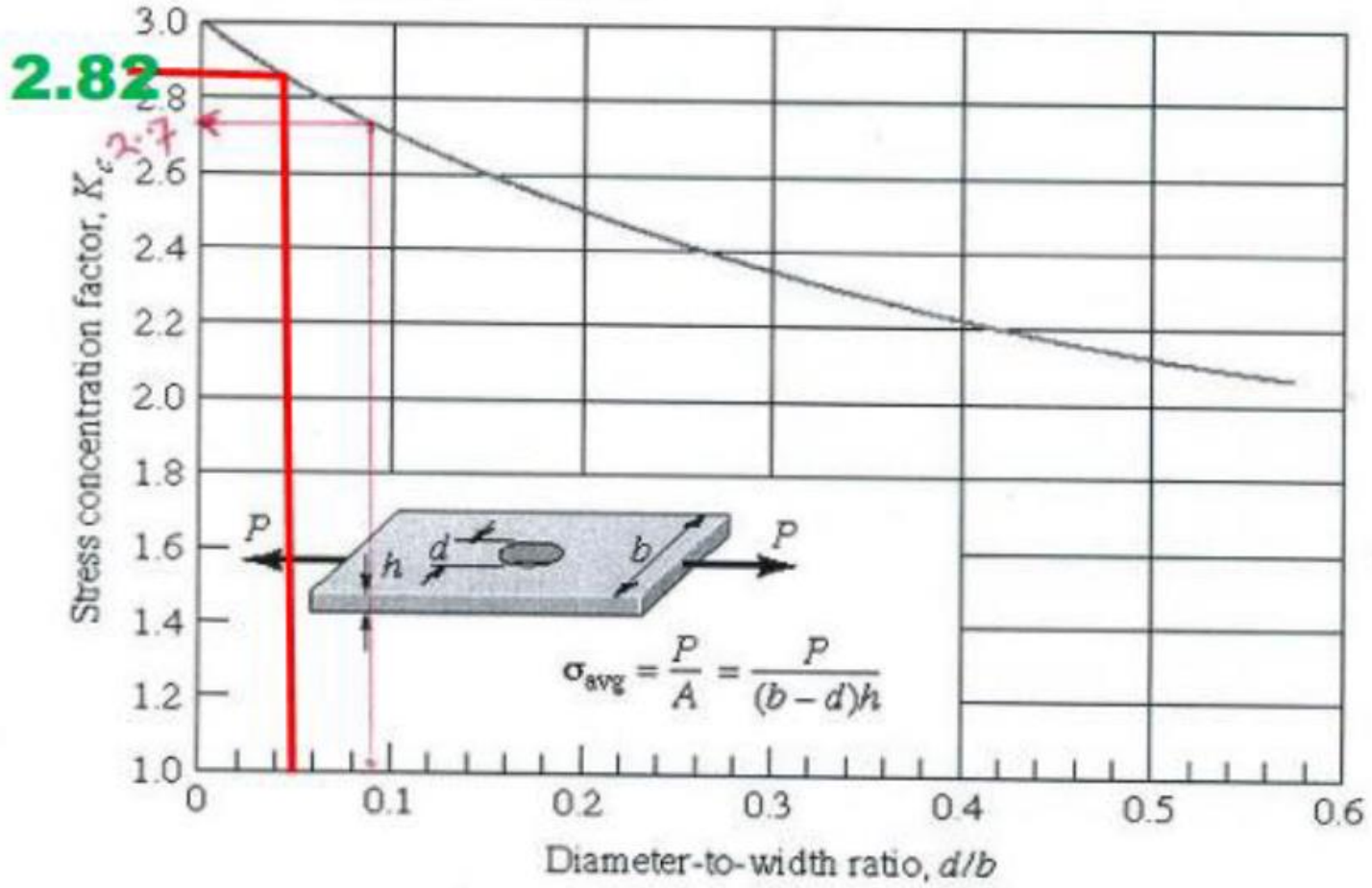
Force-couple System at A and B

F = P = 7.5 kN

Axial Stress

at the hole, $\sigma_{nom} = F/A_{net}$ = 9.58 MPa

Example 2–3 (continued)



$r/d = 0.05$; From Fig K = 2.82

$\therefore \sigma_{max} = K\sigma_{nom} = \mathbf{27\ MPa}$

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Example 2–3 (continued)

at the fillet, $\sigma_{nom} = F/A_{net}$ = 13.90 MPa

This implies $13.9K_{fillet} = 27$

Hence, $K_{fillet} = 1.945$

= 1.6;

Example 2–3 (continued)

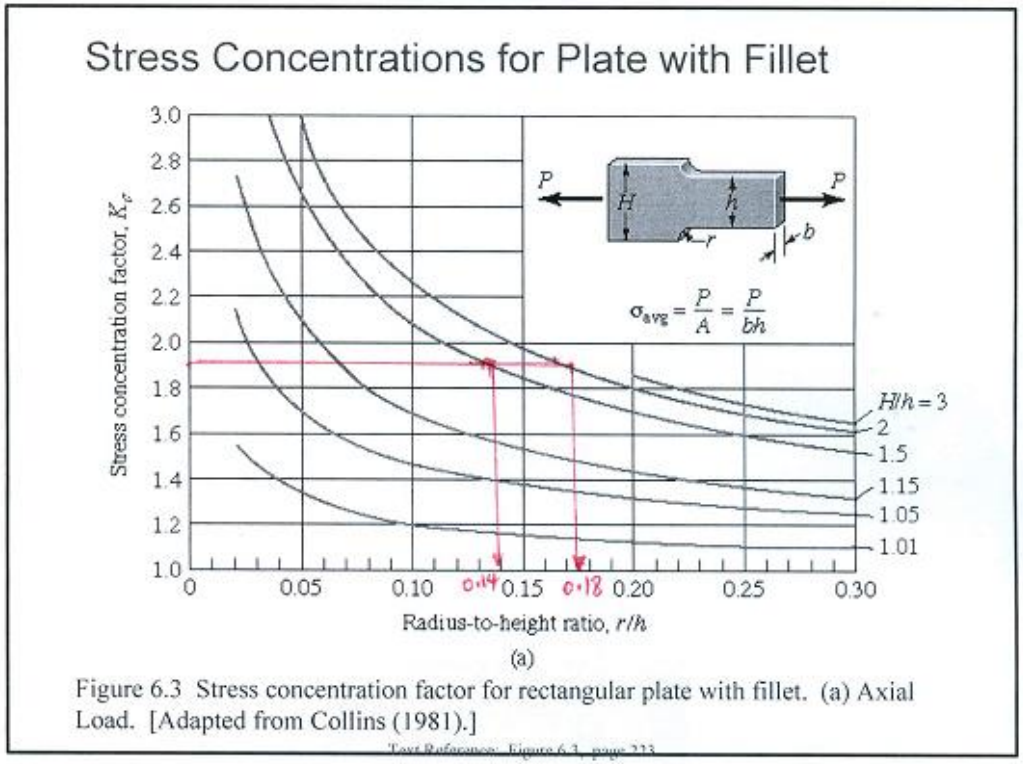


Figure 6.3 Stress concentration factor for rectangular plate with fillet. (a) Axial Load. [Adapted from Collins (1981).]

Text Reference: Figure 6.3, page 223

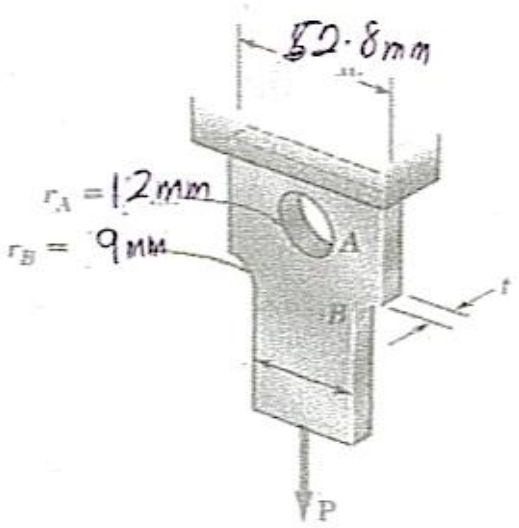
For K_t=1.945 and D/d = 1.6,

Therefore,

From Figure, r_c = **10 mm**

Example 2–4

For P=8.5kN, determine the allowable stress if the thickness t is 20 mm.



Section Properties

At the hole A, r = 12 mm;

d = 52.8 – 24 = 28.8 mm; t = 20 mm;

$A_{net} = dt$ = 5.76 x 10^{-4} m^2

At the fillet B, D = 52.8 mm; d = 38.4 mm; r_f = 9 mm;

$A_{net} = dt$ = 3.46 x 10^{-4} m^2

Example 2–4 (continued)

Force-couple System at A and B

F = P = 8.5 kN

Axial Stress

at the hole A, $\sigma_{nom} = F/A_{net}$ = 14.76 MPa

$\frac{r}{d}$ = 0.417; From Fig K = 2.22

$\therefore \sigma_{\max} = K\sigma_{nom}$ = **32.76 MPa**

at the fillet B, $\sigma_{nom} = F/A_{net}$ = 24.57 MPa

$D/d = 1.375$ and $r_f/d \approx 0.23$; From Fig K = 1.70

$\therefore \sigma_{\max} = K\sigma_{nom}$ = **55.7 MPa**

Normal Stress

Axial Stress

$$\sigma_{axial} = \pm F/A$$

- where F is applied force perpendicular to the cross-section area, A.
- The stress may be in tension (+) or compression (-)

Bending Stress

$$\sigma_{bending} = \pm M/S$$

$$M = PL \qquad S = I/y_{\max}$$

- where S is the Bending Section Modulus and I is the moment of inertia.

Normal Stress

Tangential Stress (Thin-walled Pressure Vessel)

$$\sigma_\theta = \begin{bmatrix} pr/t \text{ for cylindrical vessel} \\ pr/2t \text{ for spherical vessel} \end{bmatrix}$$

Axial Stress in (Thin-walled Pressure Vessel)

$$\sigma_{axial} = \begin{bmatrix} pr/2t \text{ for cylindrical vessel} \\ pr/2t \text{ for spherical vessel} \end{bmatrix}$$

- where p is the internal pressure, t is the wall thickness and r is the mean radius

Shearing Stress

Torsional Shearing Stress

$$\tau = T/Z \qquad Z = J/y_{\max}$$

where Z is the Torsional Section Modulus.

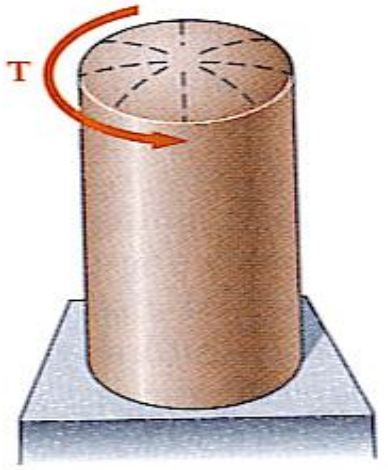
Transverse Shearing Stress

$$\tau_{xy} = \begin{bmatrix} 3V/2A \ for \ rectangular \ section \\ 4V/3A \ for \ solid \ cylindrical \ section \\ 2V/A \ for \ thin \ wall \ cylindrical \ section \end{bmatrix}$$

- τ_V is zero at maximum bending locations.

Example 2–5

- A torque of magnitude T = 12kN m is applied to the end of a tank containing compressed air under a pressure of 8 MPa. The tank has a 180-mm inner diameter and a 12-mm wall thickness, determine the normal stress and the shearing stress in the tank.



Solution

- Section Properties

$$r_i = d_i/2 = 90mm$$

$$c = r_o = r_i + t = 102mm$$

$$J = \pi/2\left(r_o^4 - r_i^4\right) == 66.97x10^{-6}\, m^4$$

Example 2–5 (continued)

Force-couple system

Pressure, p = 8 MPa and T = 12 kN.m

Stresses in Pressure Vessel

Tangential:

$$\sigma_x = pr/t = 60MPa$$

Axial:

$$\sigma_y = pr/2t = 30MPa$$

Total Stresses

σ_x = 60 MPa;

σ_y = 30 MPa;

τ_{xy} = 18.3 MPa;

Torsional Shearing Stress

$$\tau_{xy} = Tc/J = 18.3MPa$$

Working Stresses

- The average stress : $\sigma_{avg} = \frac{1}{2}(\sigma_x + \sigma_y)$
- The radius : $R = \sqrt{\tau_{xy}^2 + \left[\frac{1}{2}\left(\sigma_x - \sigma_y\right)\right]^2}$
- Maximum Principal Stress : $\sigma_1 = \sigma_{avg} + R$
- Minimum Principal Stress : $\sigma = \sigma_{avg} - R$
- Maximum Shear Stress : $\tau_{\max} = R$

Example 2–6

Determine the principal stresses, and. the maximum shearing stress in example 2-5.

Solution

From Example 2-4, the stresses are

σ_x = 60 MPa; σ_y = 30MPa; τ_{xy} = 18MPa;

Mohr's Circle

$$\sigma_{avg} = \tfrac{1}{2}(\sigma_x + \sigma_y) = 45\ MPa$$

$$R = \sqrt{\tau_{xy}^2 + \left[\tfrac{1}{2}\left(\sigma_x - \sigma_y\right)\right]^2} = 23\ MPa$$

Example 2–6 (continued)

$$\sigma_{\max} = \sigma_{avg} + R = 68\ MPa$$

$$\sigma_{\min} = \sigma_{avg} - R = 22\ MPa$$

$$\tau_{\max} = R = 23\ MPa$$

THANK YOU

Compiled by NKAY STUDY

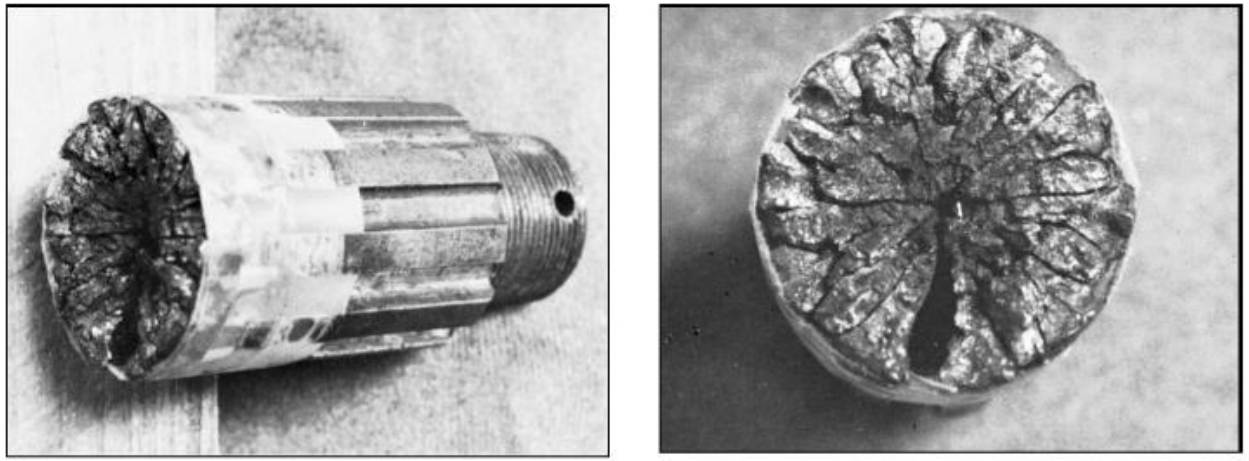
LECTURE 3

FAILURES RESULTING FROM STATIC LOADING

Lecture Outline

- Introduction-Failure Examples
- Required Background-Stress Analysis
- Required Background- Design Philosophy
- Basic Design Concepts
- Ductile Versus Brittle Behavior
- Fundamental Design Equation for Ductile Failure
- Design Principle for Brittle Materials
- Selection of Failure Criteria in Flowchart Form

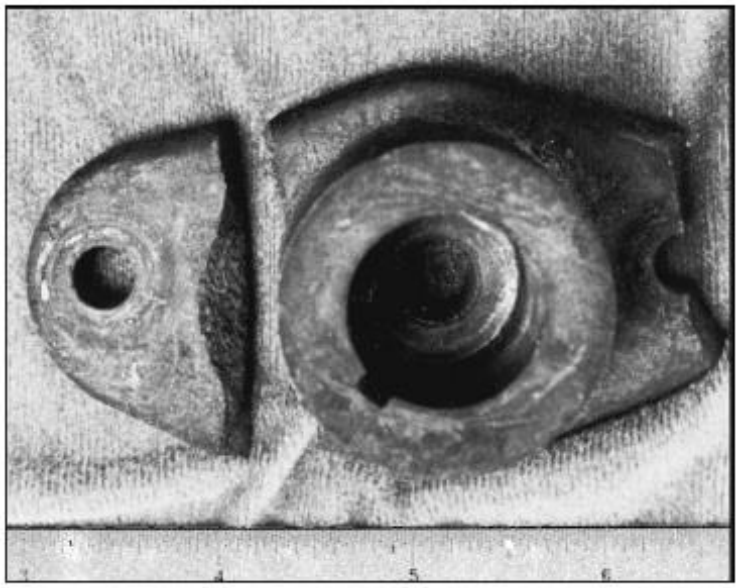
Introduction-Failure Examples



(a) (b)

- Failure of truck driveshaft spline due to corrosion fatigue

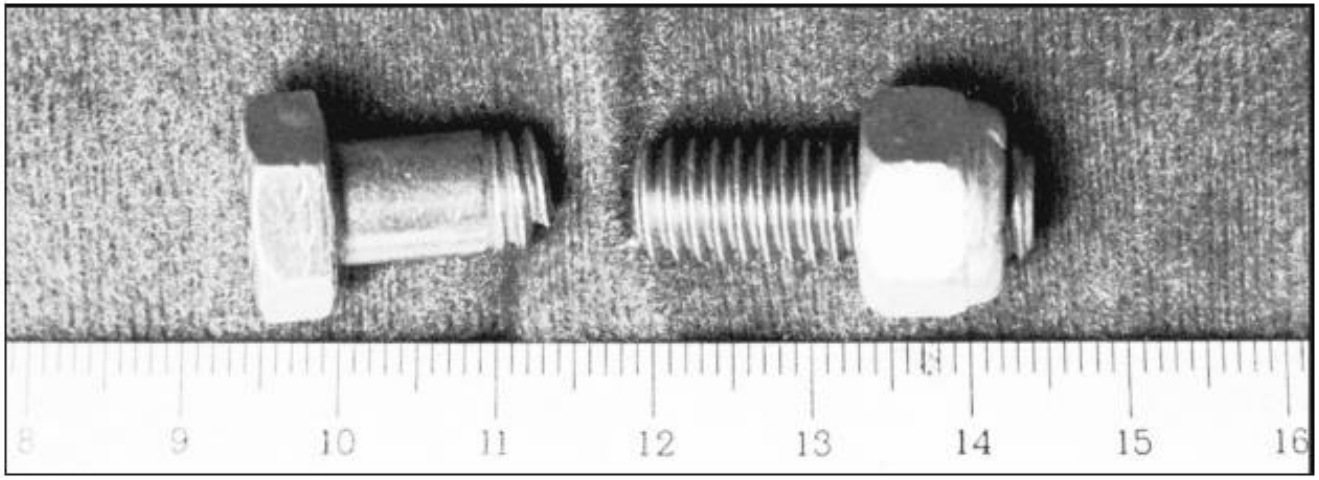
Introduction-Failure Examples



- Impact failure of a lawn-mower blade driver hub.

- The blade impacted a surveying pipe marker.

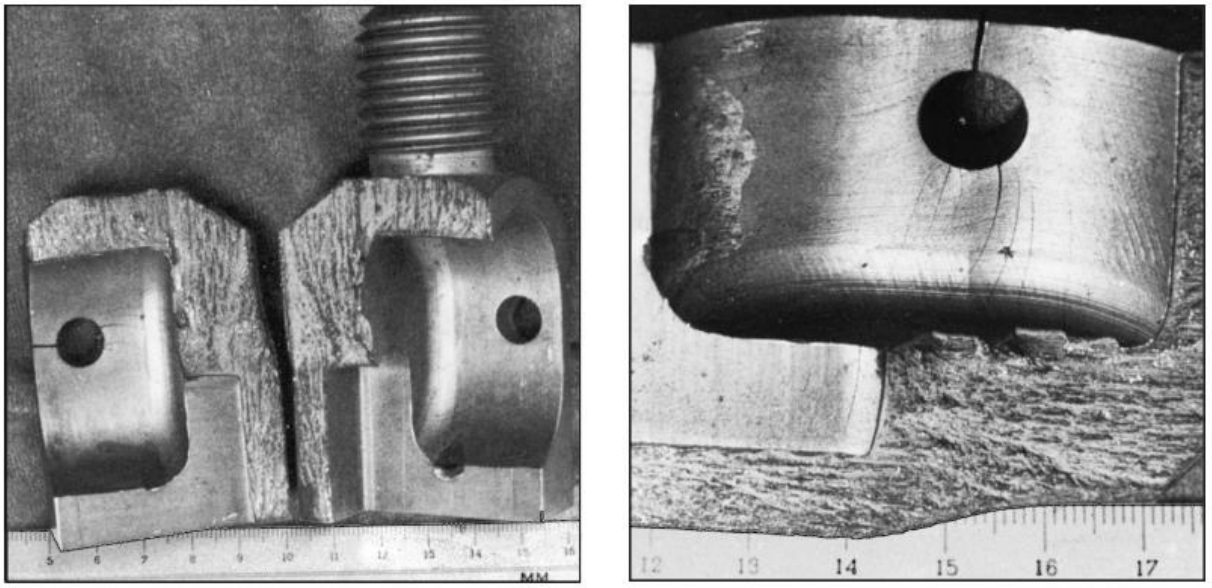
Introduction-Failure Examples



- Failure of an overhead-pulley retaining bolt on a weightlifting machine.

- A manufacturing error caused a gap that forced the bolt to take the entire moment load.

Introduction-Failure Examples



(a) (b)

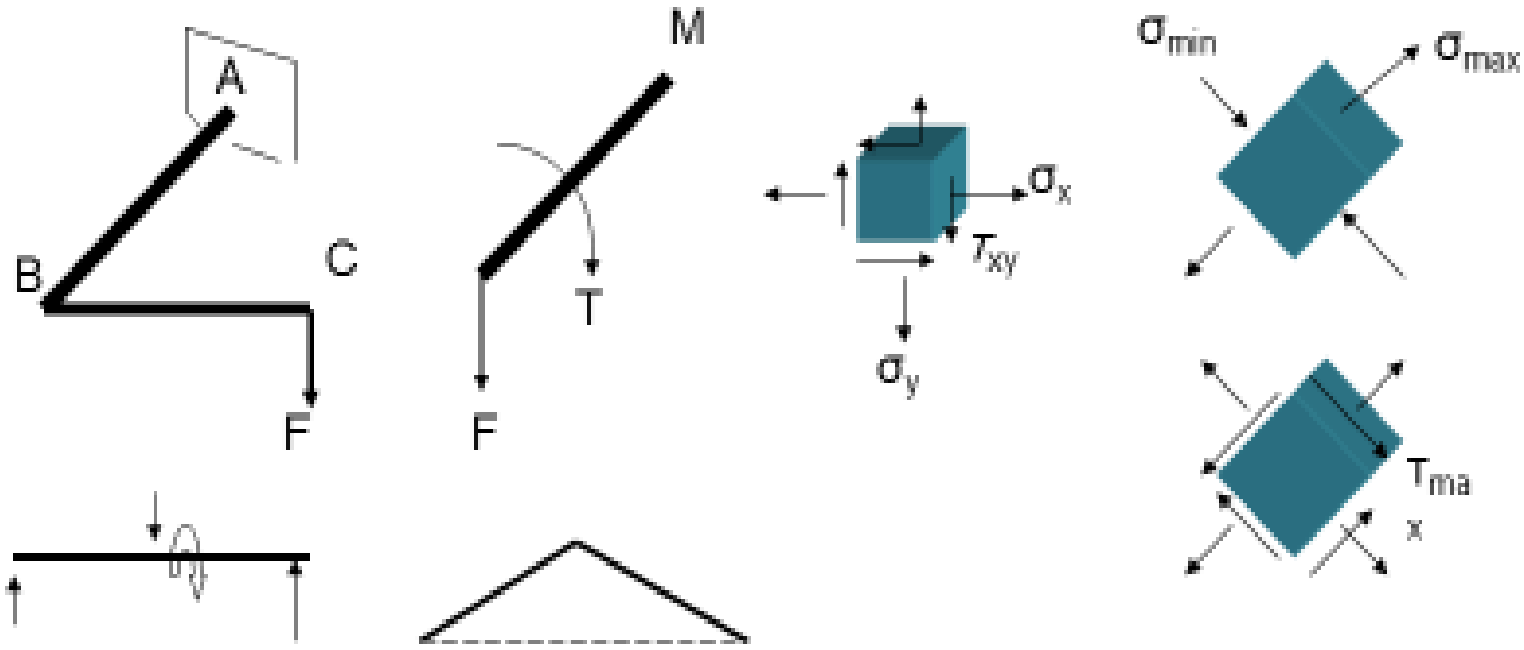
Chain test fixture that failed in one cycle.

To alleviate complaints of excessive wear, the manufacturer decided to case-harden the material

(a) Two halves showing brittle fracture initiated by stress concentration

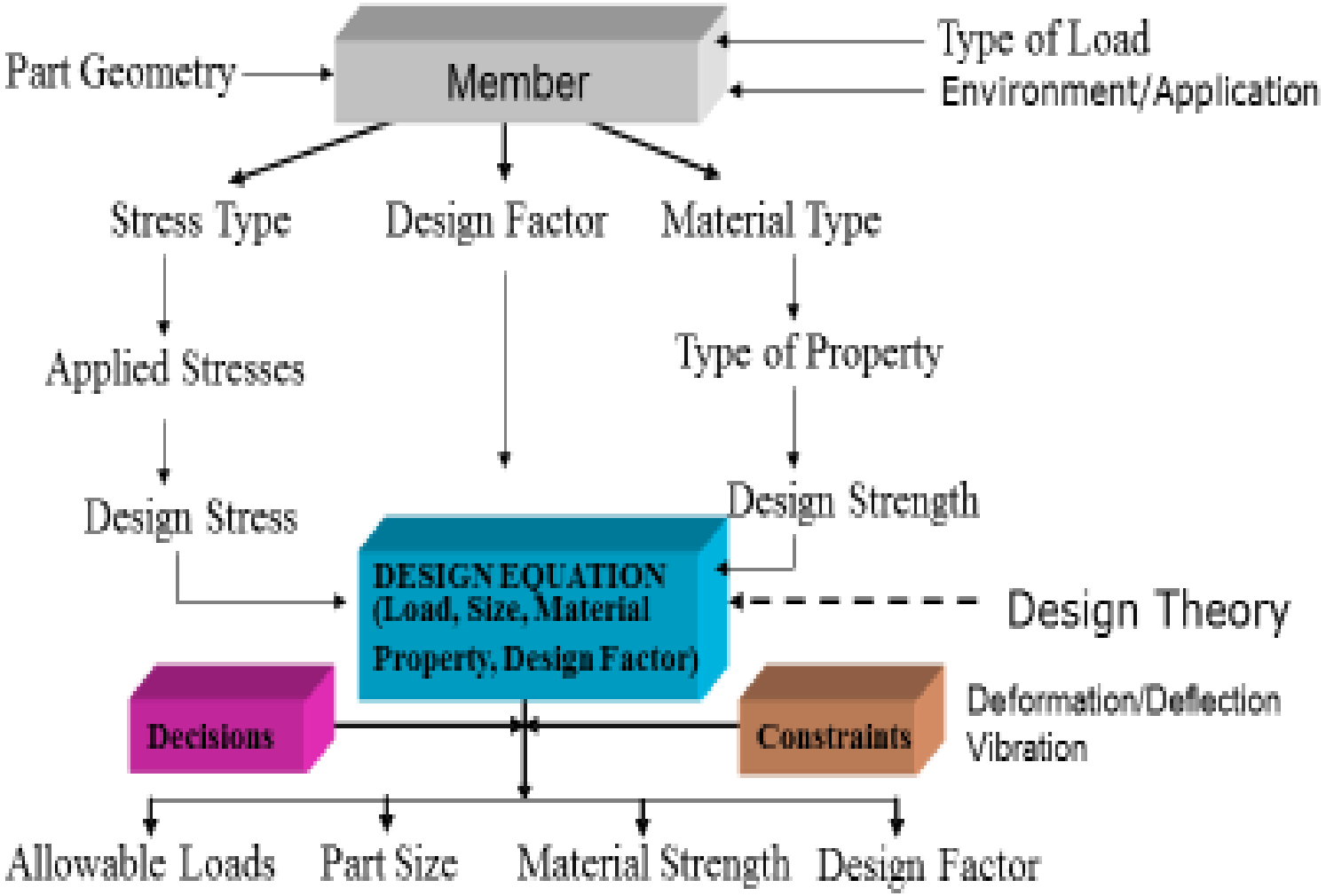
(b) Enlarged view showing cracks induced by stress concentration at the support-pin holes

Required Background-Stress Analysis



Given Mechanical System → Load Synthesis (F, T, M) → Applied Stresses (σ_x , σ_y , τ_{xy}) → Principal Stresses (σ_{max} , σ_{min} , τ_{max})

Required Background- Design Philosophy

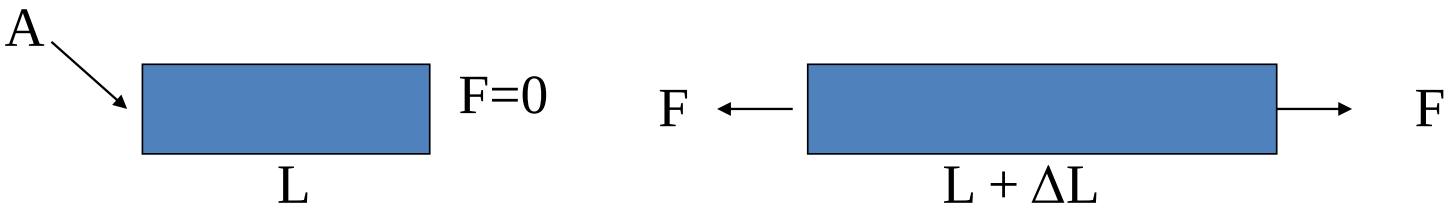


Basic Design Concepts

- Determination of a feasible part configuration (geometry), safe sizes (dimensions), and suitable material.
- *Stress* is applied to or induced in a part through loading conditions.
- *Strength* is an intrinsic property of a material. A part with no load possesses its strength.
- A part is said to have failed if it is stressed to a value that exceeds its strength.
- Failure means that the ***design stress*** in the machine member has reached the ***design strength*** of the material.

Ductile Versus Brittle Behavior

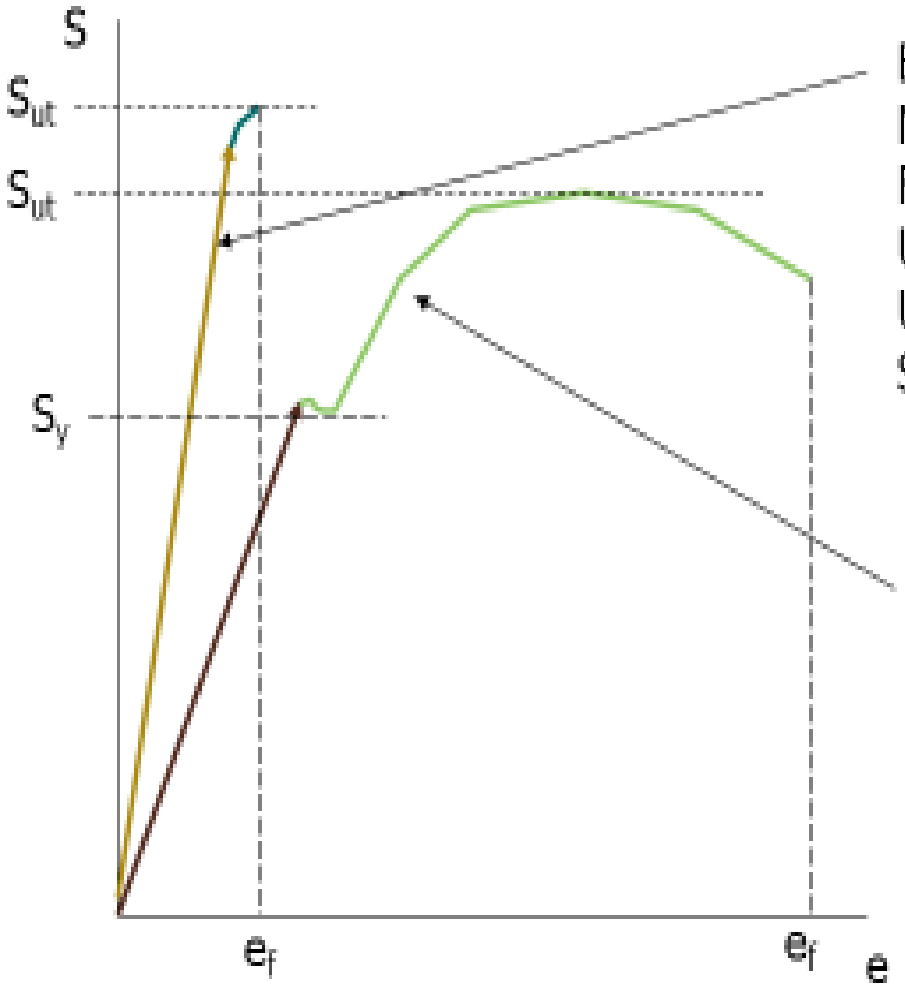
- Consider a test specimen length L and cross-section area, A
- subjected to a tensile direct normal loading, F



Engineering (Nominal) Stress, $S = F/A$ (Ignores the change in the cross-section Area)

Engineering (Nominal) Strain, $e = \Delta L/L$

Ductile Versus Brittle Behavior



Brittle Behavior
No well defined yield point
Final Strain, $e_f < 0.05$
Ultimate Tensile Strength, S_{ut}
Ultimate Compressive Strength, $S_{uc} = (3.5 \text{ to } 4)\ S_{ut}$

Ductile Behavior
Well defined yield point, S_y
Final Strain, $e_f \geq 0.05$
Ultimate Tensile Strength, S_{ut}
Ultimate Compressive Strength, $S_{uc} = S_{ut}$

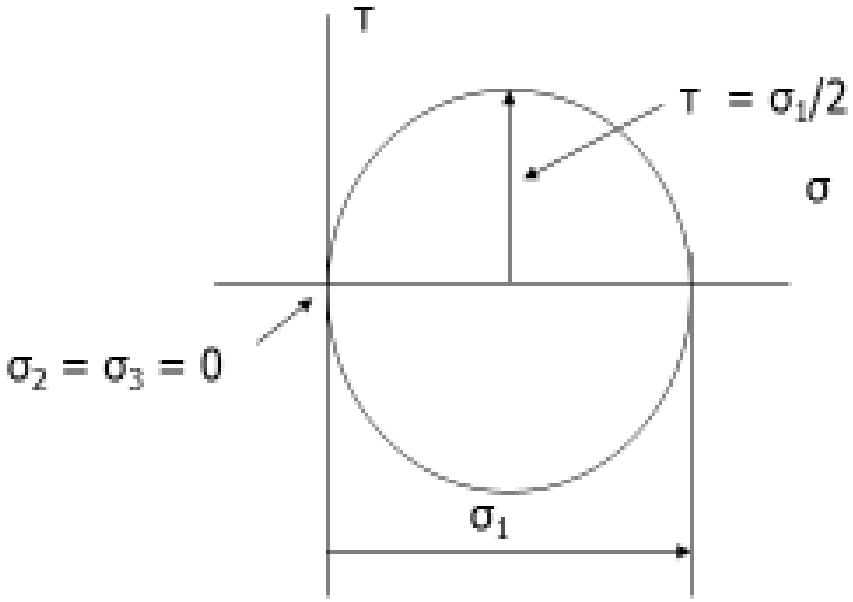
Fundamental Design Equation for Ductile Failure

- n = design factor or factor of safety (where n>1)
- Design stress = τ_{max} (maximum shear stress)
- Design strength = S_{ys} (yield strength in shear)
- At the verge of failure: $\tau_{max} = S_{ys}$
- The part is safe when: $\tau_{max} < S_{ys}$
- $n = S_{ys} / \tau_{max}$ (Fundamental Design Equation)

Maximum Shear Stress Theory (MSS)

- Based on Uniaxial Tensile Strength Test





Maximum Shear Stress Theory (MSS)

- Theory:
  - Yielding begins when the *maximum shear stress* in a stress element exceeds the maximum shear stress in a tension test specimen of the same material when that specimen begins to yield.
  - For a tension test specimen, the maximum shear stress is $s_1/2$.
  - At yielding, when $s_1 = S_y$, the maximum shear stress is $S_y/2$.

- Could restate the theory as follows:

- Theory:
  - Yielding begins when the *maximum shear stress* in a stress element exceeds $S_y/2$.

Maximum Shear Stress Theory (MSS)

- Ordering the principal stresses such that $s_1 \geq s_2 \geq s_3$,

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} \geq \frac{S_y}{2} \qquad \text{or} \qquad \sigma_1 - \sigma_3 \geq S_y$$

- Incorporating a factor of safety n

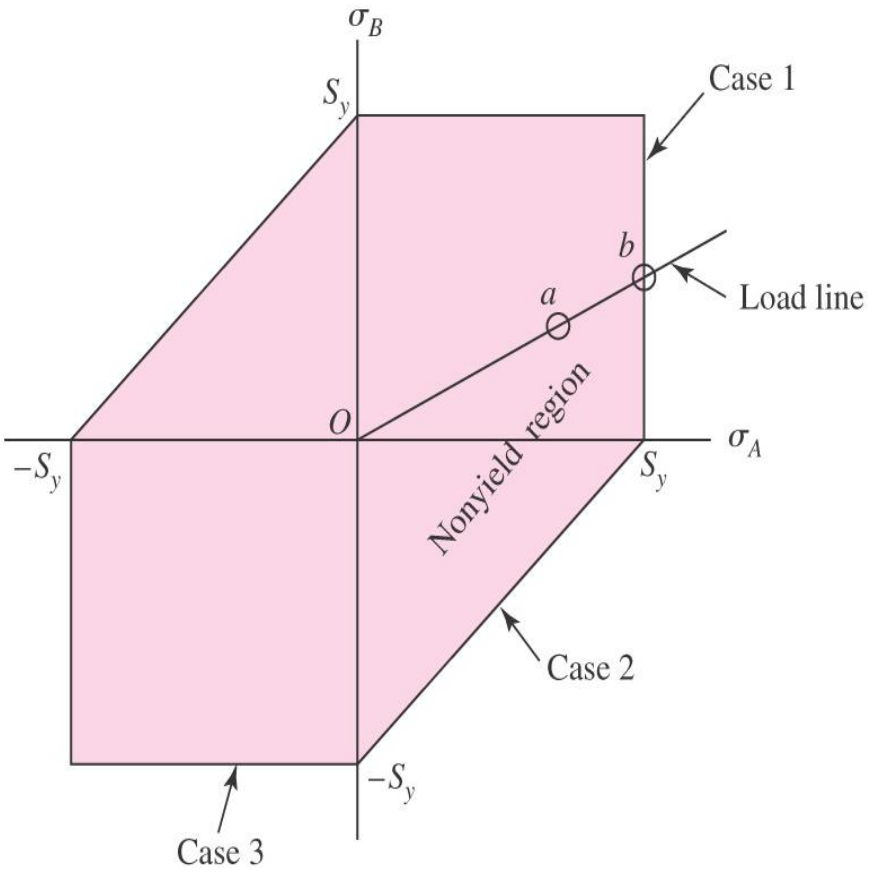
$$\tau_{\max} = \frac{S_y}{2n} \qquad \text{or} \qquad \sigma_1 - \sigma_3 = \frac{S_y}{n}$$

- Or solving for factor of safety

$$n = \frac{S_y / 2}{\tau_{\max}}$$

Maximum Shear Stress Theory (MSS)

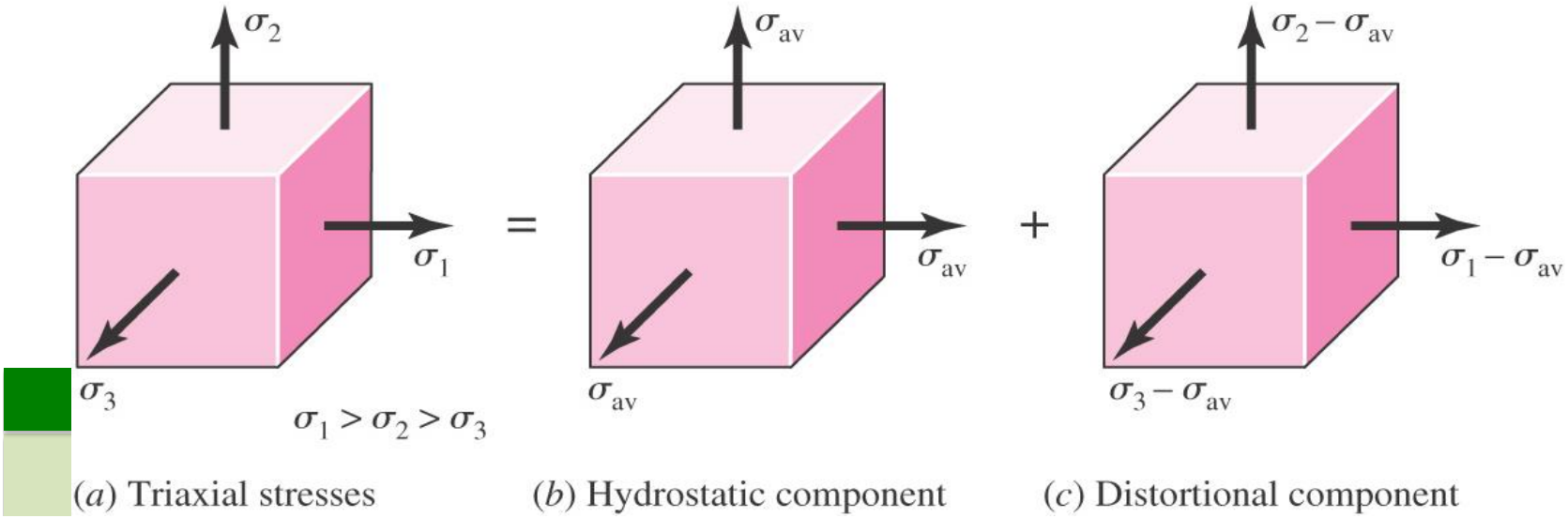
- Let s_1 and s_2 represent the two non-zero principal stresses, such that $s_1 \geq s_2$, there are three cases to consider
  - Case 1: $s_1 \geq s_2 \geq 0$
  - Case 2: $s_1 \geq 0 \geq s_2$
  - Case 3: $0 \geq s_1 \geq s_2$



Distortion Energy (DE) Failure Theory

- Originated from observation that ductile materials stressed hydrostatically (equal principal stresses) exhibited yield strengths greatly in excess of expected values.

- Theorizes that if strain energy is divided into hydrostatic volume changing energy and angular distortion energy, the yielding is primarily affected by the distortion energy.



(*a*) Triaxial stresses (*b*) Hydrostatic component (*c*) Distortional component

Distortion Energy (DE) Failure Theory

- Theory:
  - Yielding occurs when the *distortion strain energy* per unit volume reaches the distortion strain energy per unit volume for yield in simple tension or compression of the same material.

- Hydrostatic stress is average of principal stresses

$$\sigma_{av} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

- Strain energy per unit volume,

$$u = \frac{1}{2}[\epsilon_1\sigma_1 + \epsilon_2\sigma_2 + \epsilon_3\sigma_3]$$

Distortion Energy (DE) Failure Theory

- But

$$\epsilon_x = \frac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right]$$

$$\epsilon_y = \frac{1}{E}\left[\sigma_y - \nu(\sigma_x + \sigma_z)\right]$$

$$\epsilon_z = \frac{1}{E}\left[\sigma_z - \nu(\sigma_x + \sigma_y)\right]$$

- Therefore the strain energy equation is

$$u = \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

Distortion Energy (DE) Failure Theory

- Strain energy for producing only volume change is obtained by substituting s_{av} for s_1, s_2, and s_3

$$u_v = \frac{3\sigma_{av}^2}{2E}(1-2\nu)$$

- Strain energy for producing only volume change is obtained by substituting s_1, s_2, and s_3 for s_{av}

$$u_v = \frac{1-2\nu}{6E}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1)$$

- Obtain distortion energy by subtracting volume changing energy, from total strain energy

$$u_d = u - u_v = \frac{1+\nu}{3E}\left[\frac{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2}{2}\right] \quad \text{Eq. i}$$

Distortion Energy (DE) Failure Theory

- Tension test specimen at yield has $s_1 = S_y$ and $s_2 = s_3 = 0$
- Applying to U_d, distortion energy for tension test specimen is

$$u_d = \frac{1+\nu}{3E} S_y^2 \qquad \textbf{Eq. ii}$$

- DE theory predicts failure when distortion energy, Eq. (i), exceeds distortion energy of tension test specimen, Eq. (ii)

$$\left[\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}\right]^{1/2} \geq S_y \qquad \textbf{Eq. iii}$$

Distortion Energy (DE) Failure Theory

- Left hand side of Eq. iii is defined as *von Mises stress*

$$\sigma' = \left[\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}\right]^{1/2}$$

- For plane stress, simplifies to

$$\sigma' = \left(\sigma_A^2 - \sigma_A\sigma_B + \sigma_B^2\right)^{1/2}$$

- In terms of *xyz* components, in three dimensions

$$\sigma' = \frac{1}{\sqrt{2}}\left[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6\left(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2\right)\right]^{1/2}$$

Distortion Energy (DE) Failure Theory

- In terms of *xyz* components, for plane stress

$$\sigma' = \left(\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2\right)^{1/2}$$

- Distortion Energy failure theory simply compares von Mises stress to yield strength.

$$\sigma' \geq S_y$$

- Introducing a design factor, $\sigma' = \frac{S_y}{n}$

- Expressing as factor of safety, $n = \frac{S_y}{\sigma'}$

Von-Mises Theory (VMT)

- Von-Mises Theory is derived from the maximum distortion energy theory which takes all three principal stresses into consideration.
- It is less conservative than the MSST.
- It is desirable to find Sys from a pure shear test.
- If we did this we would have the following stress condition:

  $\sigma_2 = -\sigma_1$, $\sigma_3 = 0$ and $\sigma_1 = \tau_{max}$
- Substituting this in von-Mises stress we obtain $\sigma_e = \sqrt{3}.\tau_{\max}$

The MDET then becomes

$$n = S_{yt} \Big/ \sqrt{3}\,\tau_{\max} \quad \therefore n = 0.577 S_{yt} \Big/ \tau_{\max}$$

Ductile Coulomb Mohr Theory (DCMT)

- The above theories are good for most ductile materials where the tensile yield strength (**S_{yt}**) and compressive yield strength (**S_{yc}**) are equal.

If $S_{yt} \neq S_{yc}$.

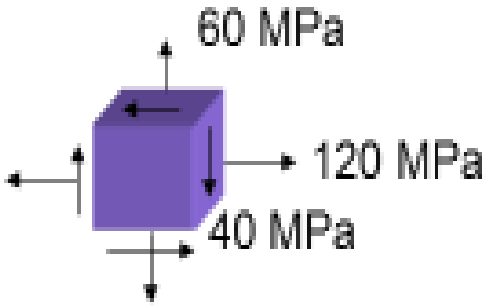
Then the DCMT suggests that $S_{ys} = S_{yt} S_{yc} / \left(S_{yt} + S_{yc} \right)$.

The design equation is $n = S_{ys} / \tau_{\max}$

Example 3-1

The stress element is obtained from the critical section of a machine member made of plain carbon steel. The part is to be designed with a factor of safety of 2.5. Specify a material for the part using:

a. Tresca theory (maximum shear stress theory)
b. Von-Mises theory, and
c. Maximum distortion



Solution

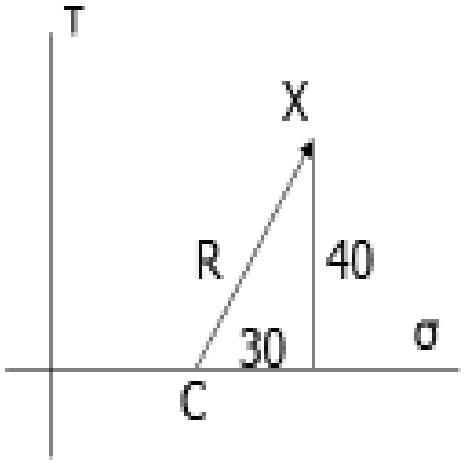
1. Stress Analysis

σ_{avg} = 90 MPa

Center, C (90,0)

Circumf. X (120,40)

R = 50 MPa



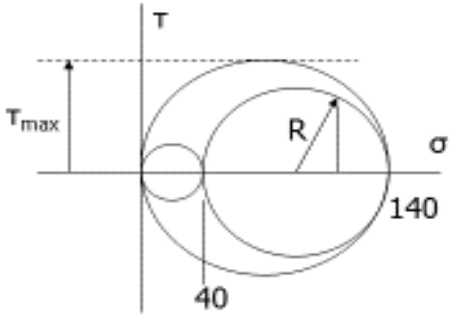
Example 3-1 (Continued)

Principal Stresses:

$\sigma_1 = \sigma_{avg} + R = 90 + 50 = 140$ MPa

$\sigma_1 = \sigma_{avg} - R = 90 - 50 = 40$ MPa

$\tau_{max} = \sigma_1 /2 = 70$ MPa



2. Design Equations and Solution

Tresca Failure Theory: $n = 0.5S_y/\tau_{max}$

Design Equation: $2.5 = 0.5S_y/70$

Solution: $S_y = 350$ MPa

ii. Von-Mises Failure Theory: $n = 0.577S_y/\tau_{max}$

Design Equation: $2.5 = 0.577S_y/70$

Solution: $S_y = 303$ MPa

Example 3-1 (Continued)

iii. Maximum Distortion Energy Theory (MDET)

Design Theory: $n = S_y / \sigma_e$

von-Mises stress: $\sigma_e = [\sigma_1^2 - \sigma_1 \sigma_2 + \sigma_2^2]^{1/2}$

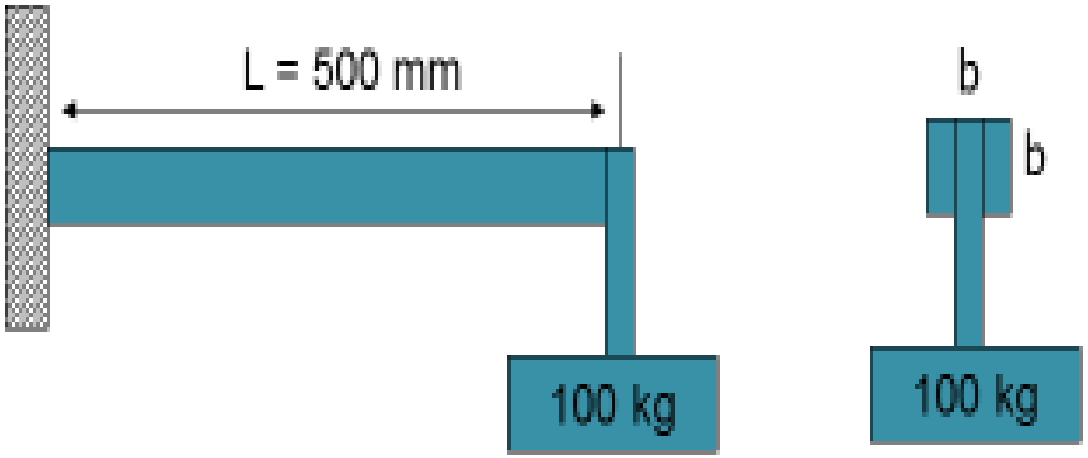
$= [140^2 - (140)(40) + 40^2]^{1/2} = 125$ MPa

Design Equation: $2.5 = S_y/125$

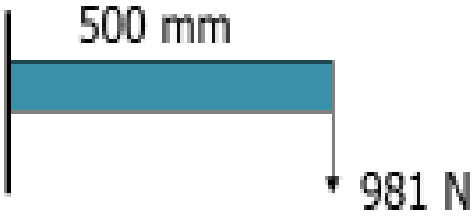
Solution: $S_y = 313$ MPa

Example 3-2

A steel rod 500 mm long and square cross-section supports a 100 kg load as shown. Complete the design of the rod by specifying the type of steel and a reasonable factor of safety.



Example 3-2 (Continued)



A. Load Analysis

$F = 100 \times 9.81$ N

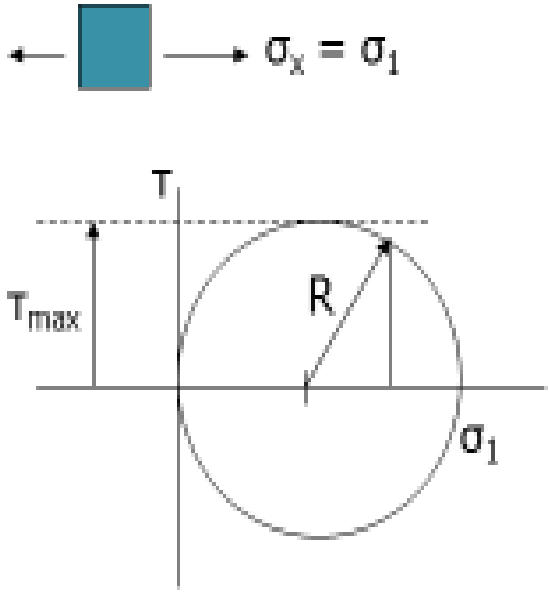
$M = M_{max} = 981 \times 500 = 490.5 \times 10^3$ N.mm

B. Stress Analysis

$\sigma_x = M/Z$ (bending)

$Z = bh^2/6 = b^3/6$ mm^3

$\sigma_x = 6M/b^3 = 2943 \times 10^3/b^3$ MPa



$T_{max} = \sigma_1 /2 = 1472 \times 10^3/b^3$

Example 3-2 (Continued)

C. Material Type: Steel → Exact Specification is Unknown
→ Ductile Material

D. Von-Mises Design Theory: $n = 0.577S_y/\tau_{max}$

Design Equation: $n = 0.577S_y/ [1472 \times 10^3/b^3]$

$n = 0.577b^3S_y/ (1472 \times 10^3)$

E. Solution: Select: AISI 1040 CD Steel, $S_y = 490$ MPa
Assume: Normal Operating Conditions, n = 2.0
Calculate: b = 21.84 mm
Specify: b = 22 mm (standard size)
Final: n = 2.04 (close to desired value-accept)

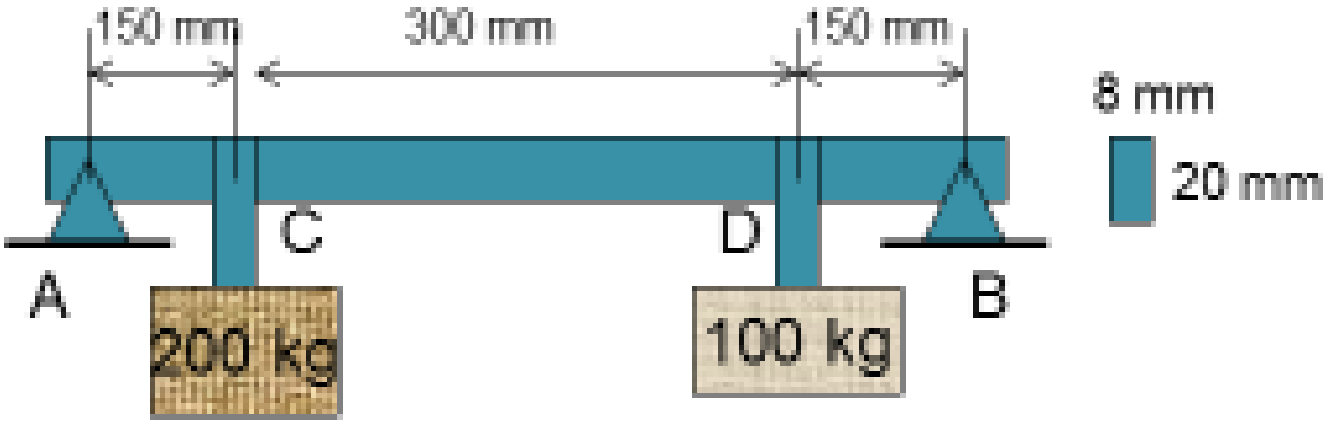
Example 3-3

Consider a simply-supported rectangular beam AB. The beam is designed for hanging loads at C and D. For the loads shown, find the factor of safety if the material is AISI CD 1040 steel.

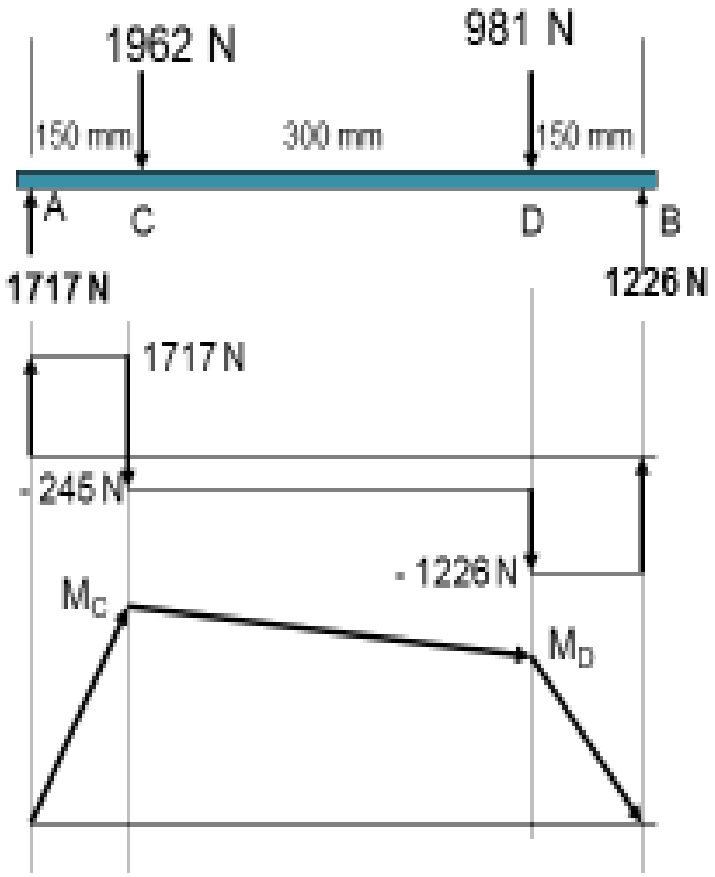
Tensile strength = 590 MPa

Yield Strength = 490 MPa

Elongation = 12%



Example 3-3 (Continued)



A. Load Analysis

Reactions: $R_A = 1717$ N

$R_B = 1226$ N

Bending Moments

$M_c = 1717 \times 150 = 257.55 \times 10^3$ N.mm

$M_D = M_C - (245 \times 300)$

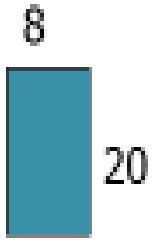
$= 184.05 \times 10^3$ N.mm

B. Stress Analysis

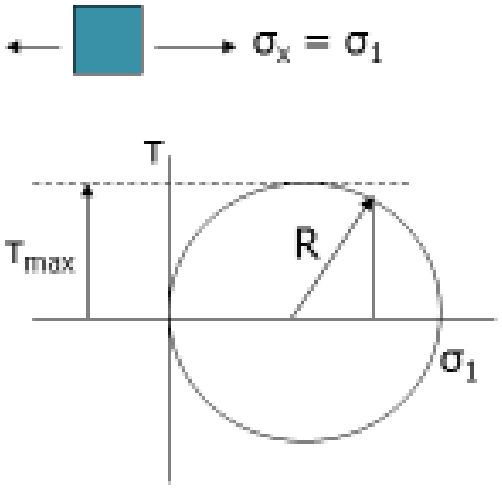
Thus $\sigma_x = 6M_c / bh^2$

$= 483$ MPa

$\tau_{xy} = 0$



Example 3-3 (Continued)



$T_{max} = \sigma_1 /2 = 241.5$ MPa

C. Material:
Elongation, $\varepsilon = 12\% > 5\%$
Thus failure mode will be ductile

D. von-Mises Design Theory:

$$n = 0.577S_y/\tau_{max}$$

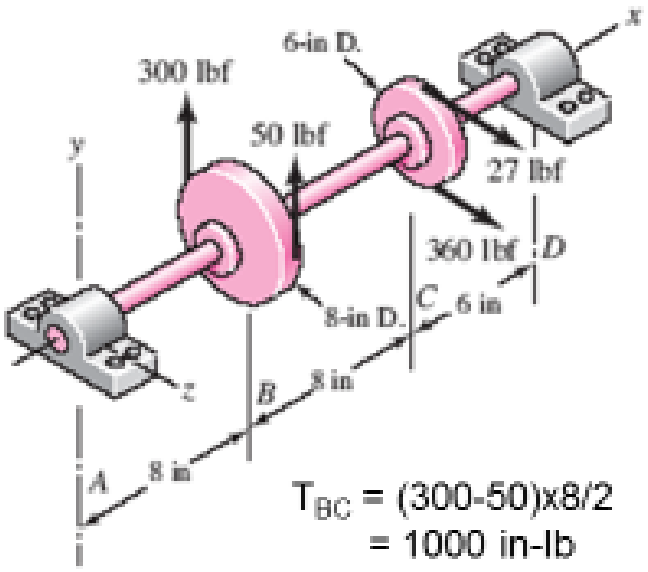
Design Equation: $n = 0.577S_y/241.5$

E. Solution: $n = 0.577 \times 490/241.5$
$= 1.17$

Comment: The design is barely safe
Consider increasing the size of the bar

Example 3-4

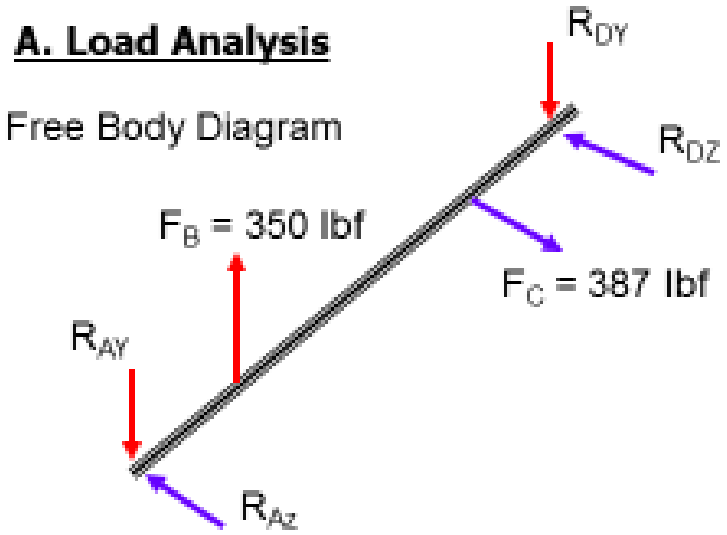
The solid cylindrical steel non-rotating shaft (axle) AB supports of two pulleys as shown. A design factor of 2.5 is desired. Provide missing information for the shaft.



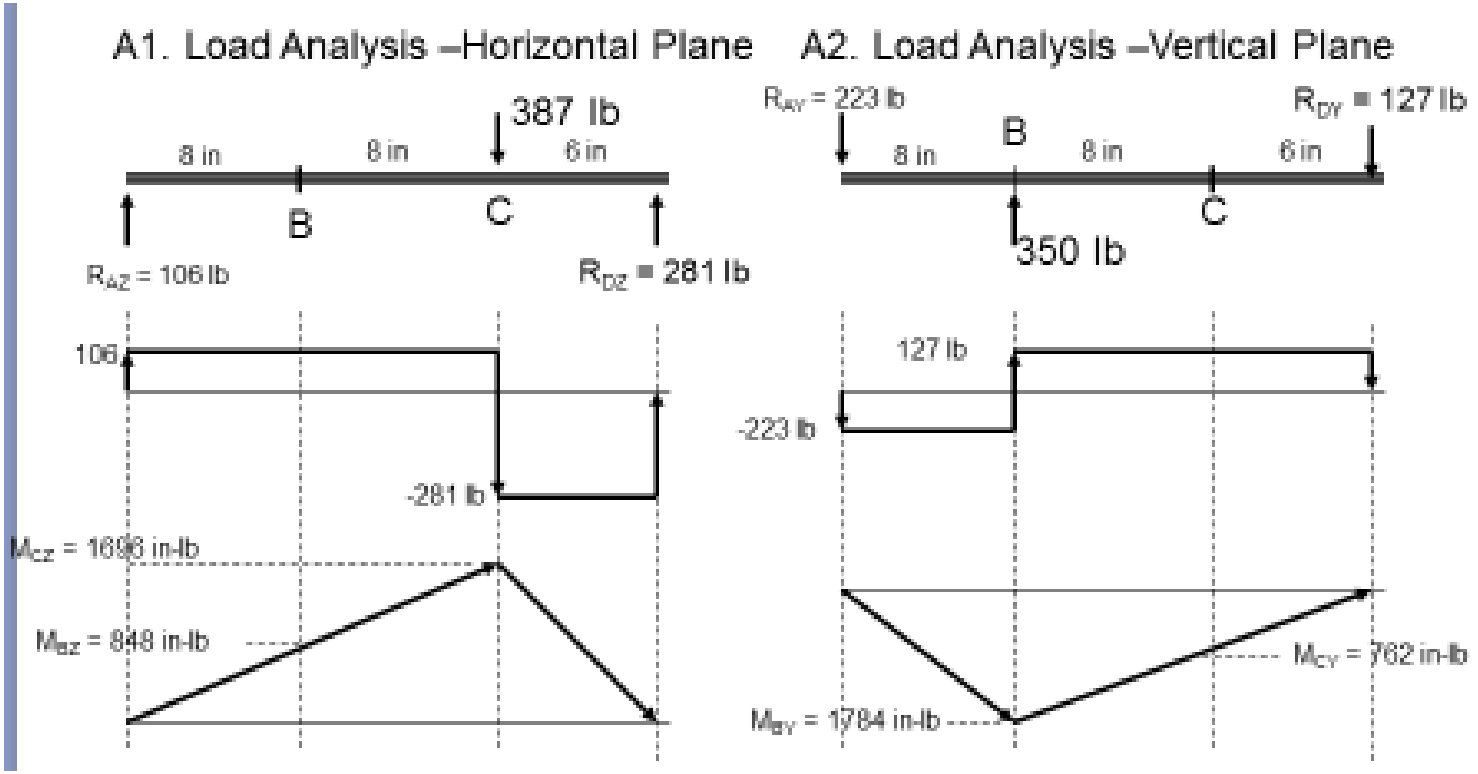
T_{BC} = (300-50)x8/2
= 1000 in-lb

A. Load Analysis

Free Body Diagram

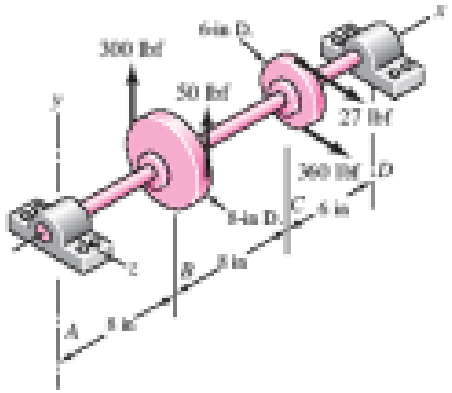


Example 3-4 (Continued)



Example 3-4 (Continued)

$M_B = (848^2+1784^2)^{½} = 1975$ in-lb
$M_C = (1696^2+762^2)^{½} = 1859$ in-lb





Torque Diagram

B. Stress Analysis

Critical section is at B

Normal Stress at Point B:

$\sigma_x = M_B / Z$
$= 32x\ M_B / \pi d^3$
$= 32x1975/\pi(d)^3$
$= 20{,}110/d^3$ lb/in^2
$= 20/d^3$ ksi

Shear Stress at B:

$T_{xy} = 16T/\pi d^3$
$= 16 \times 1000/\ \pi(d)^3$
$= 5{,}093/d^3$ lb/in^2
$= 5/d^3$ ksi

Example 3-4 (Continued)

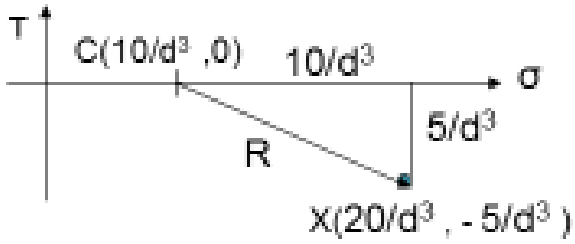
Stress Element at B:



Average stress, $\sigma_{avg} = \frac{1}{2}[20/d^3 + 0]$
$= 10/d^3$ ksi

Center Coordinate, C $(10/d^3, 0)$

Circumferential Point, X $(20/d^3, -5/d^3)$



$R = 11.2/d^3$ ksi

Max. Normal Stress, $\sigma_1 = \sigma_{avg} + R$
$= 21.2/d^3$ ksi

Min. Normal Stress, $\sigma_2 = \sigma_{avg} - R$
$= -1.2/d^3$ ksi

Max. Shear Stress, $\tau_{max} = 11.2/d^3$ ksi

B. Design Theory & Equation

von-Mises : $n = 0.577S_y/\tau_{max}$

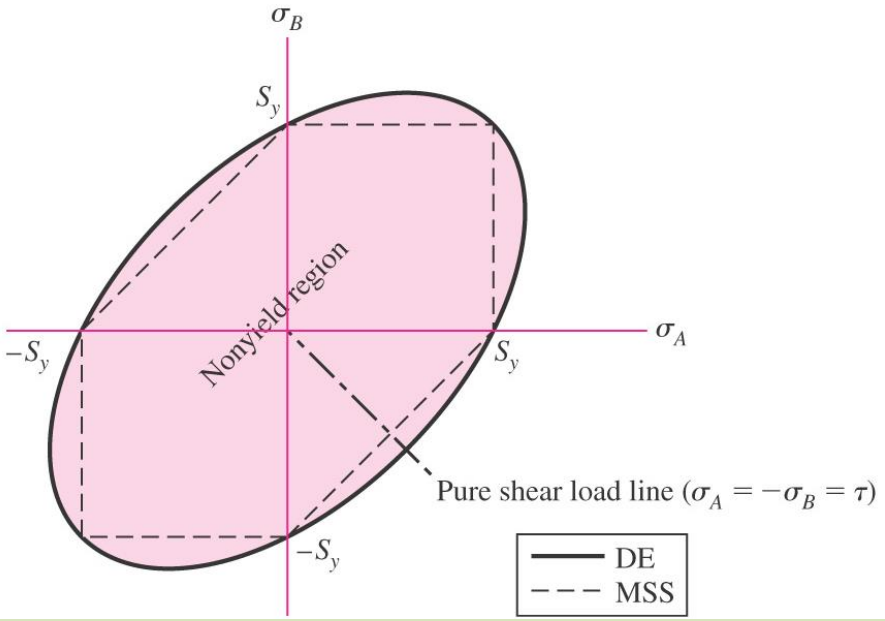
Design Equation: $2.5 = 0.577d^3S_y/11.2$

C Solution:

Select material, read S_y
Calculate d
Standardize d

Shear Strength Predictions

- For pure shear loading, Mohr's circle shows that $s_A = -s_B = t$
- Plotting this equation on principal stress axes gives load line for pure shear case
- Intersection of pure shear load line with failure curve indicates shear strength has been reached
- Each failure theory predicts shear strength to be some fraction of normal strength



Shear Strength Predictions

- For MSS theory, intersecting pure shear load line with failure line results in

$$S_{sy} = 0.5S_y$$

- For DE theory, intersection pure shear load line with failure curve

$$(3\tau_{xy}^2)^{1/2} = S_y \qquad \text{or} \qquad \tau_{xy} = \frac{S_y}{\sqrt{3}} = 0.577S_y$$

- Therefore, DE theory predicts shear strength as

$$S_{sy} = 0.577S_y$$

Example 3–5

A hot-rolled steel has a yield strength of $S_{yt} = S_{yc} = 100$ kpsi and a true strain at fracture of $\varepsilon_f = 0.55$. Estimate the factor of safety for the following principal stress states:

(*a*) $\sigma_x = 70$ kpsi, $\sigma_y = 70$ kpsi, $\tau_{xy} = 0$ kpsi

(*b*) $\sigma_x = 60$ kpsi, $\sigma_y = 40$ kpsi, $\tau_{xy} = -15$ kpsi

(*c*) $\sigma_x = 0$ kpsi, $\sigma_y = 40$ kpsi, $\tau_{xy} = 45$ kpsi

(*d*) $\sigma_x = -40$ kpsi, $\sigma_y = -60$ kpsi, $\tau_{xy} = 15$ kpsi

(*e*) $\sigma_1 = 30$ kpsi, $\sigma_2 = 30$ kpsi, $\sigma_3 = 30$ kpsi

Solution

Since $\varepsilon_f > 0.05$ and S_{yt} and S_{yc} are equal, the material is ductile and both the distortion-energy (DE) theory and maximum-shear-stress (MSS) theory apply. Both will be used for comparison. Note that cases *a* to *d* are plane stress states.

Example 3–5 (continued)

(*a*) Since there is no shear stress on this stress element, the normal stresses are equal to the principal stresses. The ordered principal stresses are $\sigma_A = \sigma_1 = 70$, $\sigma_B = \sigma_2 = 70$, $\sigma_3 = 0$ kpsi.

DE From Eq. (5–13),

$$\sigma' = [70^2 - 70(70) + 70^2]^{1/2} = 70 \text{ kpsi}$$

From Eq. (5–19),

$$n = \frac{S_y}{\sigma'} = \frac{100}{70} = 1.43$$ Answer

MSS Noting that the two nonzero principal stresses are equal, $\tau_{\max}$ will be from the largest Mohr's circle, which will incorporate the third principal stress at zero. From Eq. (3–16),

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{70 - 0}{2} = 35 \text{ kpsi}$$

From Eq. (5–3),

$$n = \frac{S_y/2}{\tau_{\max}} = \frac{100/2}{35} = 1.43$$ Answer

Example 3–5 (continued)

(*b*) From Eq. (3–13), the nonzero principal stresses are

$$\sigma_A, \sigma_B = \frac{60 + 40}{2} \pm \sqrt{\left(\frac{60 - 40}{2}\right)^2 + (-15)^2} = 68.0, 32.0 \text{ kpsi}$$

The ordered principal stresses are $\sigma_A = \sigma_1 = 68.0$, $\sigma_B = \sigma_2 = 32.0$, $\sigma_3 = 0$ kpsi.

DE $$\sigma' = [68^2 - 68(32) + 32^2]^{1/2} = 59.0 \text{ kpsi}$$

$$n = \frac{S_y}{\sigma'} = \frac{100}{59.0} = 1.70$$ Answer

MSS Noting that the two nonzero principal stresses are both positive, τ_{max} will be from the largest Mohr's circle which will incorporate the third principle stress at zero. From Eq. (3–16),

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{68.0 - 0}{2} = 34.0 \text{ kpsi}$$

$$n = \frac{S_y/2}{\tau_{max}} = \frac{100/2}{34.0} = 1.47$$ Answer

Example 3–5 (continued)

(*c*) This time, we shall obtain the factors of safety directly from the *xy* components of stress.

DE From Eq. (5–15),

$$\sigma' = (\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2)^{1/2} = [(40^2 + 3(45)^2]^{1/2} = 87.6 \text{ kpsi}$$

$$n = \frac{S_y}{\sigma'} = \frac{100}{87.6} = 1.14 \qquad \text{Answer}$$

MSS Taking care to note from a quick sketch of Mohr's circle that one nonzero principal stress will be positive while the other one will be negative, $\tau_{\max}$ can be obtained from the extreme-value shear stress given by Eq. (3–14) without finding the principal stresses.

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{0 - 40}{2}\right)^2 + 45^2} = 49.2 \text{ kpsi}$$

$$n = \frac{S_y/2}{\tau_{\max}} = \frac{100/2}{49.2} = 1.02 \qquad \text{Answer}$$

For graphical comparison purposes later in this problem, the nonzero principal stresses can be obtained from Eq. (3–13) to be 69.2 kpsi and −29.2 kpsi.

Example 3–5 (continued)

(*d*) From Eq. (3–13), the nonzero principal stresses are

$$\sigma_A, \sigma_B = \frac{-40 + (-60)}{2} \pm \sqrt{\left(\frac{-40 - (-60)}{2}\right)^2 + (15)^2} = -32.0, -68.0 \text{ kpsi}$$

The ordered principal stresses are $\sigma_1 = 0$, $\sigma_A = \sigma_2 = -32.0$, $\sigma_B = \sigma_3 = -68.0$ kpsi.

DE $$\sigma' = \left[(-32)^2 - (-32)(-68) + (-68)^2\right]^{1/2} = 59.0 \text{ kpsi}$$

$$n = \frac{S_y}{\sigma'} = \frac{100}{59.0} = 1.70$$ Answer

MSS From Eq. (3–16),

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{0 - (-68.0)}{2} = 34.0 \text{ kpsi}$$

$$n = \frac{S_y/2}{\tau_{\max}} = \frac{100/2}{34.0} = 1.47$$ Answer

Example 3–5 (continued)

(*e*) The ordered principal stresses are $\sigma_1 = 30, \sigma_2 = 30, \sigma_3 = 30$ kpsi

DE From Eq. (5–12),

$$\sigma' = \left[\frac{(30-30)^2 + (30-30)^2 + (30-30)^2}{2}\right]^{1/2} = 0 \text{ kpsi}$$

$$n = \frac{S_y}{\sigma'} = \frac{100}{0} \rightarrow \infty$$ Answer

MSS From Eq. (5–3),

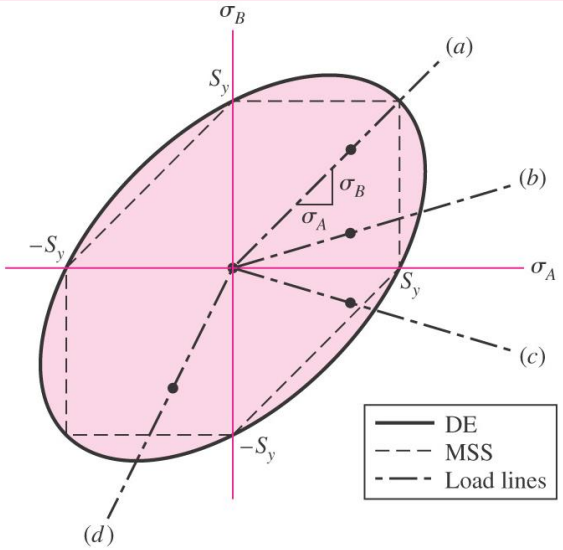
$$n = \frac{S_y}{\sigma_1 - \sigma_3} = \frac{100}{30-30} \rightarrow \infty$$ Answer

Example 3–5 (continued)

A tabular summary of the factors of safety is included for comparisons.

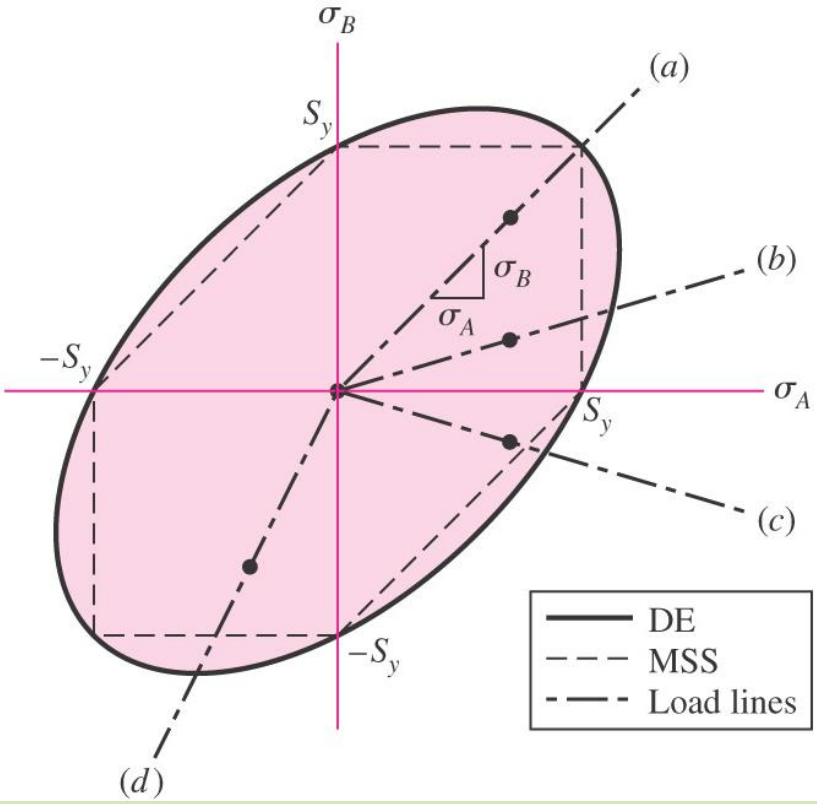
	(a)	(b)	(c)	(d)	(e)
DE	1.43	1.70	1.14	1.70	∞
MSS	1.43	1.47	1.02	1.47	∞

Since the MSS theory is on or within the boundary of the DE theory, it will always predict a factor of safety equal to or less than the DE theory, as can be seen in the table.



Example 3–5 (continued)

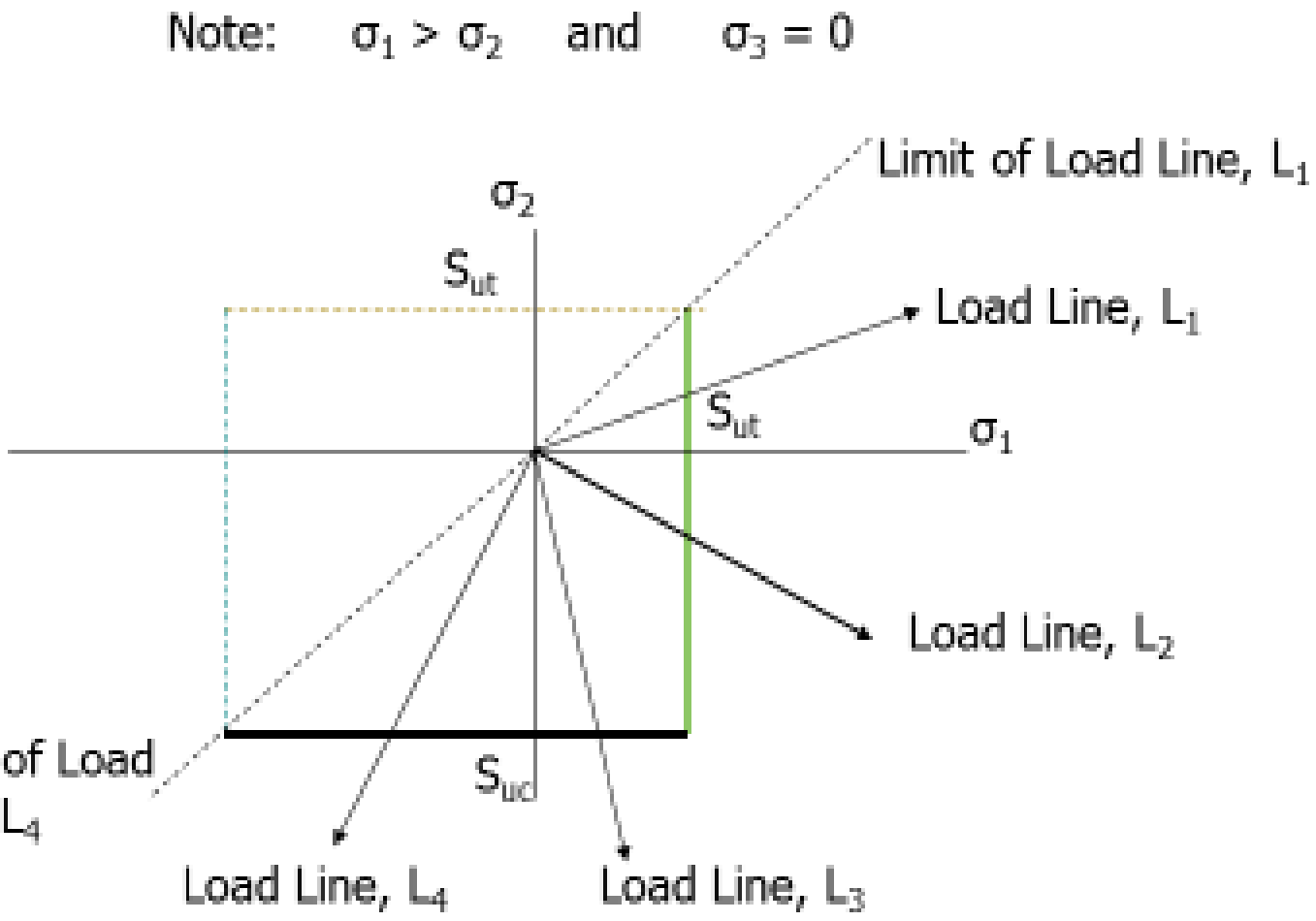
For each case, except case (*e*), the coordinates and load lines in the σ_A, σ_B plane are shown in Fig. 5–11. Case (*e*) is not plane stress. Note that the load line for case (*a*) is the only plane stress case given in which the two theories agree, thus giving the same factor of safety.



Design Principle for Brittle Materials

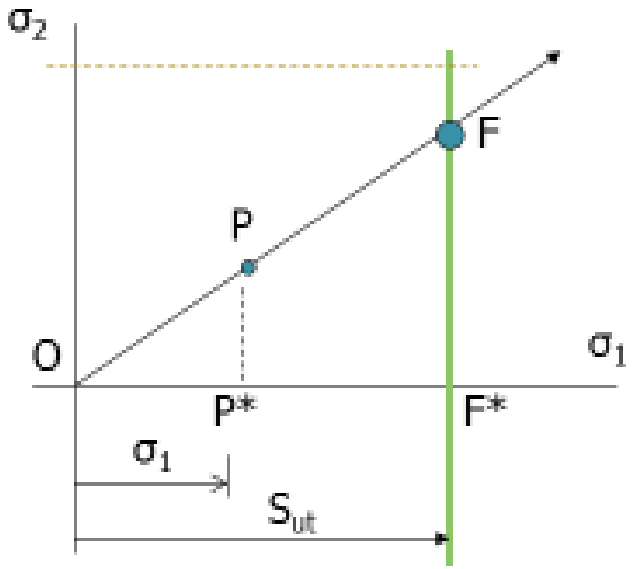
- Brittle failure is due to excessive principal normal stresses σ_1 and σ_2.
- Prevailing material properties are the ultimate tensile strength (S_{ut}) and the ultimate compressive strength (S_{uc}).
- Combination of σ_1 and σ_2 determine the failure characteristics and hence the appropriate design strength.

MNST-Effective Failure Boundaries and Load Lines



MNST Design Equation – 1st Quadrant (L_1)

1. Find σ_1 and σ_2
2. Locate Operating Point P (σ_1, σ_2)
3. Draw Load Line from O thru P to F



Factor of safety, n = OF/OP

Using similar triangles:

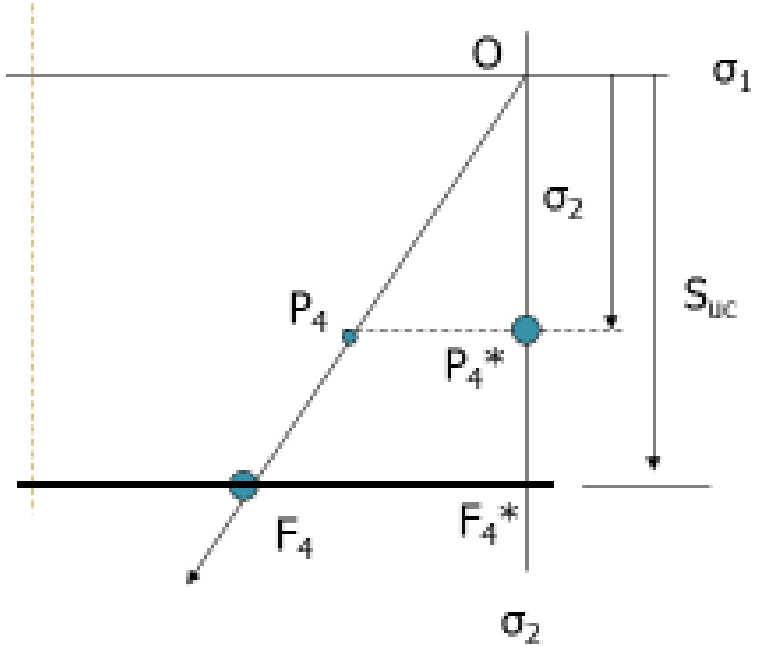
n = OF*/OP*

Design Equation for L_1:

$$n = S_{ut}/\sigma_1$$

MNST Design Equation – 3rd Quadrant (L_4)

1. Find σ_1 and σ_2
2. Locate Operating Point P_4 (σ_1, σ_2)
3. Draw Load Line from O thru P_4 to F_4



Factor of safety, $n = OF_4 / OP_4$

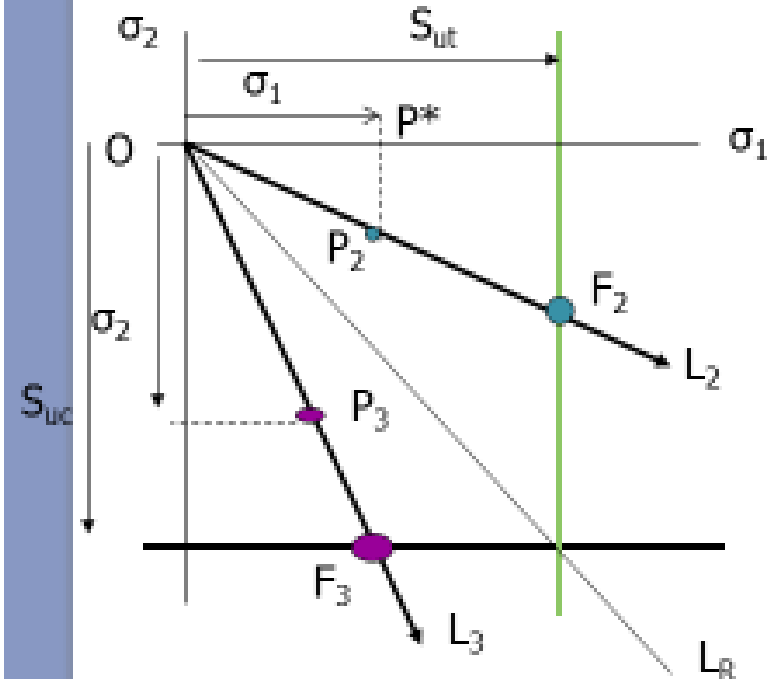
And $n = OF^*/OP^*$

Design Equation for L_4:

$$n = S_{uc}/\sigma_2$$

MSNT Design Equation – 4th Quadrant (L_2 and L_3)

1. Find σ_1 and σ_2
2. Locate Operating Point P (σ_1, σ_2)
3. Draw Load Line from O thru P to F



4. Determine slope of reference line, L_R
   $r_c = S_{uc}/S_{ut}$

5. Find magnitude of slope of Load Line
   $r_L = | \sigma_2 / \sigma_1 |$

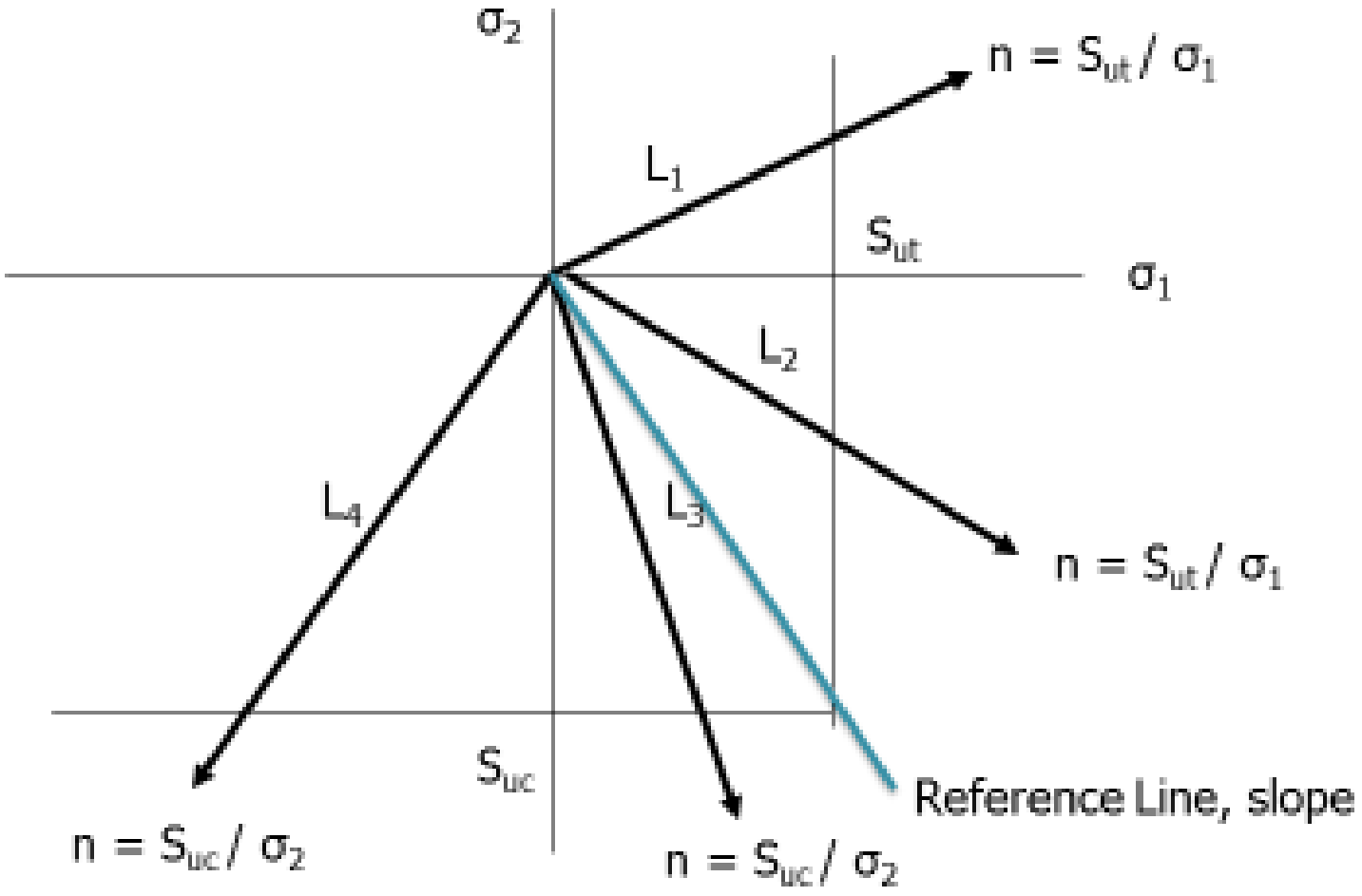
6. If $r_L < r_c$ Then Load Line is L_2
   $n = OF_2/OP_2$

   And $\boxed{n = S_{ut}/\sigma_1}$

7. If $r_L > r_c$ Then Load Line is L_3,
   $n = OF_3/OP_3$

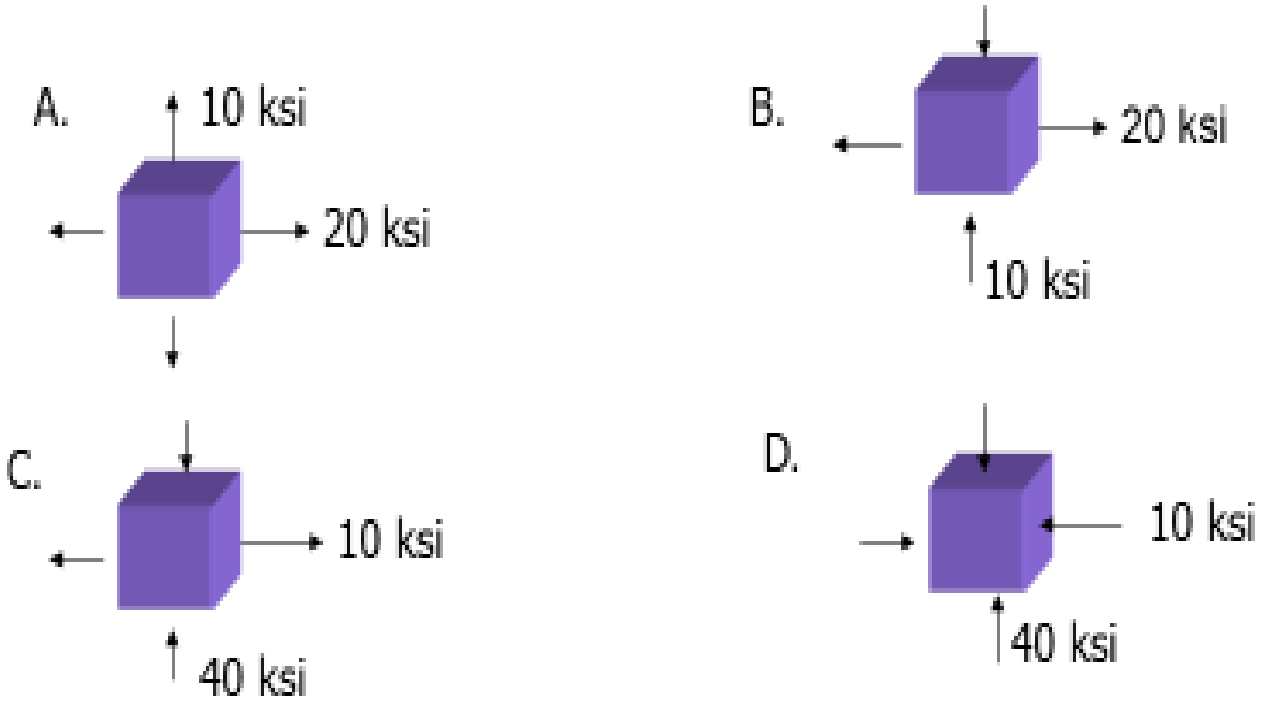
   And $\boxed{n = S_{uc}/\sigma_2}$

MNST Design Equations



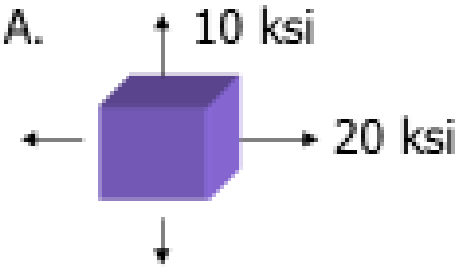
Example 3-6

Each stress element is obtained from the critical section of machine members made of ASTM Class 40 Gray Cast Iron. Find the factor of safety for each situation using MNST.



Example 3–6 (continued)

A.



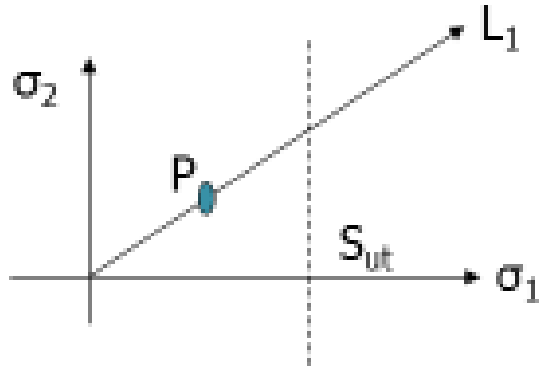
1. Principal Normal Stresses

$\sigma_1 = 20$ ksi

$\sigma_2 = 10$ ksi

2. Operating Point (20,10)

3. Load Line



4. Design Equation/Solution

For Load Line, L_1

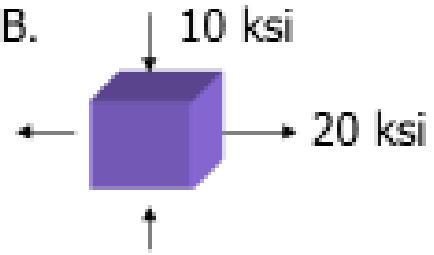
$n = S_{ut}/\sigma_1$

$n = 42.5/20$

$n = 2.12$

Example 3–6 (continued)

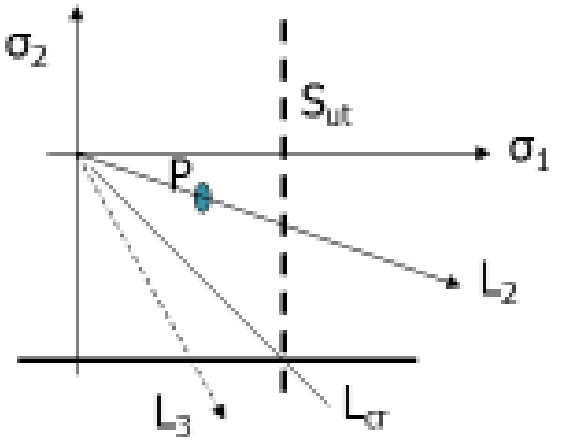
B.



1. Principal Normal Stresses
   $\sigma_1 = 20$ ksi
   $\sigma_2 = -10$ ksi

2. Operating Point (20, -10)
   4th Quadrant
   Slope of Load Line, $r_L = |\sigma_2/\sigma_1|$
   $= |-10/20| = 0.5$
   Slope of critical line, $r_c = S_{uc}/S_{ut}$
   $= 140/42.5$
   $= 3.3$

3. Load Line

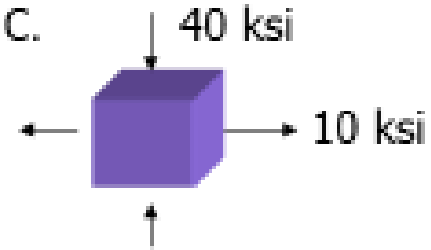


4. Design Equation/Solution

For Load Line, L_2
$n = S_{ut}/\sigma_1$
$n = 42.5/20$
$n = 2.12$

Example 3–6 (continued)

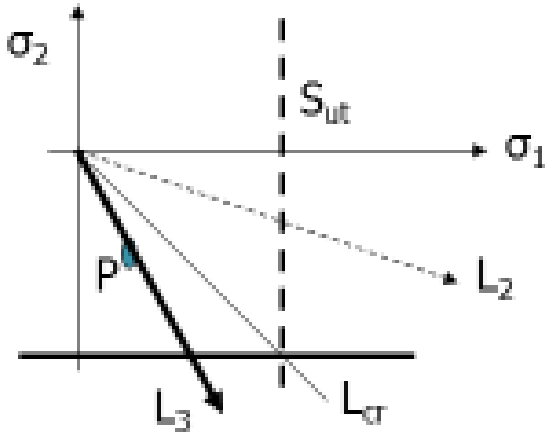
C.



1. Principal Normal Stresses
   $\sigma_1 = 10$ ksi
   $\sigma_2 = -40$ ksi

2. Operating Point (10, -40)
   4th Quadrant
   Slope of Load Line, $r_L = |\sigma_2/\sigma_1|$
   $= |-40/10| = 4.0$
   Slope of critical line, $r_c = S_{uc}/S_{ut}$
   $= 140/42.5$
   $= 3.3$

3. Load Line

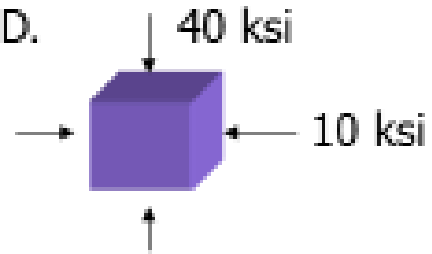


4. Design Equation/Solution

   For Load Line, L_3
   $n = S_{uc}/\sigma_2$
   $n = 140/40$
   $n = 3.5$

Example 3–6 (continued)

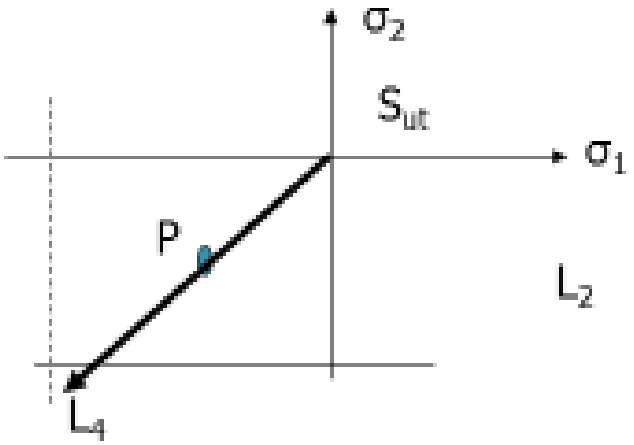
D.



1. Principal Normal Stresses
   σ_1 = -10 ksi
   σ_2 = - 40 ksi

2. Operating Point (-10, -40)
   3rd Quadrant

3. Load Line



4. Design Equation/Solution

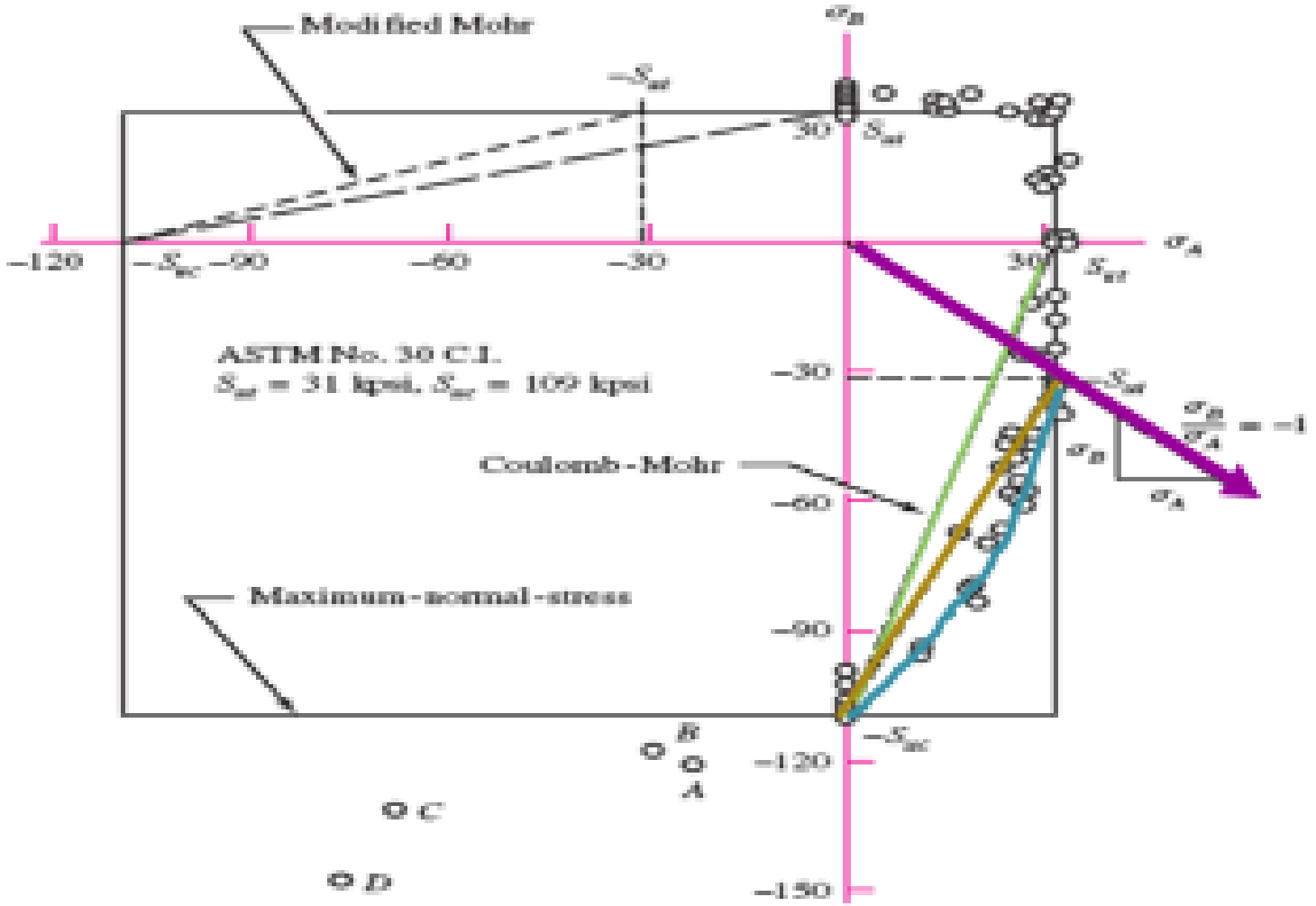
For Load Line, L_4

$n = S_{ut}/\sigma_2$

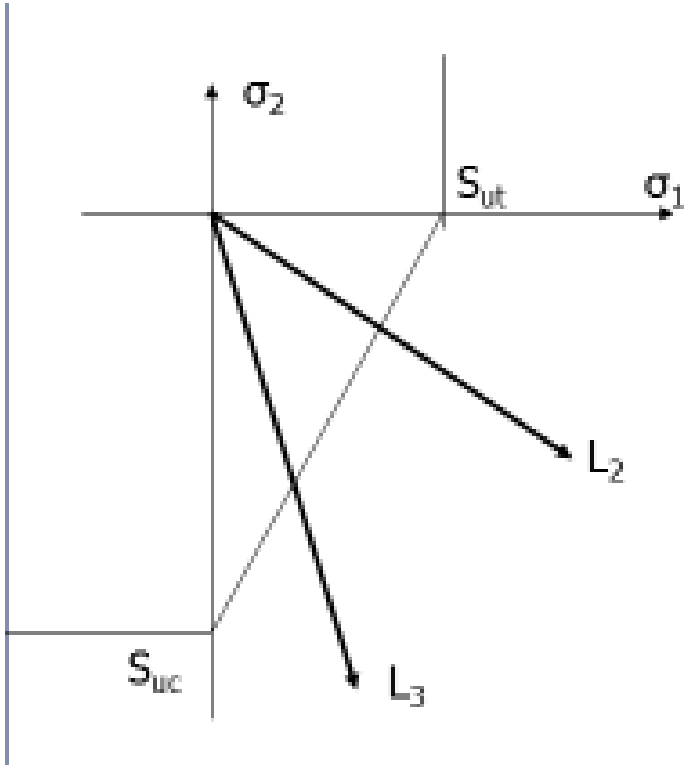
$n = 140/40$

$n = 3.5$

Experimental Verification of MNST Failure Boundaries



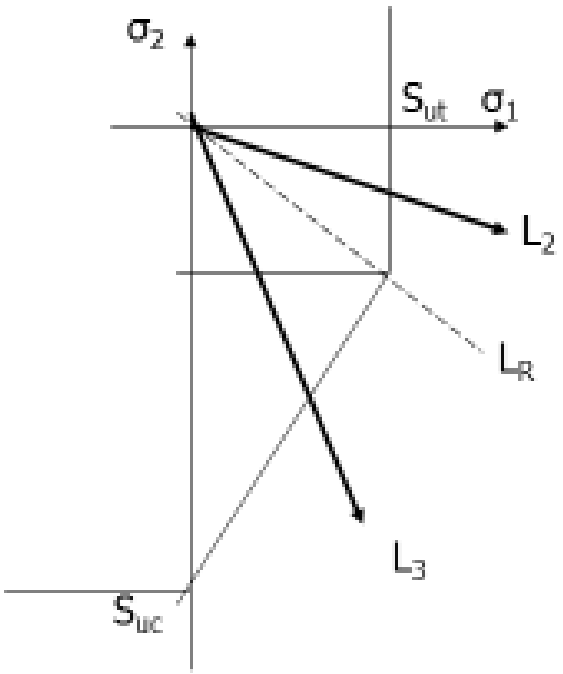
Brittle Coulomb Mohr Theory (BCMT) – 4th Quadrant



Design Equation for L_2 and L_3:

$$1/n = \sigma_1/S_{ut} - \sigma_2/S_{uc}$$

Modified Mohr – 1 Theory (MM1T) : 4th Quadrant



1. Magnitude of slope of reference line, L_R
   $r_c = 1$
2. Find magnitude of slope of Load Line
   $r_L = | \sigma_2 / \sigma_1 |$
3. If $r_L \leq 1$, Then Load Line is L_2

$$n = S_{ut}/\sigma_1$$

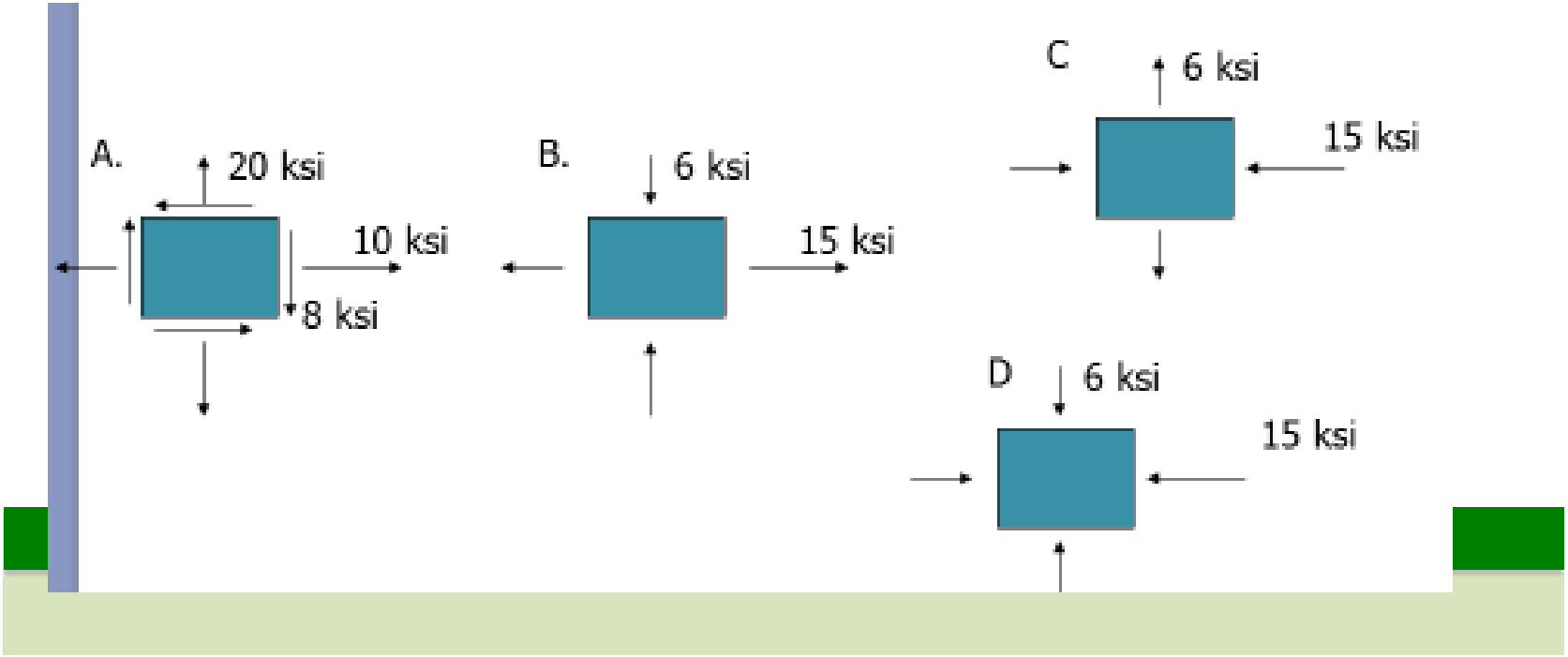
4. If $r_L > 1$, Then Load Line is L_3

$$1/n = [(S_{uc} - S_{ut})\sigma_1/S_{uc}\,S_{ut}] - \sigma_2/S_{uc}$$

Example 3-7

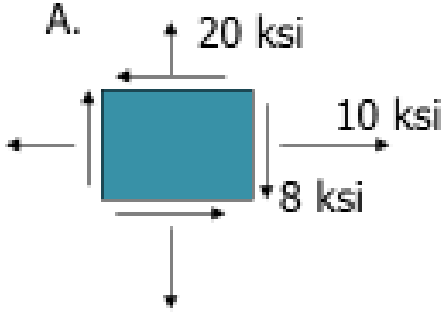
Each stress element is obtained from the critical section of machine members made of ASTM Class 40 Gray Cast Iron. Find the factor of safety for each situation using:

a. Brittle Coulomb-Mohr Theory
b. Modified Mohr-1 Theory



Example 3–7 (continued)

A.



Stress Analysis:

Applied Stresses are:

σ_x = 10 ksi σ_y = 20 ksi

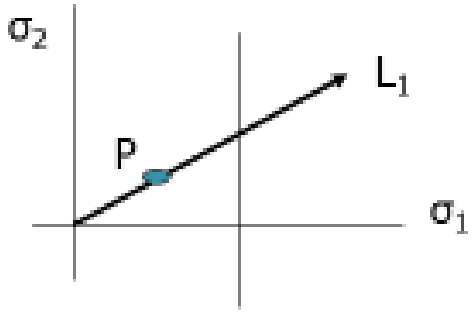
τ_{xy} = 8 ksi

The Principal Stresses are:

σ_1 = 24.4 ksi

σ_2 = 5.6 ksi

i. BCMT for P(24.4, 5.6)



Design Equation for L_1 : $n = S_{ut}/\sigma_1$

Solution: $n = 42.5/24.4$

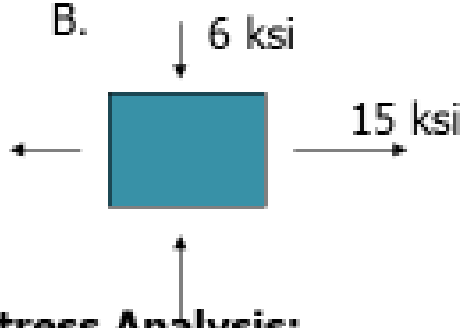
$n = 1.74$

ii. Modified Mohr-1 Theory for P(24.4, 5.6)

L_1 → SAME SOLUTION FOR ALL THEORIES

Example 3–7 (continued)

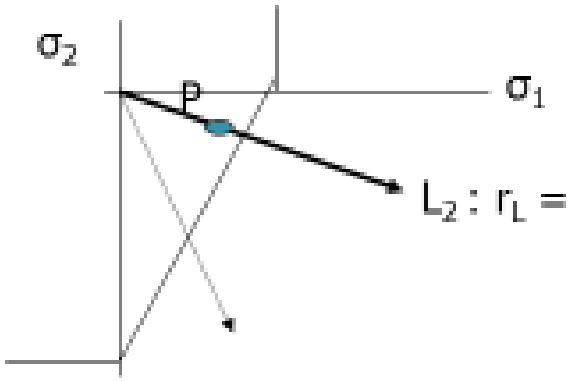
B.



Stress Analysis:
Applied Stresses are Principal Stresses:

$\sigma_x = \sigma_1 = 15$ ksi
$\sigma_y = \sigma_2 = -6$ ksi

i. BCMT for P(15, - 6)



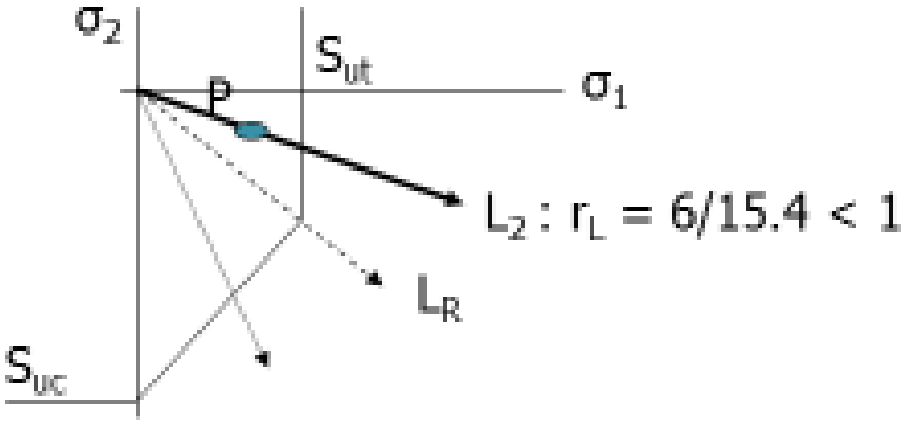
Design Equation: $1/n = \sigma_1/S_{ut} - \sigma_2/S_{uc}$

Solution: $1/n = 15/42.5 - (-6)/140$

$n = 2.53$

Example 3–7 (continued)

ii. Modified Mohr-1 Theory for P(15, -6)

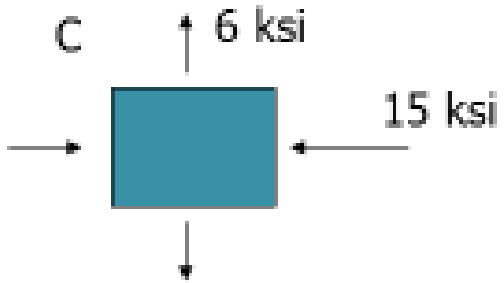


Design Equation: $n = S_{ut}/\sigma_1$

Solution: $n = 42.5/15$

$n = 2.83$

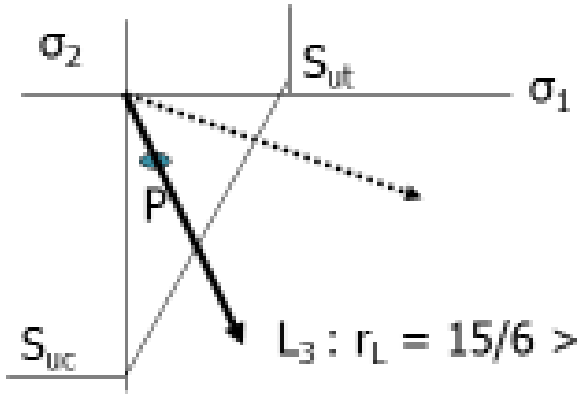
Example 3–7 (continued)



Stress Analysis:
Applied Stresses are Principal Stresses:

$\sigma_Y = \sigma_1 = \quad 6$ ksi
$\sigma_X = \sigma_2 = -15$ ksi

i. BCMT for P(6, - 15)



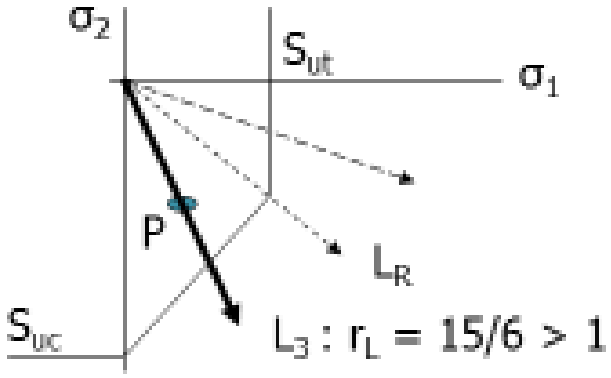
Design Equation: $1/n = \sigma_1/S_{ut} - \sigma_2/S_{uc}$

Solution: $1/n = 6/42.5 - (-15)/140$

$n = 4.03$

Example 3–7 (continued)

ii. Modified Mohr-1 Theory for P(6, -15)

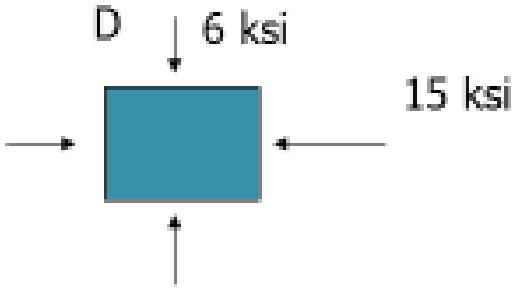


Design Equation for L_3: $1/n = [(S_{uc} - S_{ut})\sigma_1/S_{uc}\,S_{ut}] - \sigma_2/S_{uc}$

Solution: $1/n = [(140 - 42.5)\,6/(140)(42.5)] - (-15)/140$

$n = 4.87$

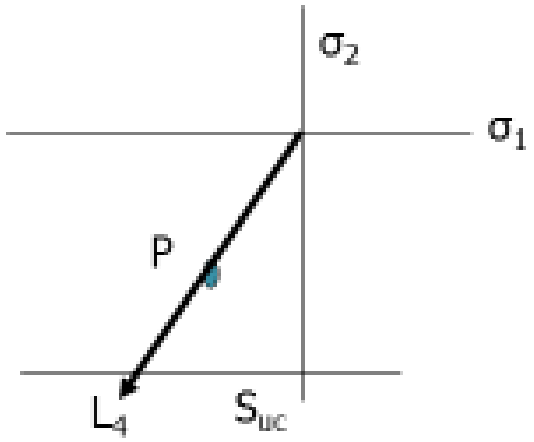
Example 3–7 (continued)



Stress Analysis:
Applied Stresses are Principal Stresses:

$\sigma_Y = \sigma_1 = -6$ ksi
$\sigma_X = \sigma_2 = -15$ ksi

i. BCMT for P(-6, - 15)

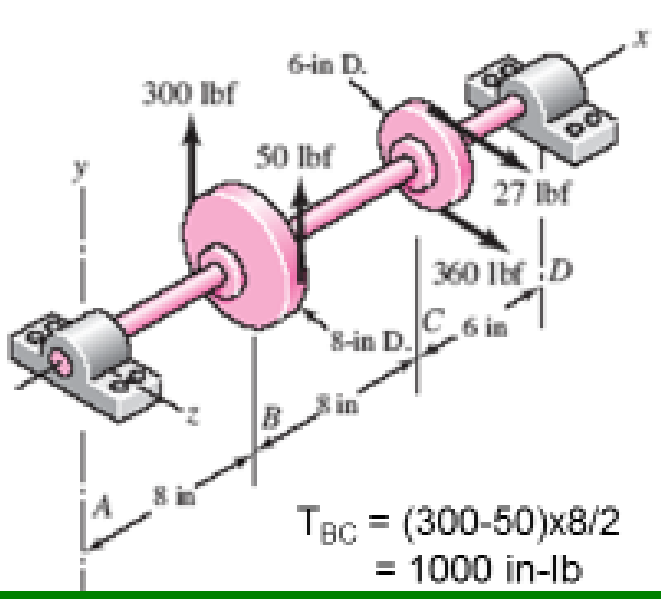


Design Equation for L_4 : $n = S_{uc}/\sigma_2$
Solution: $n = 140/15$
$n = 9.33$ (Over-design)

ii. Modified Mohr-1 Theory for P(-6, -15)
L_4 → SAME SOLUTION FOR ALL THEORIES

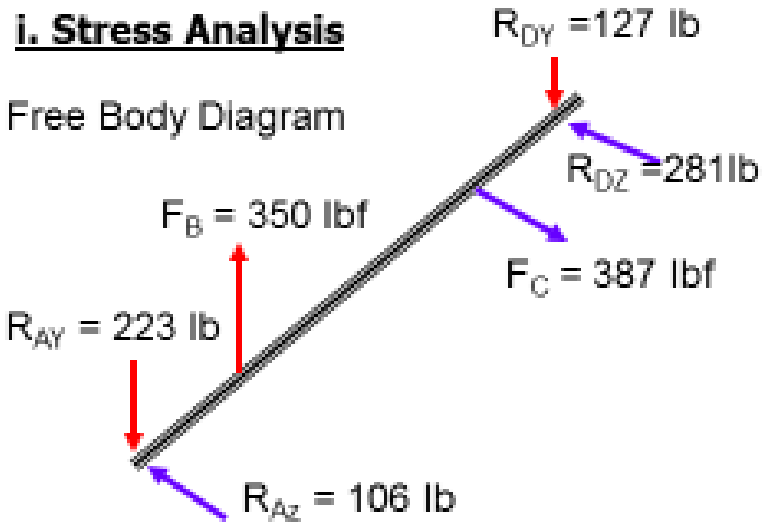
Example 3–8

The solid cylindrical shaft AB supports of two pulleys as shown. The material is cast iron. A design factor of 2.5 is desired. Assume steady conditions. Provide missing information to complete the design of the shaft.



i. Stress Analysis

Free Body Diagram



Example 3–8 (continued)

From Example:

M_B = 1975 in-lb

M_C = 1859 in-lb

Stress Element at B:

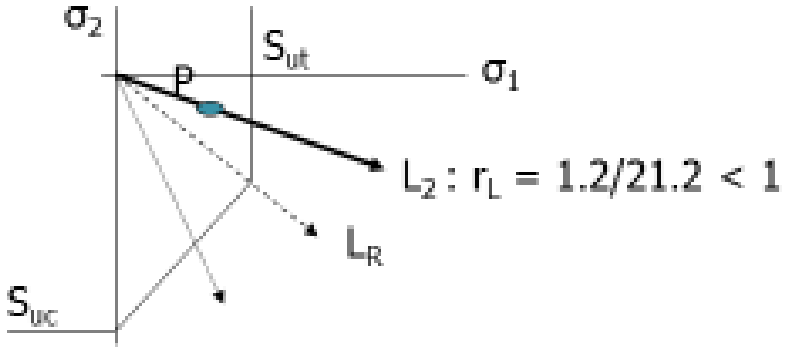


Principal Stresses are:

$\sigma_1 = 21.2/d^3$ ksi

$\sigma_2 = -1.2/d^3$ ksi

ii. Modified Mohr-1 Theory for P(21.2, -1.2)



Design Equation for L_2: $n = S_{ut}/\sigma_1$

$2.5 = S_{ut}\, d^3/21.2$

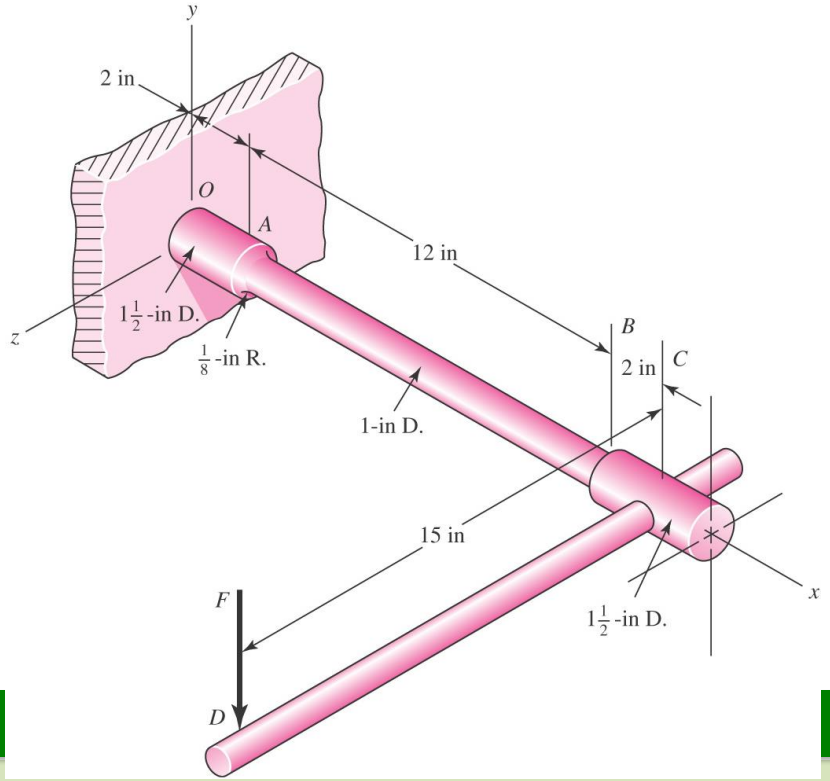
Select ASTM Class 25 Gray Cast Iron, S_{ut} = 31 ksi

d = 1.196 in

Specify d = 1.20 in

Example 3–9

Consider the wrench in Ex. 5–3, Fig. 5–16, as made of cast iron, machined to dimension. The force F required to fracture this part can be regarded as the strength of the component part. If the material is ASTM grade 30 cast iron, find the force F with
(a) Coulomb-Mohr failure model.
(b) Modified Mohr failure model.



Example 3–9 (continued)

We assume that the lever *DC* is strong enough, and not part of the problem. Since grade 30 cast iron is a brittle material *and* cast iron, the stress-concentration factors K_t and K_{ts} are set to unity. From Table A–24, the tensile ultimate strength is 31 kpsi and the compressive ultimate strength is 109 kpsi. The stress element at *A* on the top surface will be subjected to a tensile bending stress and a torsional stress. This location, on the 1-in-diameter section fillet, is the weakest location, and it governs the strength of the assembly. The normal stress σ_x and the shear stress at *A* are given by

$$\sigma_x = K_t \frac{M}{I/c} = K_t \frac{32M}{\pi d^3} = (1)\frac{32(14F)}{\pi(1)^3} = 142.6F$$

$$\tau_{xy} = K_{ts} \frac{Tr}{J} = K_{ts} \frac{16T}{\pi d^3} = (1)\frac{16(15F)}{\pi(1)^3} = 76.4F$$

From Eq. (3–13) the nonzero principal stresses σ_A and σ_B are

$$\sigma_A, \sigma_B = \frac{142.6F + 0}{2} \pm \sqrt{\left(\frac{142.6F - 0}{2}\right)^2 + (76.4F)^2} = 175.8F, -33.2F$$

This puts us in the fourth-quadrant of the σ_A, σ_B plane.

Example 3–9 (continued)

(*a*) For BCM, Eq. (5–31*b*) applies with $n = 1$ for failure.

$$\frac{\sigma_A}{S_{ut}} - \frac{\sigma_B}{S_{uc}} = \frac{175.8F}{31(10^3)} - \frac{(-33.2F)}{109(10^3)} = 1$$

Solving for F yields

$$F = 167 \text{ lbf}$$ Answer

(*b*) For MM, the slope of the load line is $|\sigma_B/\sigma_A| = 33.2/175.8 = 0.189 < 1$. Obviously, Eq. (5–32*a*) applies.

$$\frac{\sigma_A}{S_{ut}} = \frac{175.8F}{31(10^3)} = 1$$

$$F = 176 \text{ lbf}$$ Answer

As one would expect from inspection of Fig. 5–19, Coulomb-Mohr is more conservative.

Selection of Failure Criteria

First determine ductile vs. brittle

For ductile

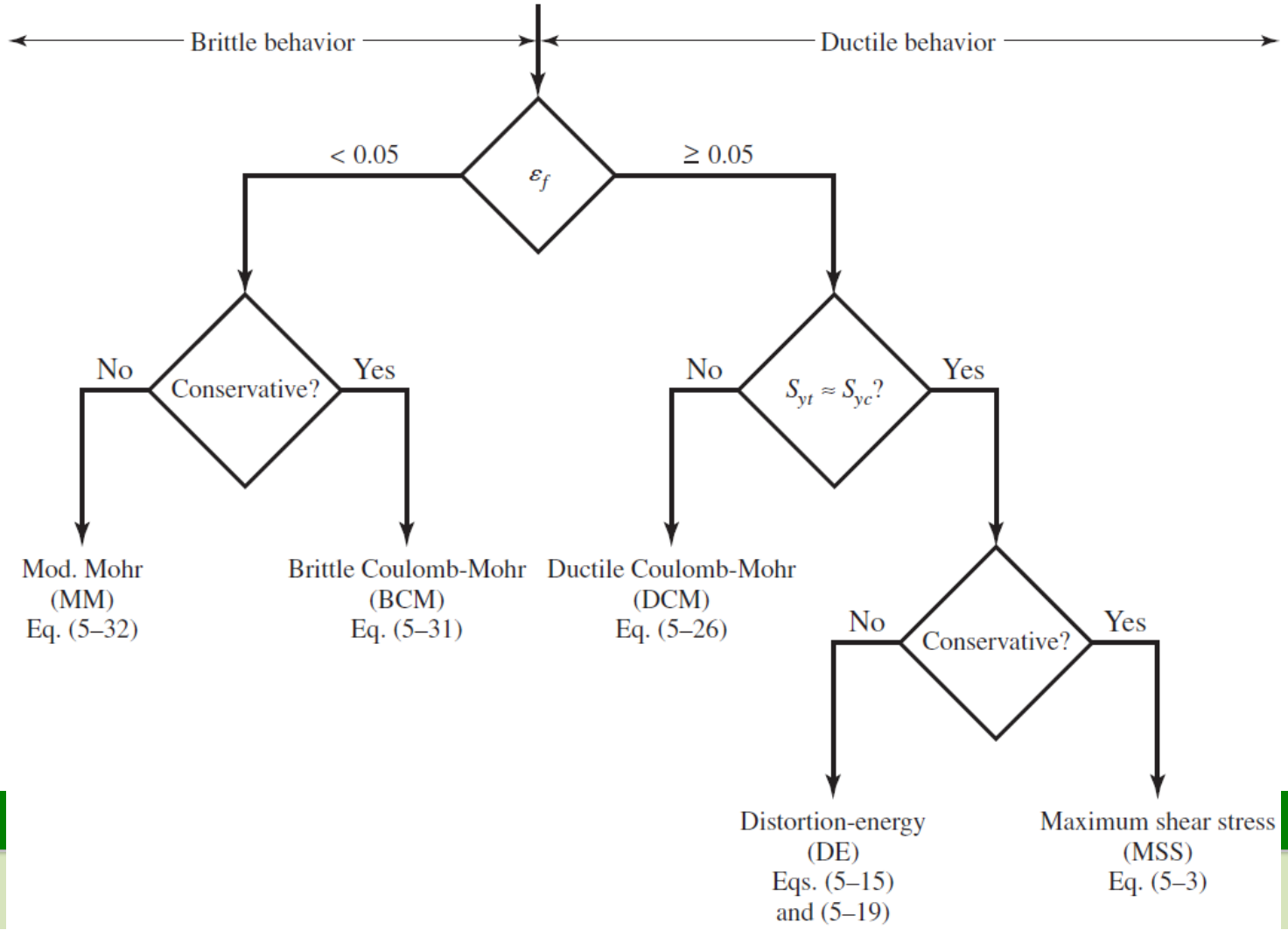
- MSS is conservative, often used for design where higher reliability is desired
- DE is typical, often used for analysis where agreement with experimental data is desired
- If tensile and compressive strengths differ, use Ductile Coulomb-Mohr

For brittle

- Mohr theory is best, but difficult to use
- Brittle Coulomb-Mohr is very conservative in 4th quadrant
- Modified Mohr is still slightly conservative in 4th quadrant, but closer to typical

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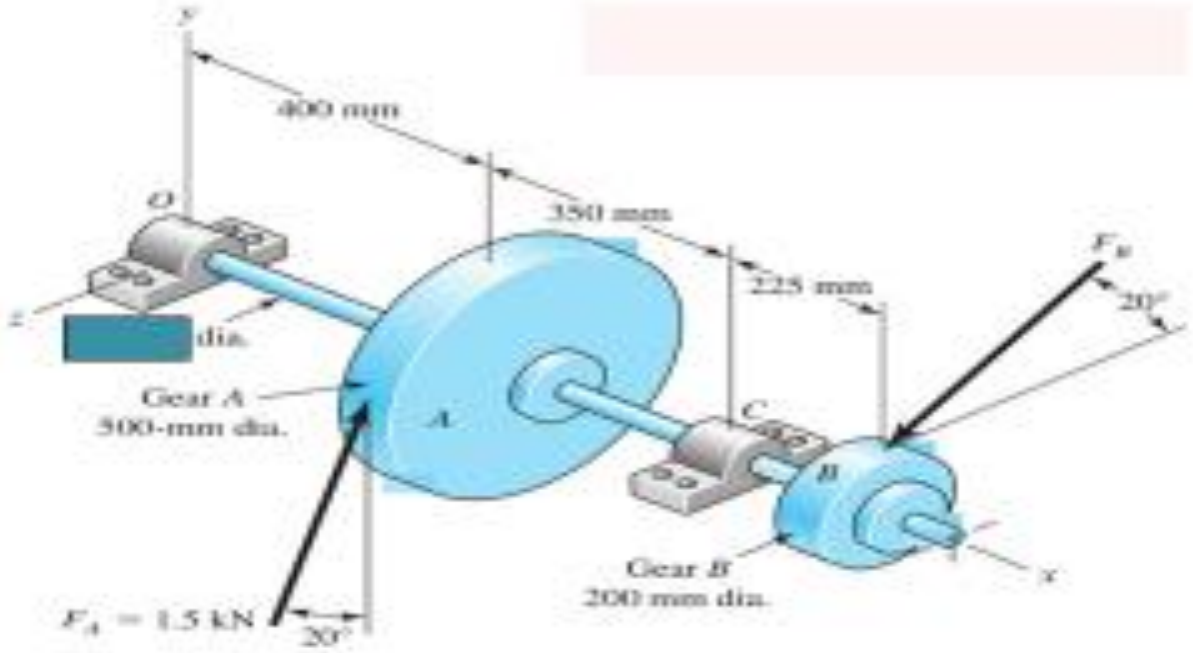
Selection of Failure Criteria



Assignment 1- Design of Axle

The axle shown above is to be designed to have the same diameter. Using the design process, provide complete specification for:

a. steel axle and
b. for cast iron axle



Problem 3–72*

THANK YOU

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LECTURE 4

FATIGUE FAILURES RESULTING FROM VARYING LOADING

Lecture Outline

- Introduction to Fatigue in Metals
- Stages of Fatigue Failure
- Schematics of Fatigue Fracture Surfaces
- Fatigue Fracture Examples
- Fatigue-Life Methods
- The Endurance Limit
- Stress Concentration and Notch Sensitivity
- Characterizing Fluctuating Stresses
- Fatigue Loading
- Failure Criteria for Ductile Materials
- Fatigue Criteria for Brittle Materials

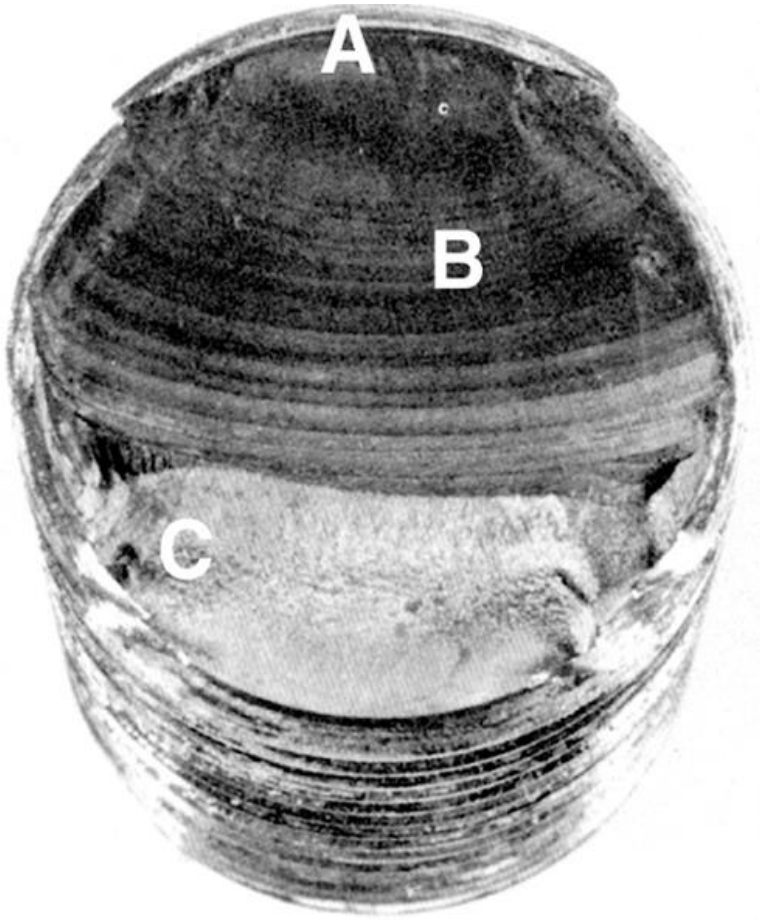
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Introduction to Fatigue in Metals

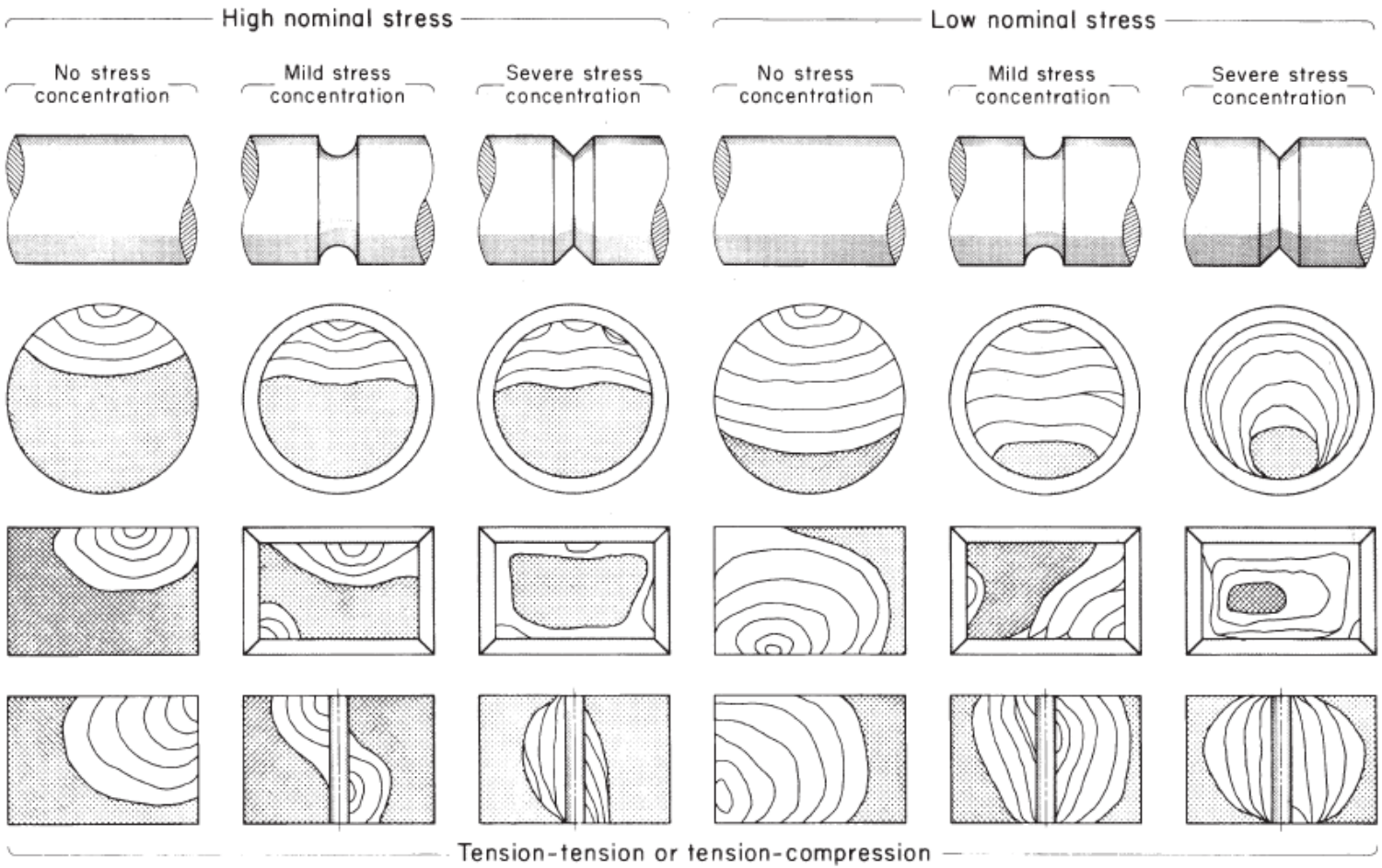
- Loading produces stresses that are variable, repeated, alternating, or fluctuating
- Maximum stresses well below yield strength
- Failure occurs after many stress cycles
- Failure is by sudden ultimate fracture
- No visible warning in advance of failure

Stages of Fatigue Failure

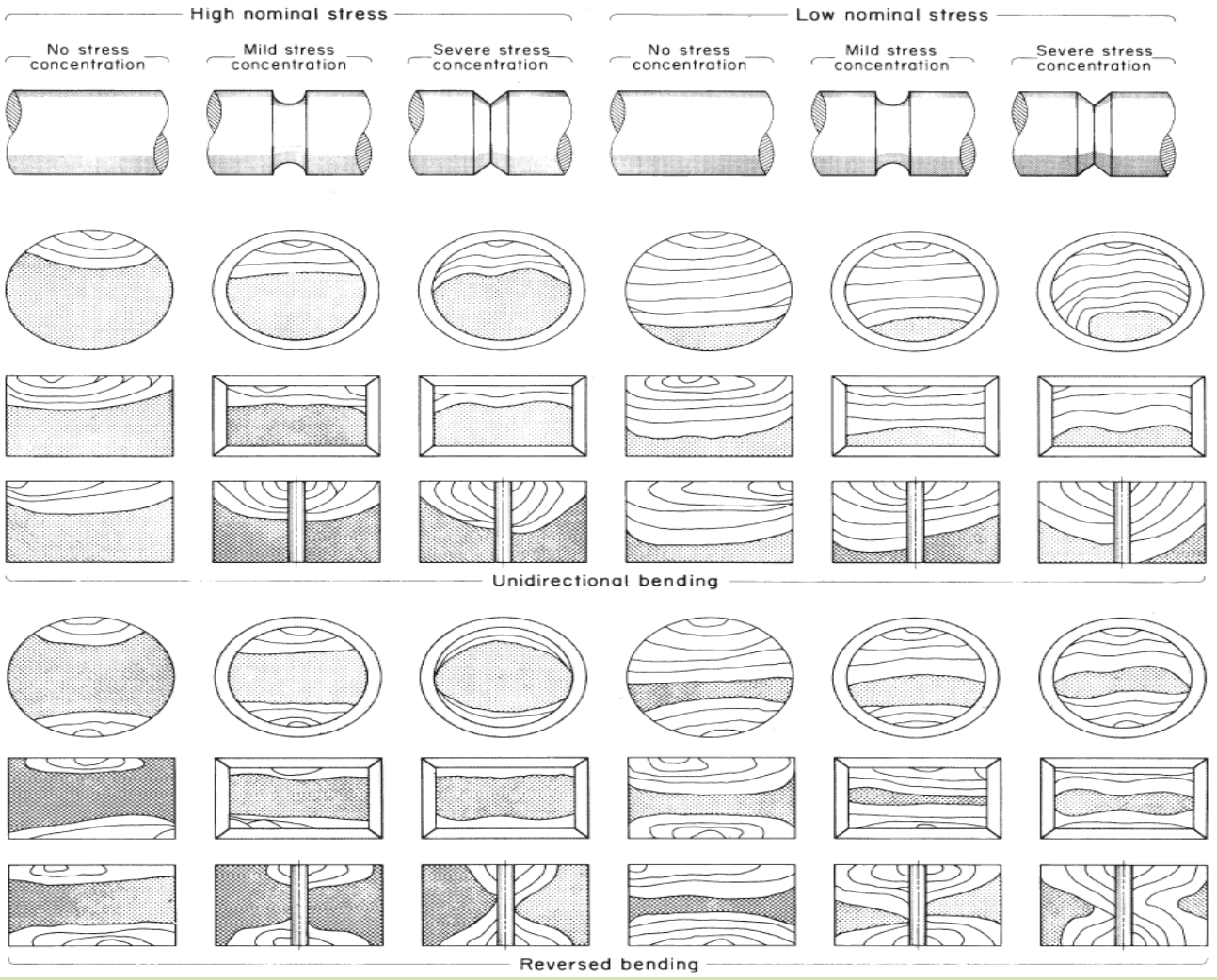
- *Stage I* – Initiation of micro-crack due to cyclic plastic deformation
- *Stage II* – Progresses to macro-crack that repeatedly opens and closes, creating bands called *beach marks*
- *Stage III* – Crack has propagated far enough that remaining material is insufficient to carry the load, and fails by simple ultimate failure



Schematics of Fatigue Fracture Surfaces

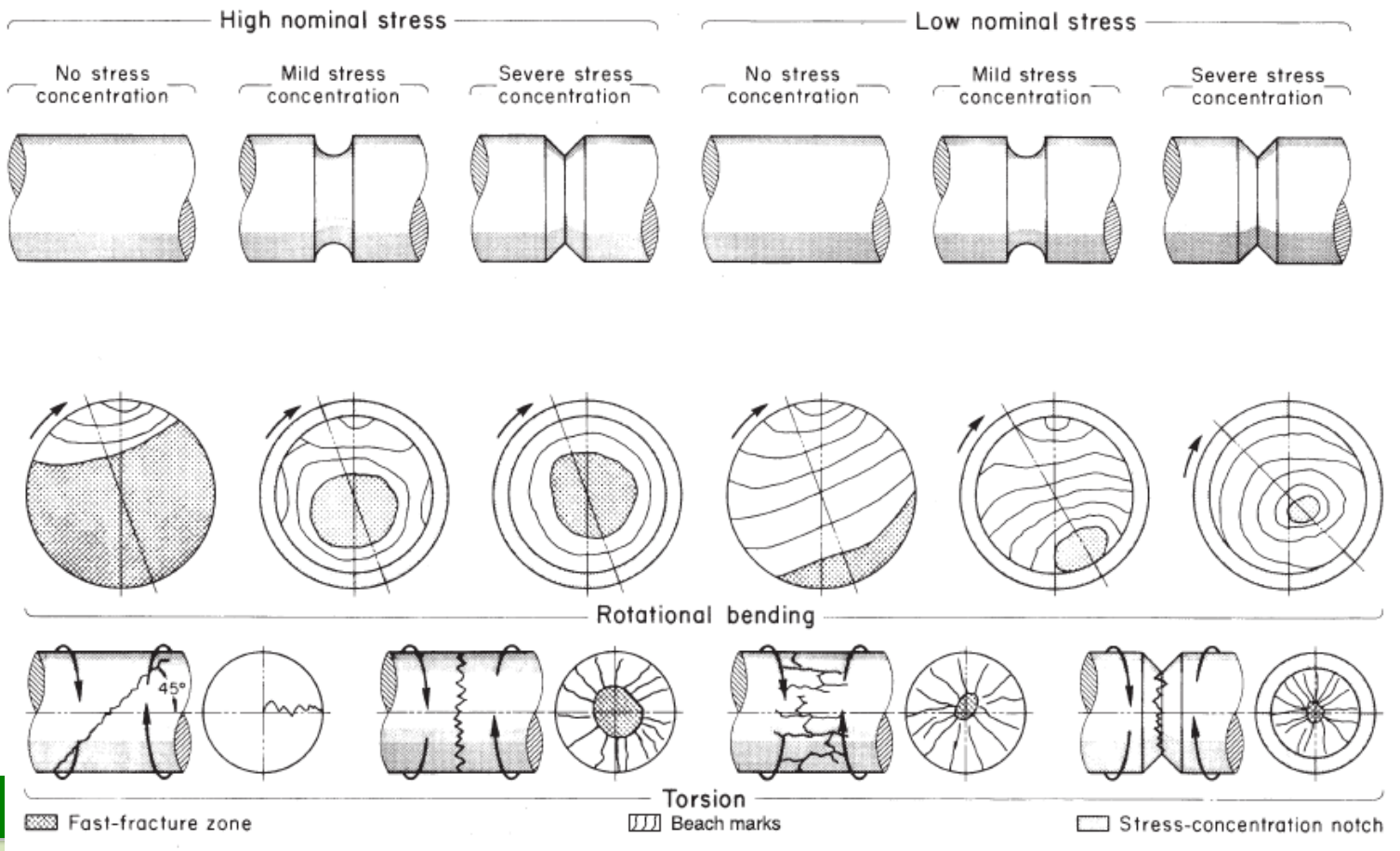


Schematics of Fatigue Fracture Surfaces



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Schematics of Fatigue Fracture Surfaces



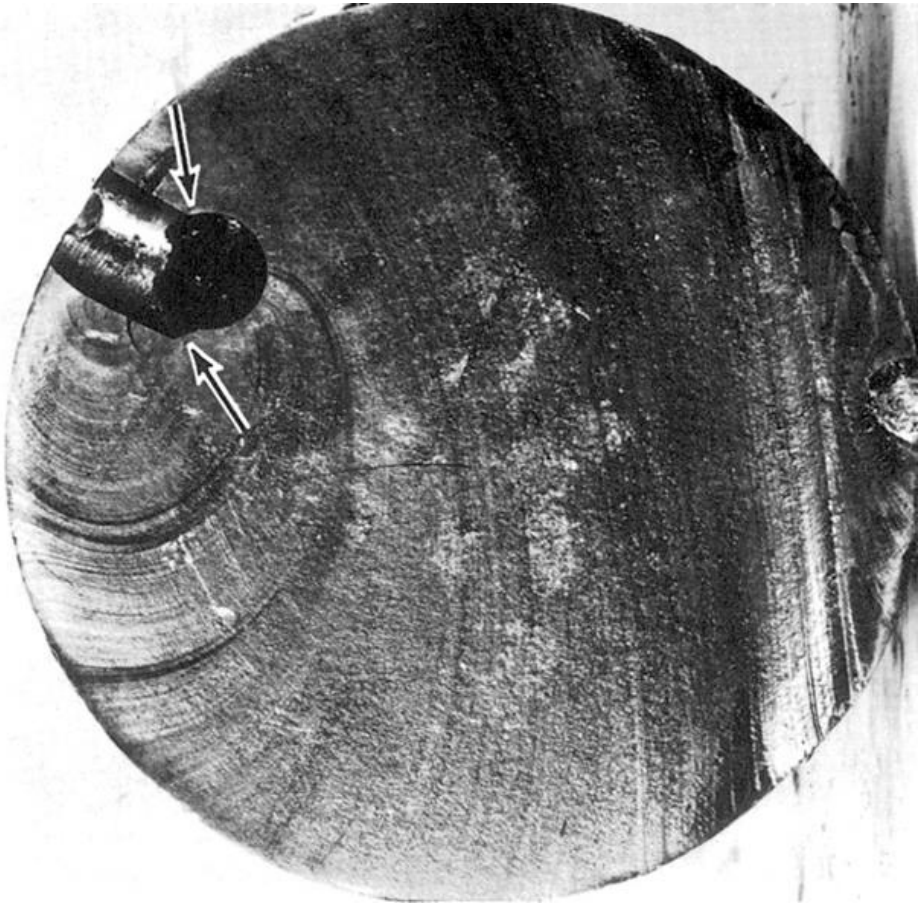
Fatigue Fracture Examples

- AISI 4320 drive shaft
- B– crack initiation at stress concentration in keyway
- C– Final brittle failure



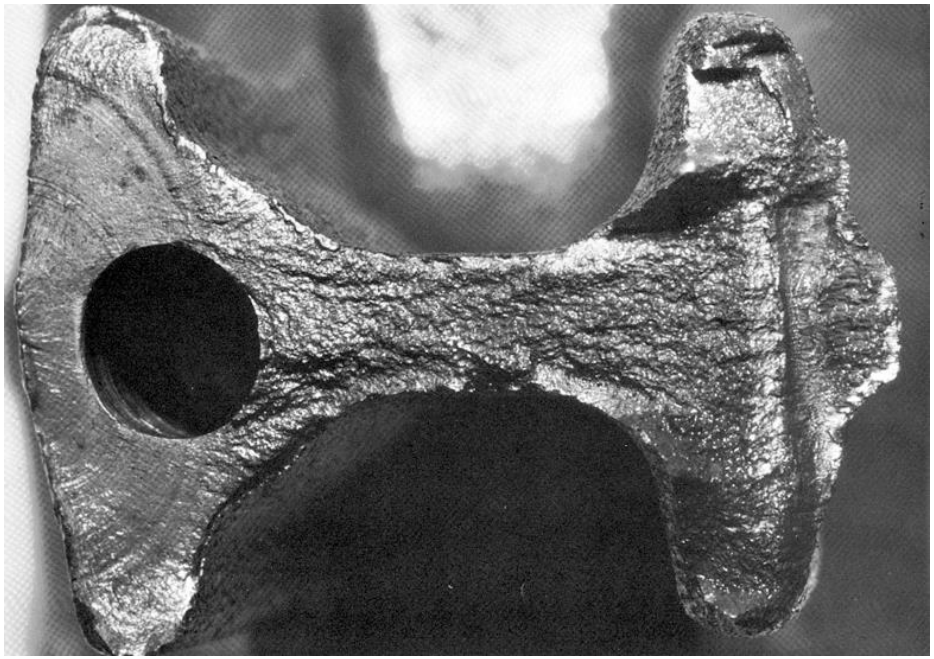
Fatigue Fracture Examples

- Fatigue failure initiating at mismatched grease holes
- Sharp corners (at arrows) provided stress concentrations



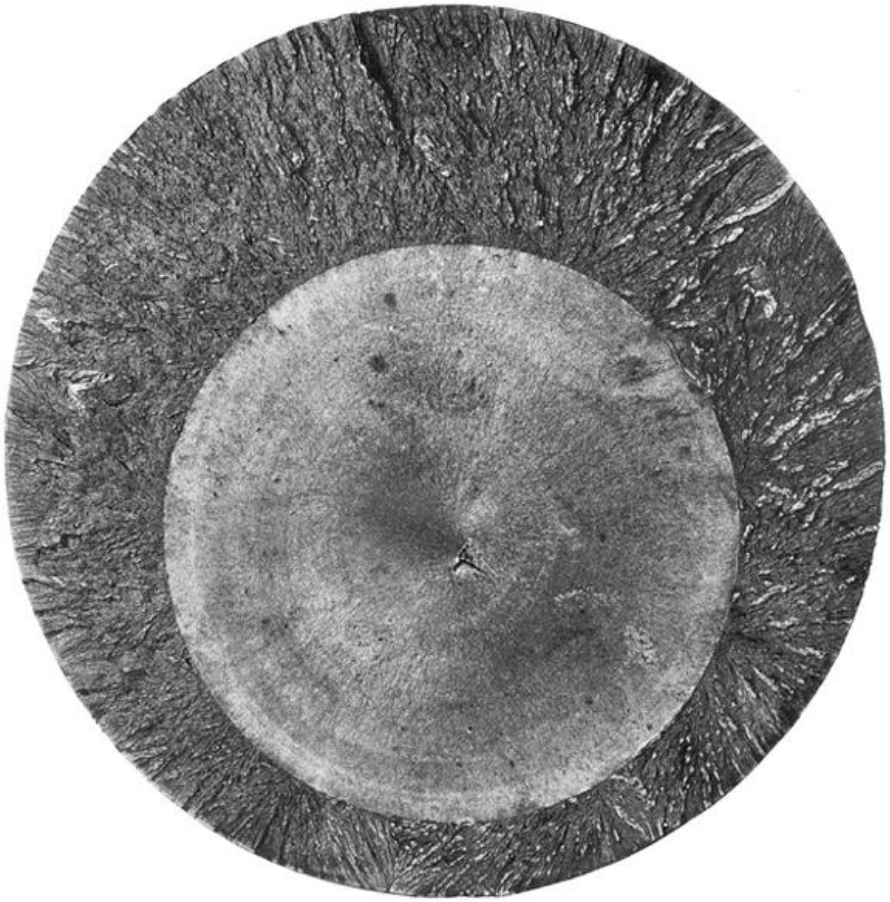
Fatigue Fracture Examples

- Fatigue failure of forged connecting rod
- Crack initiated at flash line of the forging at the left edge of picture
- Beach marks show crack propagation halfway around the hole before ultimate fracture



Fatigue Fracture Examples

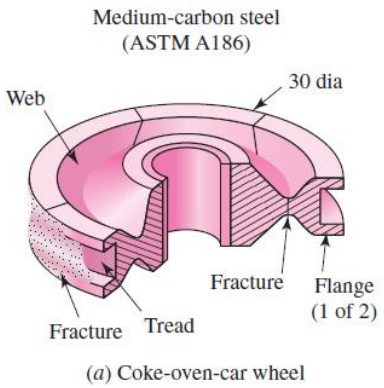
- Fatigue failure of a 200-mm diameter piston rod of an alloy steel steam hammer
- Loaded axially
- Crack initiated at a forging flake internal to the part
- Internal crack grew outward symmetrically



Fatigue Fracture Examples

- Double-flange trailer wheel

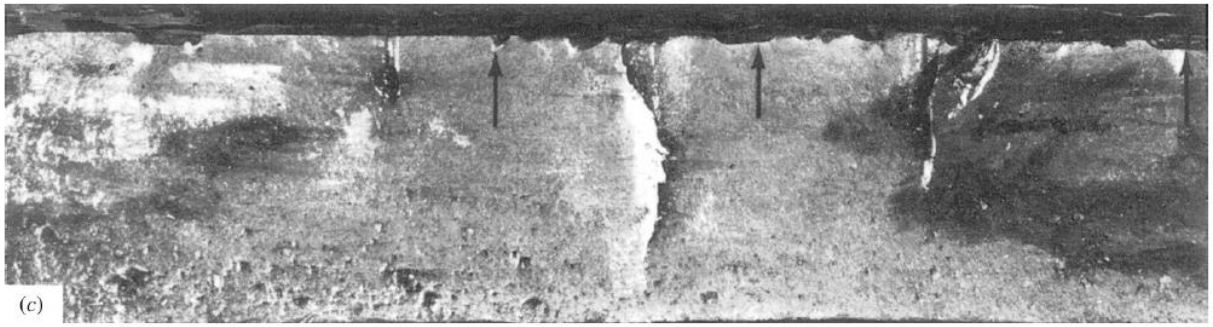
- Cracks initiated at stamp marks



(a) Coke-oven-car wheel

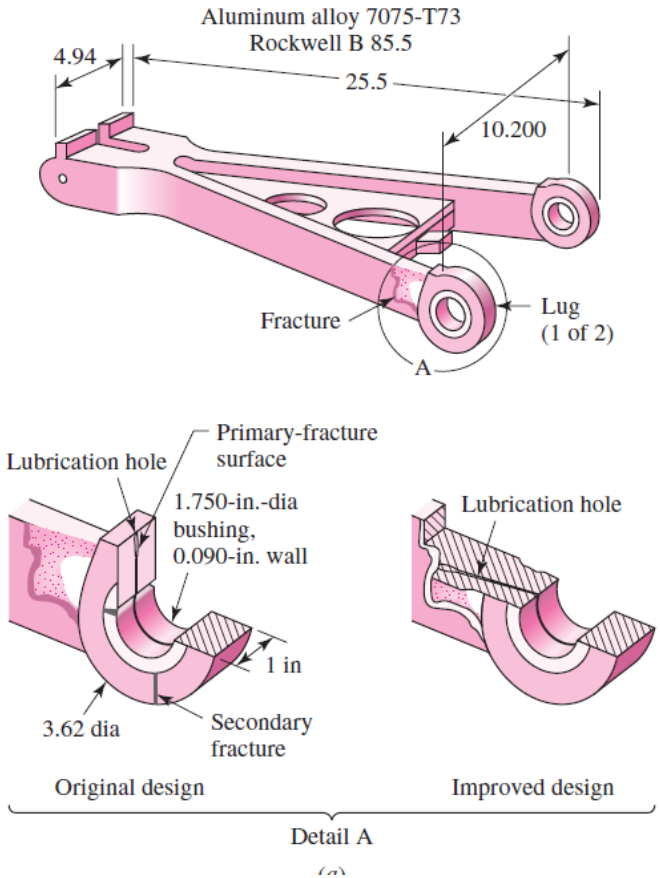


(b)


(c)

Fatigue Fracture Examples

- Aluminum allow landing-gear torque-arm assembly redesign to eliminate fatigue fracture at lubrication hole



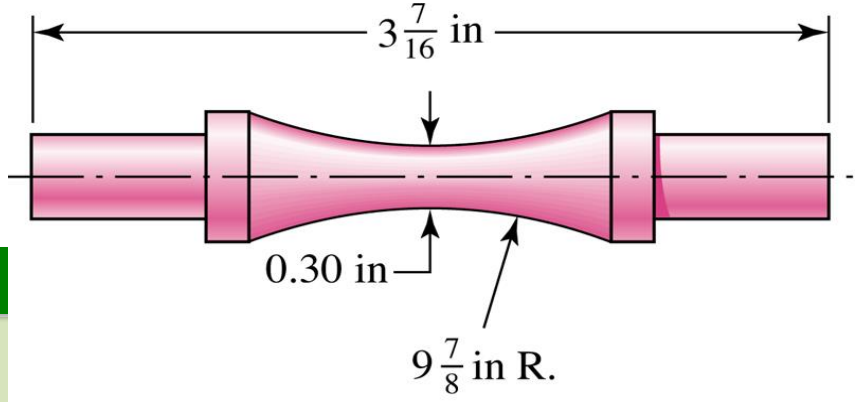


Fatigue-Life Methods

- Three major fatigue life models
- Methods predict life in number of cycles to failure, *N*, for a specific level of loading
- Stress-life method
  - ✓ Least accurate, particularly for low cycle applications
  - ✓ Most traditional, easiest to implement
- Strain-life method
  - ✓ Detailed analysis of plastic deformation at localized regions
  - ✓ Several idealizations are compounded, leading to uncertainties in results

Fatigue-Life Methods: Stress-Life Method

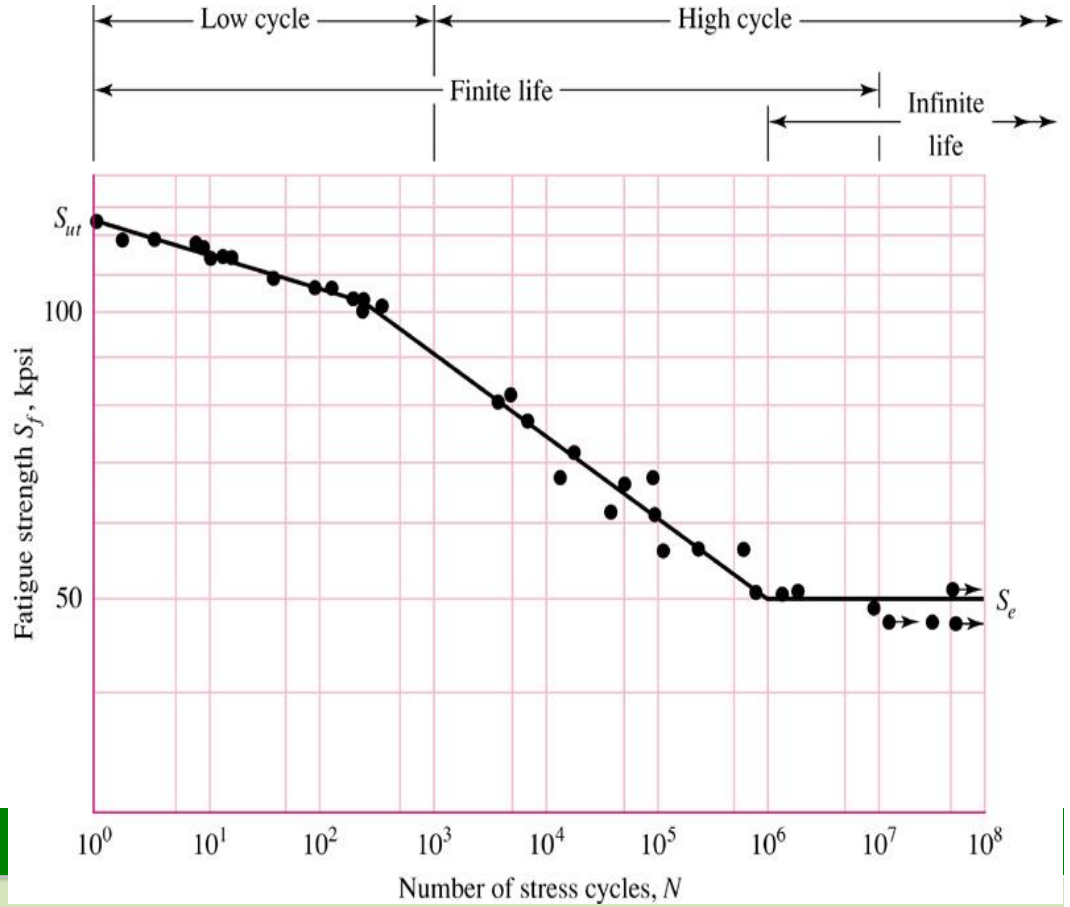
- Test specimens are subjected to repeated stress while counting cycles to failure
- Most common test machine is R. R. Moore high-speed rotating-beam machine
- Subjects specimen to pure bending with no transverse shear
- As specimen rotates, stress fluctuates between equal magnitudes of tension and compression, known as *completely reversed* stress cycling
- Specimen is carefully machined and polished



Fatigue-Life Methods: Stress-Life Method

S-N Diagram

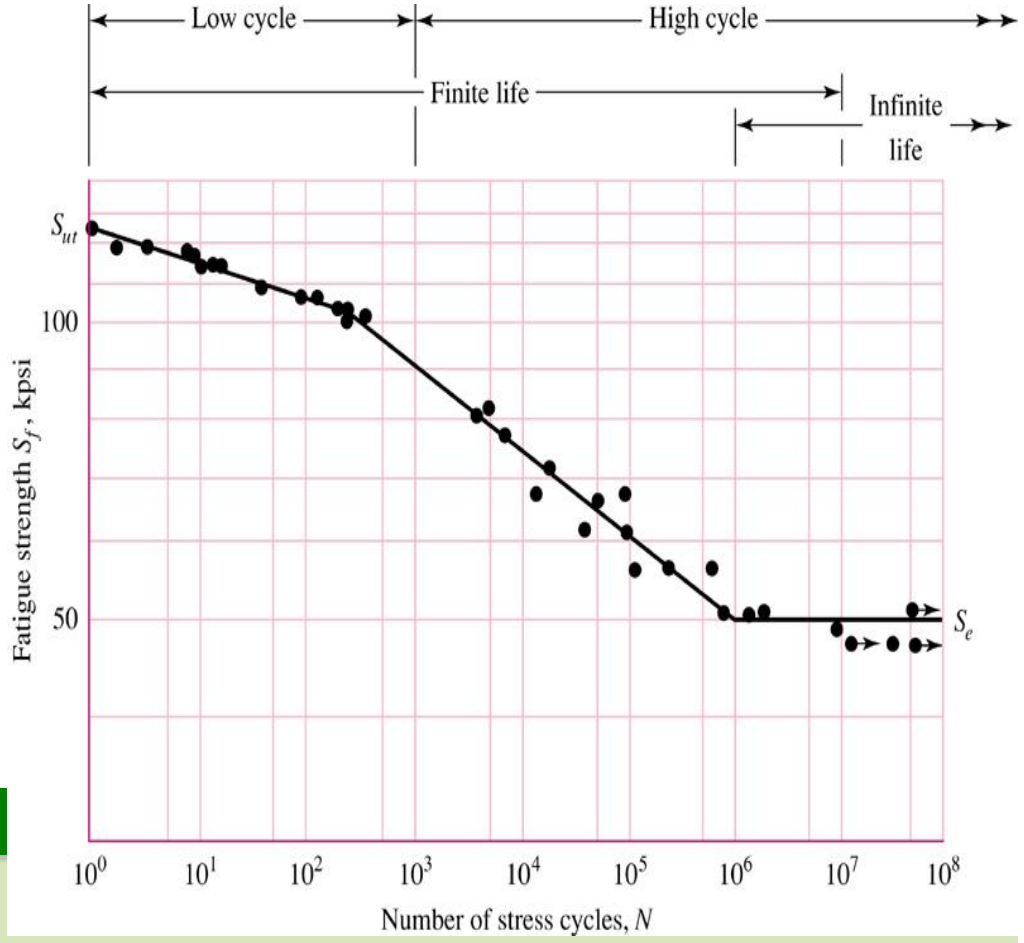
- Number of cycles to failure at varying stress levels is plotted on log-log scale
- For steels, a knee occurs near 10^6 cycles
- Strength corresponding to the knee is called *endurance limit* S_e



Fatigue-Life Methods: Stress-Life Method

S-N Diagram for Steel

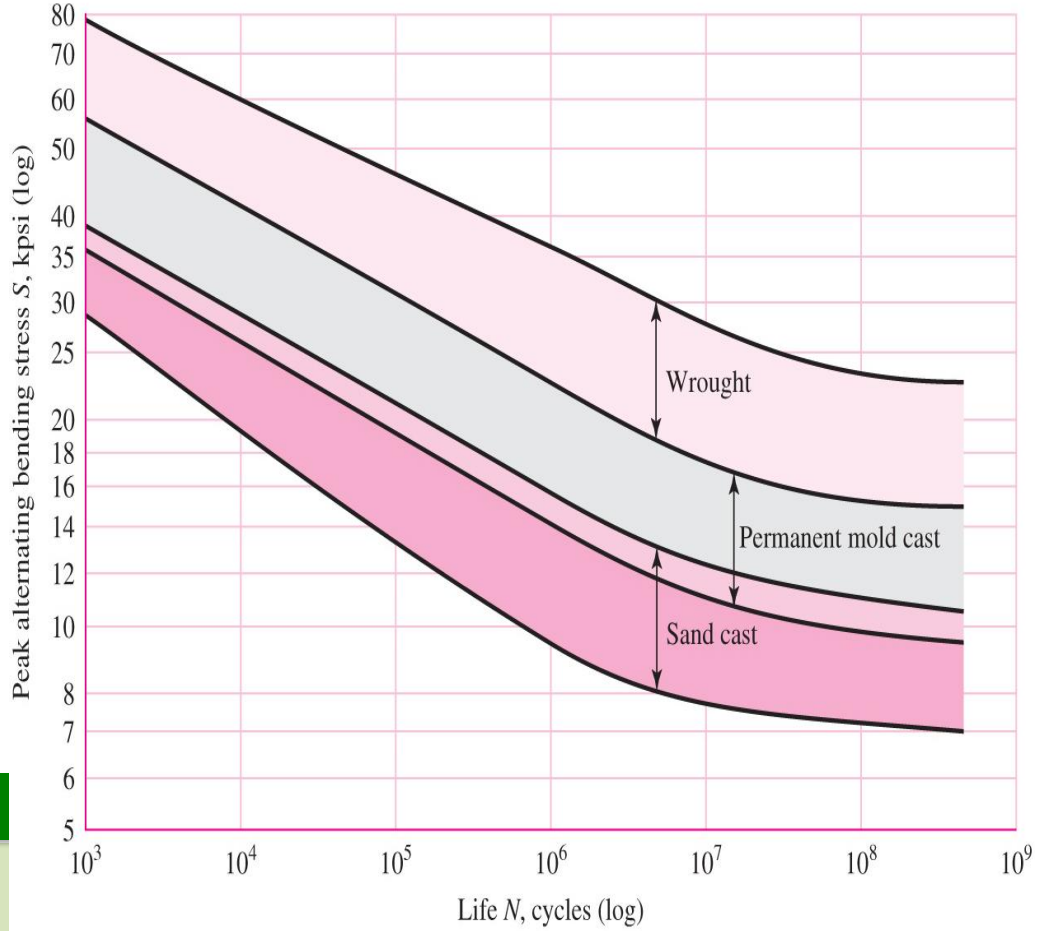
- Stress levels below S_e predict infinite life
- Between 10^3 and 10^6 cycles, finite life is predicted
- Below 10^3 cycles is known as *low cycle*, and is often considered quasi-static. Yielding usually occurs before fatigue in this zone.



Fatigue-Life Methods: Stress-Life Method

S-N Diagram for Nonferrous Metals

- Nonferrous metals often do not have an endurance limit.
- Fatigue strength S_f is reported at a specific number of cycles
- Figure shows typical *S-N* diagram for aluminums

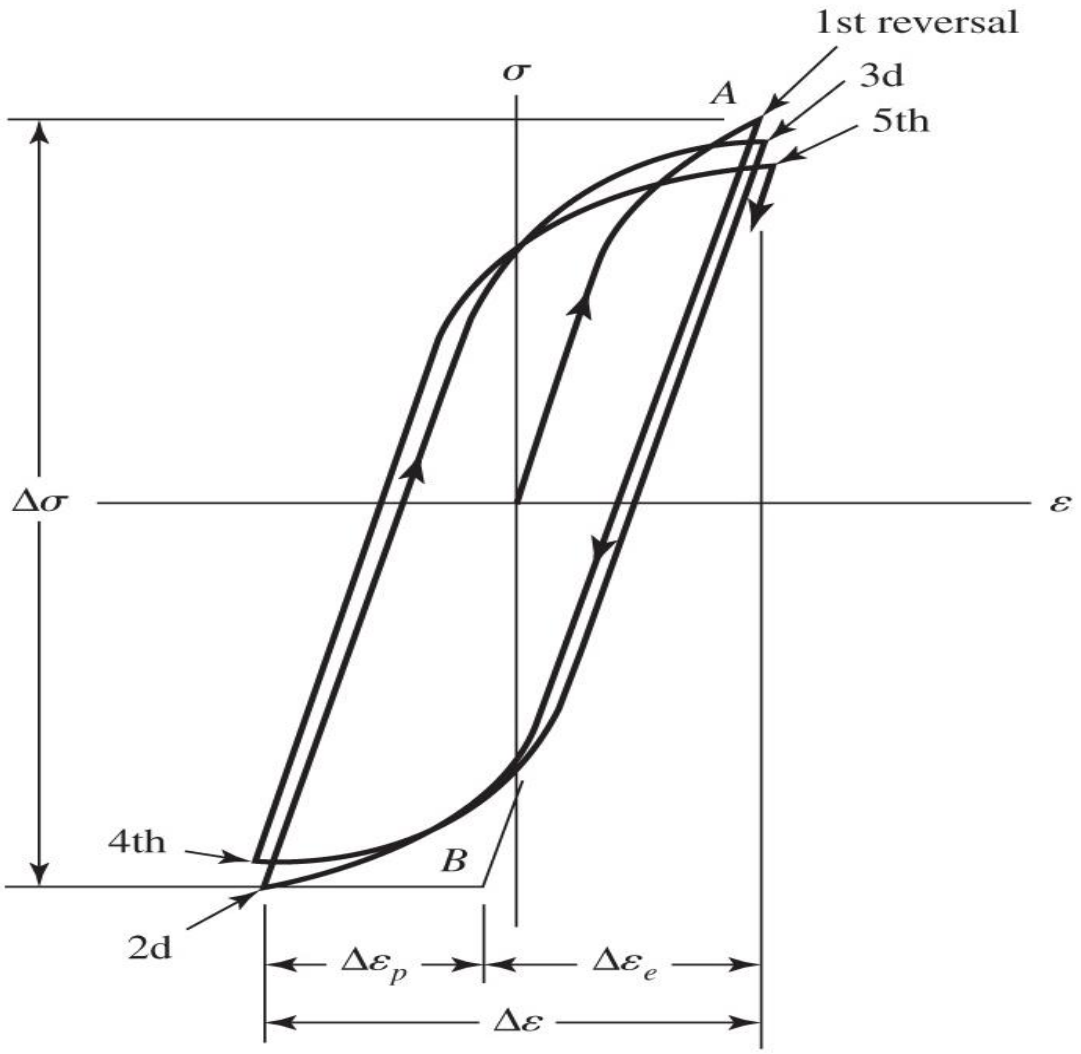


Fatigue-Life Methods: Strain-Life Method

- Method uses detailed analysis of plastic deformation at localized regions
- Compounding of several idealizations leads to significant uncertainties in numerical results
- Useful for explaining nature of fatigue
- Fatigue failure almost always begins at a local discontinuity
- When stress at discontinuity exceeds elastic limit, plastic strain occurs
- Cyclic plastic strain can change elastic limit, leading to fatigue

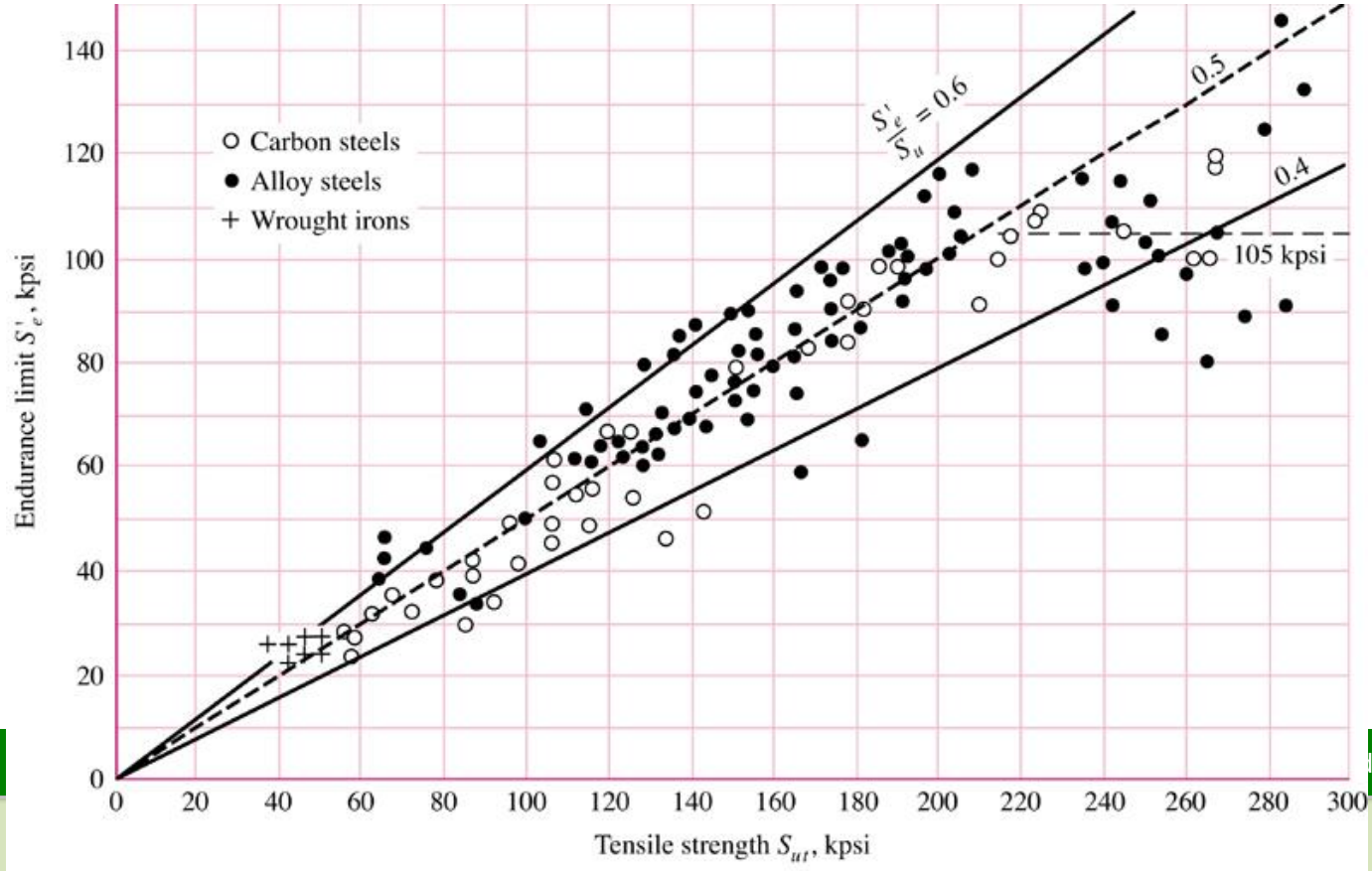
Fatigue-Life Methods: Strain-Life Method

- Fig. shows true stress-true strain hysteresis loops of the first five stress reversals



The Endurance Limit

- The endurance limit for steels has been experimentally found to be related to the ultimate strength



The Endurance Limit: Steel ($S_e' = S_{10}^6$)

- Simplified estimate of endurance limit for steels for the rotating-beam specimen, S'_e

US Units

For $S_{ut} \leq 200$ Ksi: $S_e' = 0.5S_{ut}$ Ksi

For $S_{ut} > 200$ Ksi: $S_e' = 100$ Ksi

SI Units

For $S_{ut} \leq 1400$ MPa: $S_e' = 0.5S_{ut}$ MPa

For $S_{ut} > 1400$ MPa: $S_e' = 700$ MPa

Note: For Steel, $S_{ut} = 0.5H_B$ (Ksi) $= 3.41H_B$ (MPa)

The Endurance Limit: Wrought Al-Alloys ($S_e' = S_{5x10^8}$)

- Simplified estimate of endurance limit for Wrought Al-Alloy for the rotating-beam specimen, S'_e

US Units

For $S_{ut} \leq 47$ Ksi

$$S_e' = 0.4S_{ut} \text{ Ksi}$$

For $S_{ut} > 47$ Ksi

$$S_e' = 19 \text{ Ksi}$$

SI Units

For $S_{ut} \leq 325$ MPa

$$S_e' = 0.4S_{ut} \text{ MPa}$$

For $S_{ut} > 325$ MPa

$$S_e' = 130 \text{ MPa}$$

Endurance Limit: Modifying Factors

- Endurance limit S'_e is for carefully prepared and tested specimen
- If warranted, S_e is obtained from testing of actual parts

$$S_e = k_a k_b k_c k_d k_e k_f S'_e$$

k_a = surface condition modification factor

k_b = size modification factor

k_c = load modification factor

k_d = temperature modification factor

k_e = reliability factor[13]

k_f = miscellaneous-effects modification factor

S'_e = rotary-beam test specimen endurance limit

S_e = endurance limit at the critical location of a machine part in the geometry and condition of use

Endurance Limit: Modifying Factors

Surface Factor k_a

- Stresses tend to be high at the surface
- Surface finish has an impact on initiation of cracks at localized stress concentrations
- Surface factor is a function of ultimate strength. Higher strengths are more sensitive to rough surfaces.

$$k_a = a S_{ut}^b$$

Surface Finish	Factor a S_{ut}, kpsi	Factor a S_{ut}, MPa	Exponent b
Ground	1.34	1.58	−0.085
Machined or cold-drawn	2.70	4.51	−0.265
Hot-rolled	14.4	57.7	−0.718
As-forged	39.9	272.	−0.995

Endurance Limit: Modifying Factors

Size Factor k_b

- Larger parts have greater surface area at high stress levels
- Likelihood of crack initiation is higher
- Size factor is obtained from experimental data with wide scatter
- For bending and torsion loads, the trend of the size factor data is given by

$$k_b = \begin{cases} (d/0.3)^{-0.107} = 0.879d^{-0.107} & 0.11 \le d \le 2 \text{ in} \\ 0.91d^{-0.157} & 2 < d \le 10 \text{ in} \\ (d/7.62)^{-0.107} = 1.24d^{-0.107} & 2.79 \le d \le 51 \text{ mm} \\ 1.51d^{-0.157} & 51 < d \le 254 \text{ mm} \end{cases}$$

- Applies only for round, rotating diameter
- For axial load, there is no size effect, so $k_b = 1$

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Endurance Limit: Modifying Factors

Size Factor k_b

- For parts that are not round and rotating, an equivalent round rotating diameter is obtained.
- Equate the volume of material stressed at and above 95% of the maximum stress to the same volume in the rotating-beam specimen.
- Lengths cancel, so equate the areas.
- For a rotating round section, the 95% stress area is the area of a ring,

$$A_{0.95\sigma} = \frac{\pi}{4}[d^2 - (0.95d)^2] = 0.0766d^2 \qquad \textbf{Eq. ii}$$

- Equate 95% stress area for other conditions to Eq. ii and solve for d as the equivalent round rotating diameter

Endurance Limit: Modifying Factors

Size Factor k_b

- For non-rotating round, $A_{0.95\sigma} = 0.01046d^2$ **Eq. iii**

- Equating to Eq. ii and solving for equivalent diameter,

$$d_e = 0.370d \quad \textbf{Eq. iv}$$

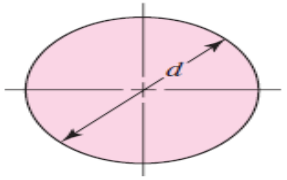
- Similarly, for rectangular section h x b, $A_{95s} = 0.05\,hb$. Equating to Eq. ii

$$d_e = 0.808(hb)^{1/2} \quad \textbf{Eq. v}$$

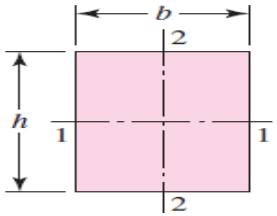
- Other common cross sections are given in Table

Endurance Limit: Modifying Factors

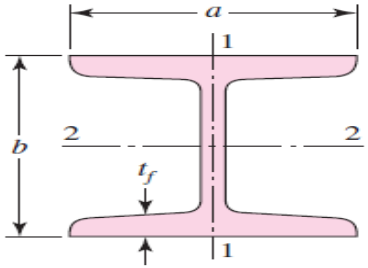
Size Factor k_b



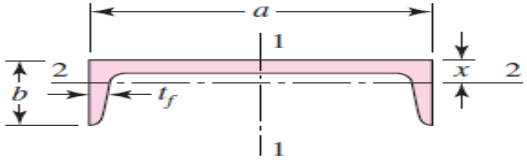
$$A_{0.95\sigma} = 0.01046d^2$$
$$d_e = 0.370d$$



$$A_{0.95\sigma} = 0.05hb$$
$$d_e = 0.808\sqrt{hb}$$



$$A_{0.95\sigma} = \begin{cases} 0.10at_f & \text{axis 1-1} \\ 0.05ba \quad t_f > 0.025a & \text{axis 2-2} \end{cases}$$



$$A_{0.95\sigma} = \begin{cases} 0.05ab & \text{axis 1-1} \\ 0.052xa + 0.1t_f(b-x) & \text{axis 2-2} \end{cases}$$

Endurance Limit: Modifying Factors

Loading Factor k_c

- Accounts for changes in endurance limit for different types of fatigue loading.
- Only to be used for single load types.
- Use Combination Loading method when more than one load type is present.

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion}^{17} \end{cases}$$

Endurance Limit: Modifying Factors

Temperature Factor k_d

- Endurance limit appears to maintain same relation to ultimate strength for elevated temperatures as at room temperature
- This relation is summarized in Table

Temperature, °C	S_T/S_{RT}	Temperature, °F	S_T/S_{RT}
20	1.000	70	1.000
50	1.010	100	1.008
100	1.020	200	1.020
150	1.025	300	1.024
200	1.020	400	1.018
250	1.000	500	0.995
300	0.975	600	0.963
350	0.943	700	0.927
400	0.900	800	0.872
450	0.843	900	0.797
500	0.768	1000	0.698
550	0.672	1100	0.567
600	0.549		

Endurance Limit: Modifying Factors

Temperature Factor k_d

- If ultimate strength is known for operating temperature, then just use that strength. Let $k_d = 1$ and proceed as usual.

- If ultimate strength is known only at room temperature, then use Table to estimate ultimate strength at operating temperature. With that strength, let $k_d = 1$ and proceed as usual.

- Alternatively, use ultimate strength at room temperature and apply temperature factor from Table to the endurance limit.

$$k_d = \frac{S_T}{S_{RT}} \quad \text{Eq. i}$$

- A fourth-order polynomial curve fit of the underlying data of Table can be used in place of the table, if desired.

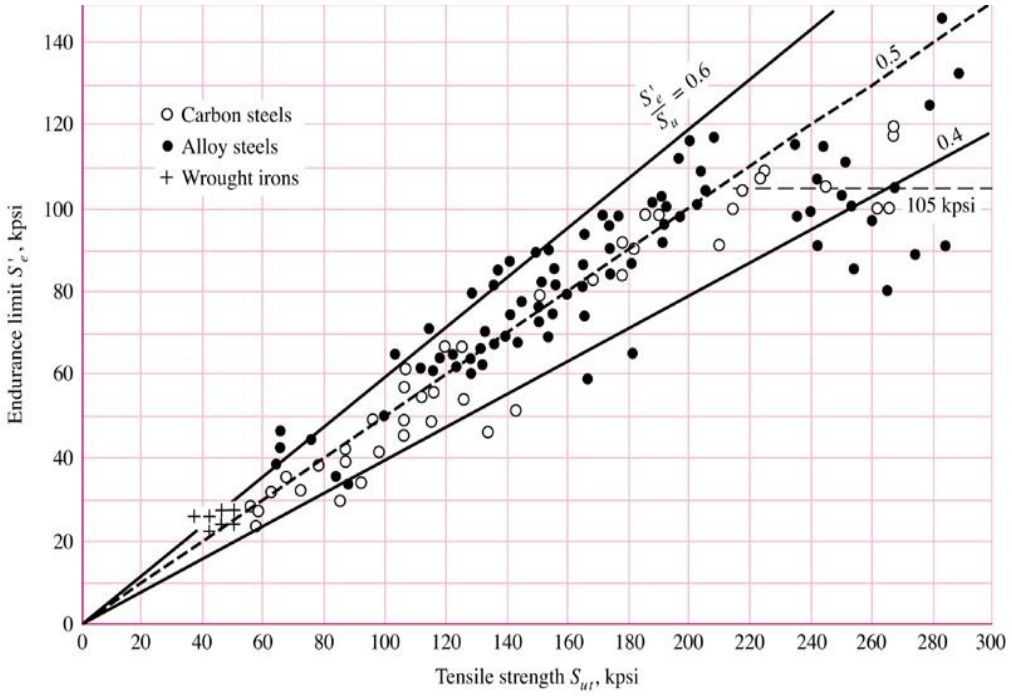
$$k_d = 0.975 + 0.432(10^{-3})T_F - 0.115(10^{-5})T_F^2 + 0.104(10^{-8})T_F^3 - 0.595(10^{-12})T_F^4 \quad \text{Eq. ii}$$

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Endurance Limit: Modifying Factors

Reliability Factor k_e

- From Fig., $S'_e = 0.5 S_{ut}$ is typical of the data and represents 50% reliability.
- Reliability factor adjusts to other reliabilities.
- *Only* adjusts Fig. assumption. *Does not* imply overall reliability.



Endurance Limit: Modifying Factors

Reliability Factor k_e

- Simply obtain k_e for desired reliability from Table.

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

Endurance Limit: Modifying Factors

Miscellaneous-Effects Factor k_f

- Reminder to consider other possible factors.
  - Residual stresses
  - Directional characteristics from cold working
  - Case hardening
  - Corrosion
  - Surface conditioning, e.g. electrolytic plating and metal spraying
  - Cyclic Frequency
  - Frettage Corrosion

- Limited data is available.

- May require research or testing.

Endurance Strength: 10^3 Cycles

- The endurance strength at 10^3 is given by

$$S_{10}^{3} = f\, S_{ut}$$

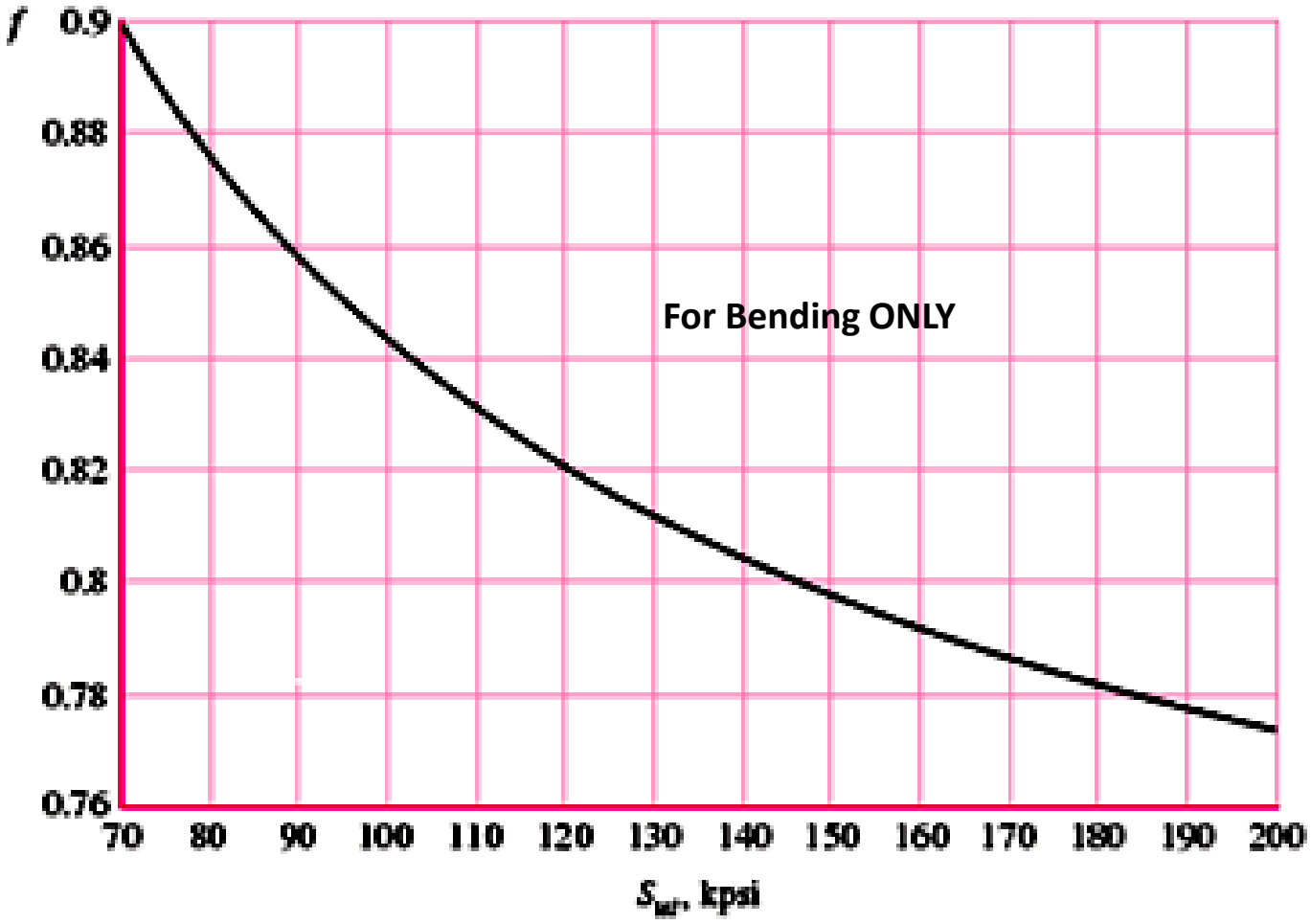
- Where f is

$$f = \begin{cases} 0.9 \text{ for Bending} \\ 0.75 \text{ for Axial/Direct Load} \\ 0.72 \text{ for Torsion (Steels)} \\ 0.63 \text{ for Torsion (All other ductile materials)} \end{cases}$$

- See Fig. to find the value of "f" (fatigue strength fraction)

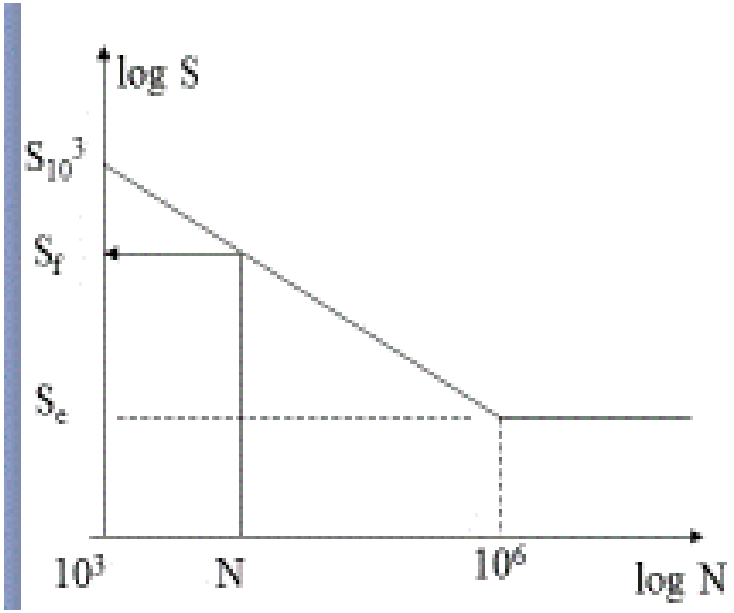
Compiled by NKAY STUDY

Endurance Strength: 10^3 Cycles



Endurance Strength: N Cycles Life

1. Find S_{10}^3
2. Find S_e
3. Make S-N Plot
4. Use extrapolation to find S_f
5. Results of Interpolation : $S_f = S_N$



It can be shown that:

$$S_f = aN^b$$

Where:

$$a = (S_{10}^3)^2 / S_e$$

$$b = (-1/3) \log [S_{10}^3 / S_e]$$

Example 4–1

A steel has a minimum ultimate strength of 520 MPa and a machined surface. Estimate k_a.

Solution From Table 6–2, $a = 4.51$ and $b = -0.265$. Then, from Eq. (6–19)

Answer $$k_a = 4.51(520)^{-0.265} = 0.860$$

Example 4–2

A steel shaft loaded in bending is 32 mm in diameter, abutting a filleted shoulder 38 mm in diameter. The shaft material has a mean ultimate tensile strength of 690 MPa. Estimate the Marin size factor k_b if the shaft is used in

(a) A rotating mode.

(b) A nonrotating mode.

Solution (a) From Eq. (6–20)

Answer
$$k_b = \left(\frac{d}{7.62}\right)^{-0.107} = \left(\frac{32}{7.62}\right)^{-0.107} = 0.858$$

(b) From Table 6–3,

$$d_e = 0.37d = 0.37(32) = 11.84 \text{ mm}$$

From Eq. (6–20),

Answer
$$k_b = \left(\frac{11.84}{7.62}\right)^{-0.107} = 0.954$$

Example 4–3

A 1035 steel has a tensile strength of 70 kpsi and is to be used for a part that sees 450°F in service. Estimate the Marin temperature modification factor and $(S_e)_{450^\circ}$ if
(*a*) The room-temperature endurance limit by test is $(S'_e)_{70^\circ} = 39.0$ kpsi.
(*b*) Only the tensile strength at room temperature is known.

Solution (*a*) First, from Eq. (6–27),

$$k_d = 0.975 + 0.432(10^{-3})(450) - 0.115(10^{-5})(450^2) + 0.104(10^{-8})(450^3) - 0.595(10^{-12})(450^4) = 1.007$$

Thus,

Answer $$(S_e)_{450^\circ} = k_d(S'_e)_{70^\circ} = 1.007(39.0) = 39.3 \text{ kpsi}$$

Example 4–3 (continued)

(*b*) Interpolating from Table 6–4 gives

$$(S_T/S_{RT})_{450^\circ} = 1.018 + (0.995 - 1.018)\frac{450 - 400}{500 - 400} = 1.007$$

Thus, the tensile strength at 450°F is estimated as

$$(S_{ut})_{450^\circ} = (S_T/S_{RT})_{450^\circ}\,(S_{ut})_{70^\circ} = 1.007(70) = 70.5 \text{ kpsi}$$

From Eq. (6–8) then,

Answer $$(S_e)_{450^\circ} = 0.5\,(S_{ut})_{450^\circ} = 0.5(70.5) = 35.2 \text{ kpsi}$$

Part *a* gives the better estimate due to actual testing of the particular material.

Example 4–4

Estimate the design endurance limit at 50% reliability for the 50-mm diameter machined bar loaded as shown. The material is AISI 1050 CD Steel. The part is non-rotating will be used at room temperature.



Solution

Material: S_y =580 MPa S_{ut} = 690 MPa

Specimen Endurance Limit, $S_e' = 0.5S_{ut} = 0.5\ (690) = 345$ MPa

Design Endurance Limit, $S_e = k_a k_b k_c k_d k_e S_e' = k_a k_b k_c k_d k_e\ (345)$

Surface factor (machined), $k_a = aS_{ut}^{\ b} = \mathbf{4.51\ (690)^{-0.265} = 0.798}$

Size factor (non-rotating), d = 50mm

$d_e = 0.370d = 18.5$ (Table)

Bending $k_b = 1.24d_e^{-0.107} = 0.907$

Load Factor (Bending), $k_c = 1.0$

Example 4–4 (continued)

Temperature (Room), $k_d = 1.0$

Reliability (50%), $k_e = 1.0$

Finally, $S_e = (0.798)(0.907)(1.0)(1.0)(1.0)(0.5x690)$

$= 250$ MPa

Example 4–5

A machine part is made from AISI 1035 Steel. It has ground finish and the load is variable and torsional. The material is heat-treated to a tensile strength of 710 MPa. The part diameter is 32 mm and will be rotating during service. It will be used in a machine at temperature of 300 F. Estimate: (a) the endurance limit at 95% reliability and (b) endurance strength at 10^4 cycles and show it on S-N Plot

S_{ut} = 710 MPa S_e' =0.5S_{ut} = 355 MPa

A.

Design Endurance Limit, $S_e = k_a k_b k_c k_d k_e S_e'$

Surface factor (ground), $k_a = aS_{ut}^{b}$ **= 1.58 (710)$^{-0.085}$ = 0.904**

Size factor (rotating), d = 32mm

Torsion $k_b = 1.24d^{-0.107} = 0.856$

Example 4–5 (continued)

Load Factor (Torsion), $k_c = 0.59$

Temperature (300F), $k_d = 1.024$

Reliability (95%), $k_e = 0.868$

Finally,

$S_e = (0.904)(0.856)(0.59)(1.024)(0.868)(355)$

$= \underline{144 \text{ MPa}}$

Example 4–5 (continued)

B. To find S_{10}^4

$S_f = S_{10}^4 = aN^b = a(10^4)^b$

Where $a = (S_{10}^3)^2 / S_e$

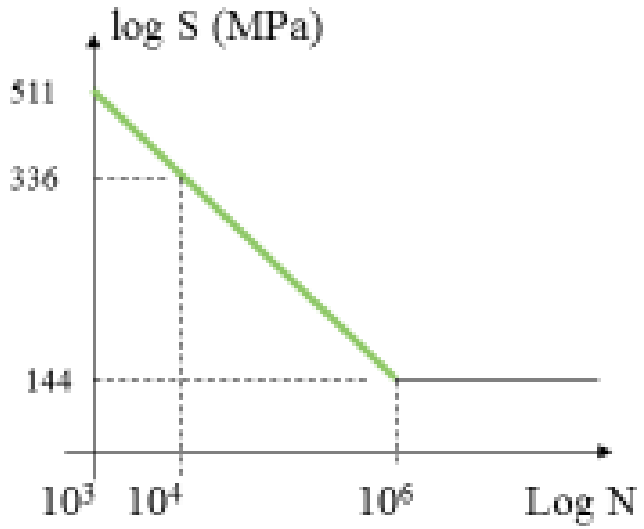
$S_{10}^3 = f\,S_{ut} = 0.72\,S_{ut} = 511$ MPa

$a = (511)^2 / S_e = 1813$ Mpa

$b = (-1/3) \log [S_{10}^3 / S_e]$

$= (-1/3) \log [511/ 144]$

$= -0.183$



Thus $S_{10}^4 = 1813(10^4)^{-0.183} = 336$ MPa

Stress Concentration and Notch Sensitivity

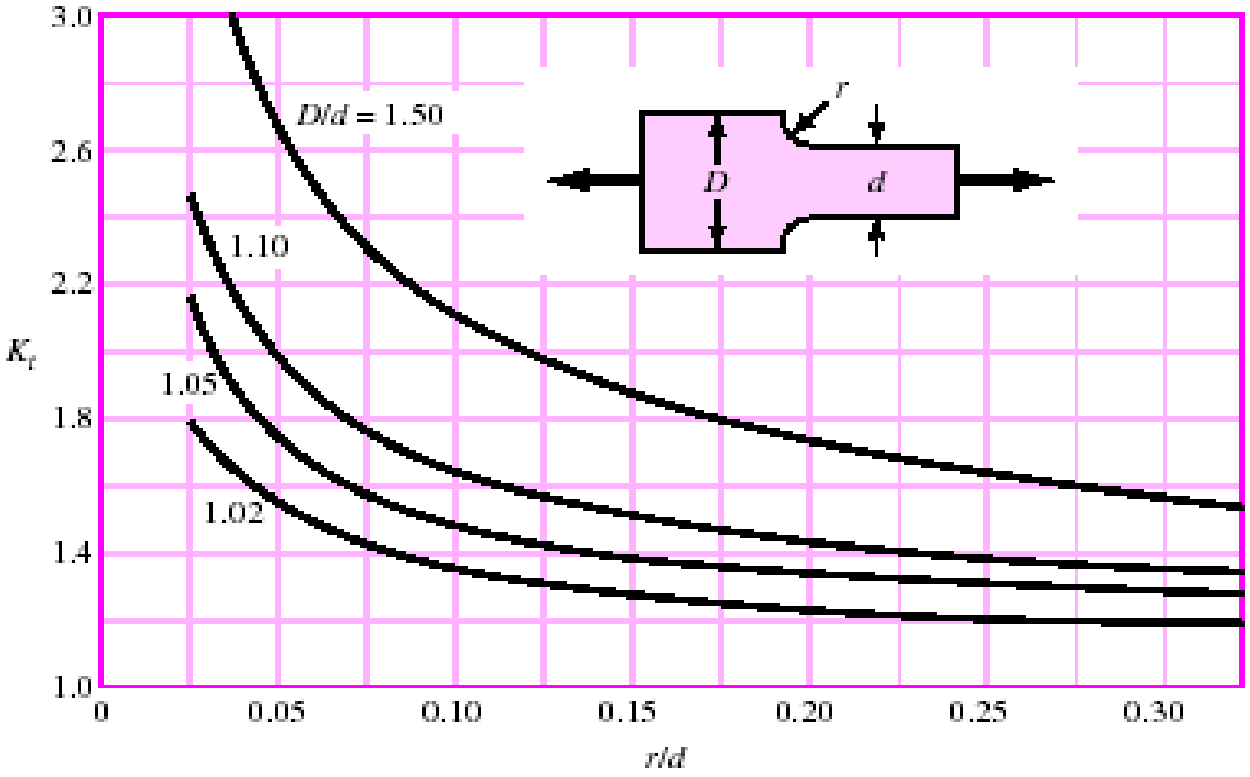
- For dynamic loading, stress concentration effects must be applied.
- Obtain K_t as usual (e.g. Appendix A–15)
- For fatigue, some materials are not fully sensitive to K_t so a reduced value can be used.
- Define K_f as the *fatigue stress-concentration factor*.
- Define q as *notch sensitivity*, ranging from 0 (not sensitive) to 1 (fully sensitive).

$$q = \frac{K_f - 1}{K_t - 1} \qquad \text{or} \qquad q_{\text{shear}} = \frac{K_{fs} - 1}{K_{ts} - 1}$$

- For $q = 0$, $K_f = 1$
- For $q = 1$, $K_f = K_t$

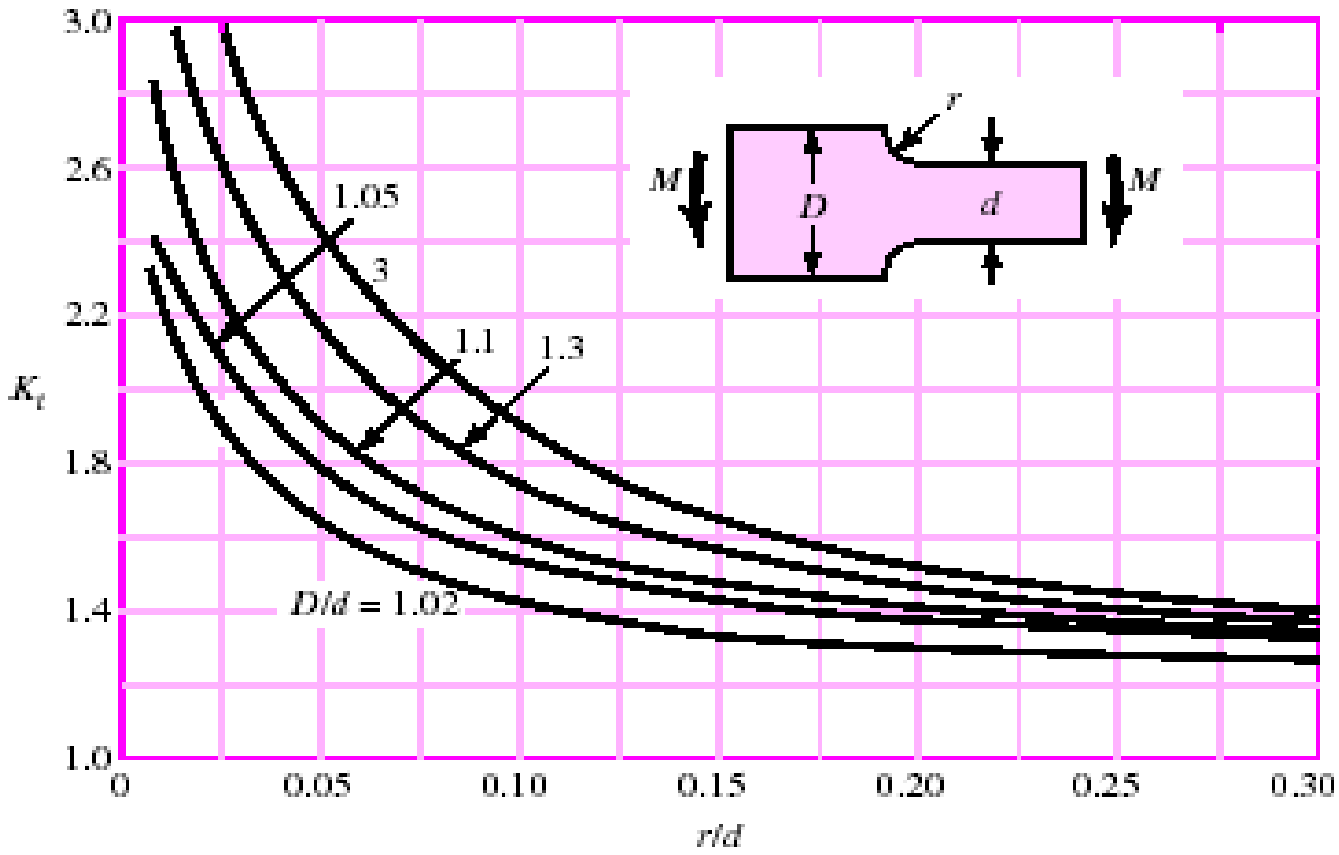
Stress Concentration and Notch Sensitivity

- Theoretical Stress Concentration Factor, K_t for Axial Loading



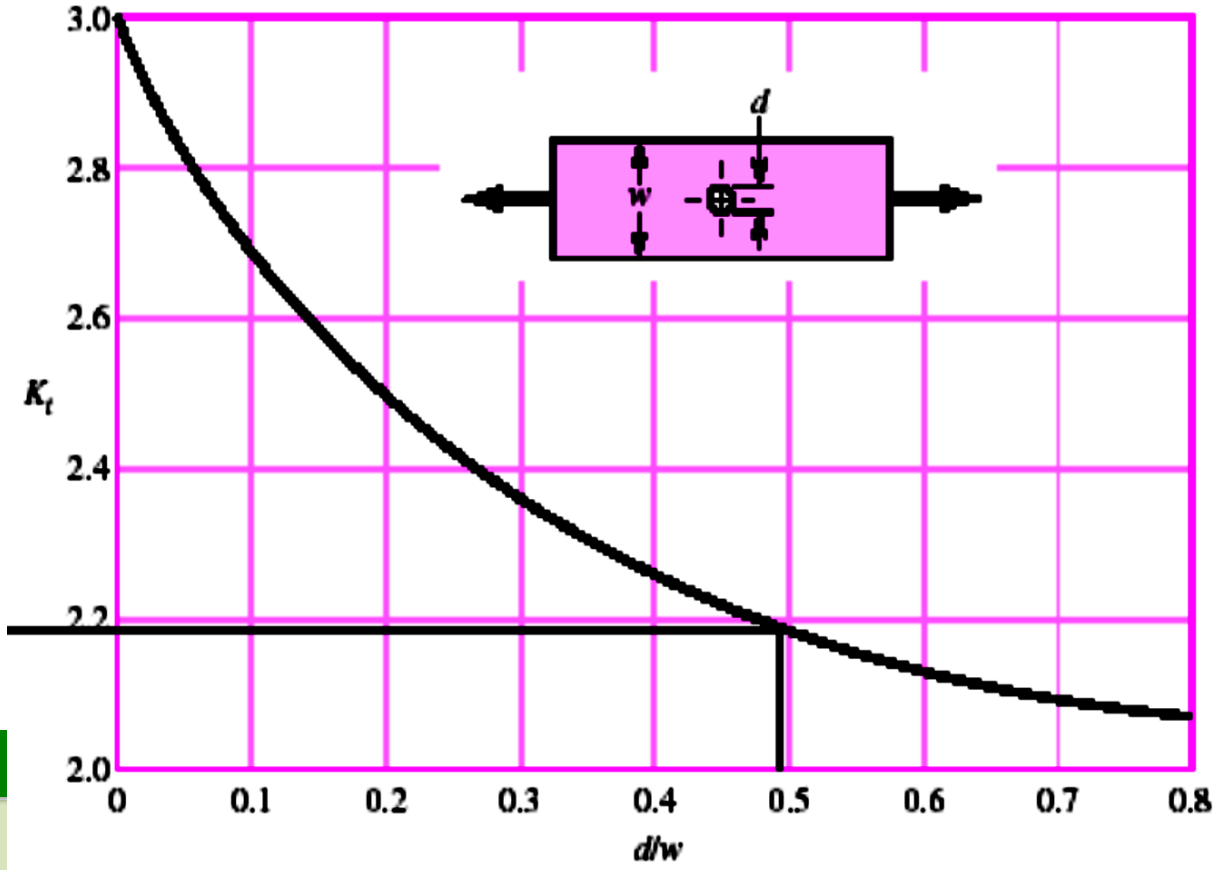
Stress Concentration and Notch Sensitivity

- Theoretical Stress Concentration Factor, K_t for Bending Loading



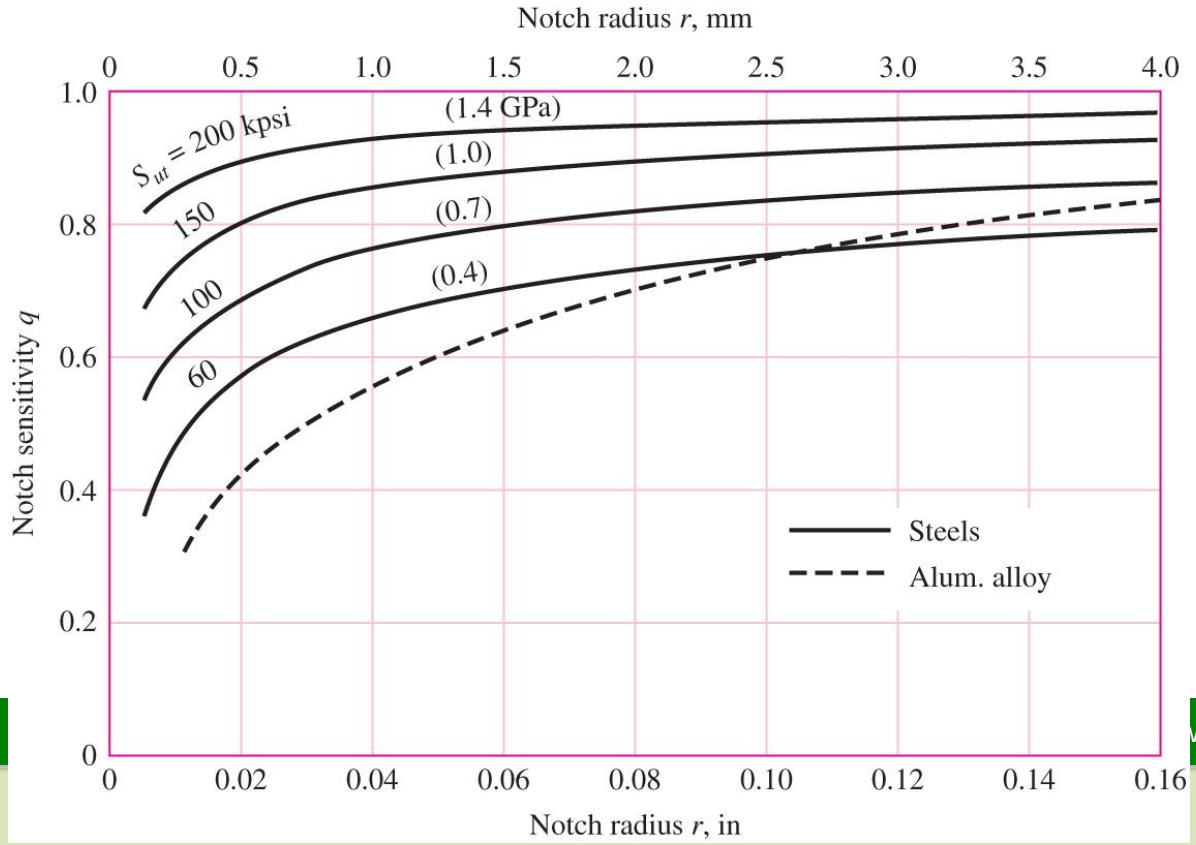
Stress Concentration and Notch Sensitivity

- Theoretical Stress Concentration Factor, K_t for Axial Loading



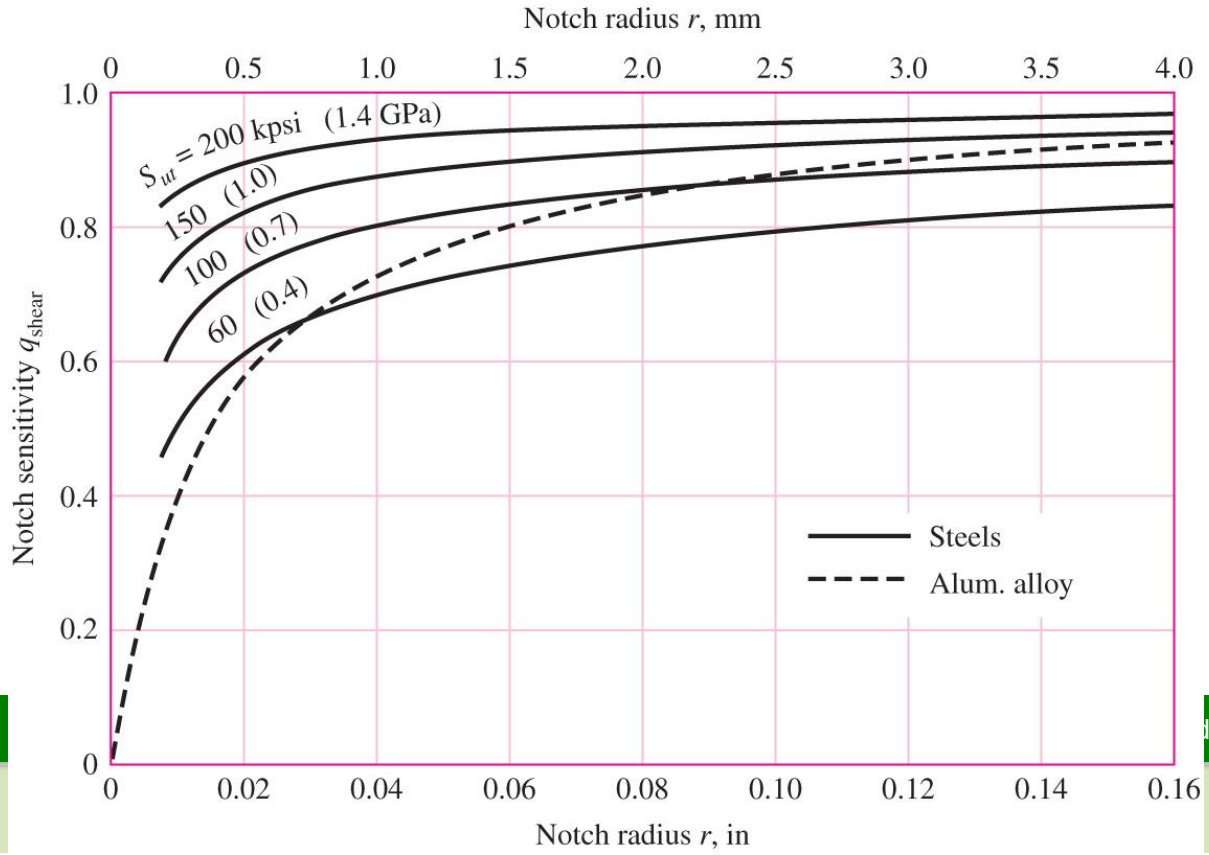
Stress Concentration and Notch Sensitivity

- Obtain q for bending or axial loading from Fig. Then get K_f from Eq. (6–32): $K_f = 1 + q(K_t - 1)$



Stress Concentration and Notch Sensitivity

- Obtain q_s for torsional loading from Fig. 6–21.
- Then get K_{fs} from Eq. (6–32): $K_{fs} = 1 + q_s(K_{ts} - 1)$



Stress Concentration and Notch Sensitivity

- Alternatively, can use curve fit equations to get notch sensitivity, or go directly to K_f .

$$q = \frac{1}{1 + \frac{\sqrt{a}}{\sqrt{r}}} \qquad K_f = 1 + \frac{K_t - 1}{1 + \sqrt{a/r}}$$

- For steels, with S_{ut} in kpsi

- Bending or axial:

$$\sqrt{a} = 0.246 - 3.08(10^{-3})S_{ut} + 1.51(10^{-5})S_{ut}^2 - 2.67(10^{-8})S_{ut}^3$$

- Torsion:

$$\sqrt{a} = 0.190 - 2.51(10^{-3})S_{ut} + 1.35(10^{-5})S_{ut}^2 - 2.67(10^{-8})S_{ut}^3$$

Stress Concentration and Notch Sensitivity

- Cast irons are already full of discontinuities, which are included in the strengths.
- Additional notches do not add much additional harm.
- Recommended to use $q = 0.2$ for cast irons.

Application of Fatigue Stress Concentration Factor

- Use K_f as a multiplier to increase the nominal stress.
- Some designers (and previous editions of textbook) sometimes applied $1/K_f$ as a Marin factor to reduce S_e.
- For infinite life, either method is equivalent, since

$$n_f = \frac{S_e}{K_f \sigma} = \frac{\left(1/K_f\right) S_e}{\sigma}$$

- For finite life, increasing stress is more conservative. Decreasing S_e applies more to high cycle than low cycle.

Example 4–6

A steel shaft in bending has an ultimate strength of 690 MPa and a shoulder with a fillet radius of 3 mm connecting a 32-mm diameter with a 38-mm diameter. Estimate K_f using:
(*a*) Figure 6–20.
(*b*) Equations (6–33) and (6–35).

Solution

From Fig. A–15–9, using $D/d = 38/32 = 1.1875$, $r/d = 3/32 = 0.093\,75$, we read the graph to find $K_t = 1.65$.
(*a*) From Fig. 6–20, for $S_{ut} = 690$ MPa and $r = 3$ mm, $q = 0.84$. Thus, from Eq. (6–32)

$$K_f = 1 + q(K_t - 1) = 1 + 0.84(1.65 - 1) = 1.55 \qquad \text{Answer}$$

(*b*) From Eq. (6–35*a*) with $S_{ut} = 690\,\text{MPa} = 100\,\text{kpsi}$, $\sqrt{a} = 0.0622\sqrt{\text{in}} = 0.313\sqrt{\text{mm}}$. Substituting this into Eq. (6–33) with $r = 3$ mm gives

$$K_f = 1 + \frac{K_t - 1}{1 + \sqrt{a/r}} = 1 + \frac{1.65 - 1}{1 + \dfrac{0.313}{\sqrt{3}}} = 1.55 \qquad \text{Answer}$$

Example 4–7

For the step-shaft of Ex. 6–6, it is determined that the fully corrected endurance limit is $S_e = 280$ MPa. Consider the shaft undergoes a fully reversing nominal stress in the fillet of $(\sigma_{rev})_{nom} = 260$ MPa. Estimate the number of cycles to failure.

Solution

From Ex. 6–6, $K_f = 1.55$, and the ultimate strength is $S_{ut} = 690$ MPa $= 100$ kpsi. The maximum reversing stress is

$$(\sigma_{rev})_{max} = K_f(\sigma_{rev})_{nom} = 1.55(260) = 403 \text{ MPa}$$

From Fig. 6–18, $f = 0.845$. From Eqs. (6–14), (6–15), and (6–16)

$$a = \frac{(f S_{ut})^2}{S_e} = \frac{[0.845(690)]^2}{280} = 1214 \text{ MPa}$$

$$b = -\frac{1}{3}\log\frac{f S_{ut}}{S_e} = -\frac{1}{3}\log\left[\frac{0.845(690)}{280}\right] = -0.1062$$

$$N = \left(\frac{\sigma_{rev}}{a}\right)^{1/b} = \left(\frac{403}{1214}\right)^{1/-0.1062} = 32.3(10^3) \text{ cycles}$$ Answer

Example 4–8

A 1015 hot-rolled steel bar has been machined to a diameter of 1 in. It is to be placed in reversed axial loading for 70 000 cycles to failure in an operating environment of 550°F. Using ASTM minimum properties, and a reliability of 99 percent, estimate the endurance limit and fatigue strength at 70 000 cycles.

Solution

From Table A–20, $S_{ut} = 50$ kpsi at 70°F. Since the rotating-beam specimen endurance limit is not known at room temperature, we determine the ultimate strength at the elevated temperature first, using Table 6–4. From Table 6–4,

$$\left(\frac{S_T}{S_{RT}}\right)_{550^\circ} = \frac{0.995 + 0.963}{2} = 0.979$$

The ultimate strength at 550°F is then

$$(S_{ut})_{550^\circ} = (S_T/S_{RT})_{550^\circ}\,(S_{ut})_{70^\circ} = 0.979(50) = 49.0 \text{ kpsi}$$

The rotating-beam specimen endurance limit at 550°F is then estimated from Eq. (6–8) as

$$S'_e = 0.5(49) = 24.5 \text{ kpsi}$$

Example 4–8 (continued)

Next, we determine the Marin factors. For the machined surface, Eq. (6–19) with Table 6–2 gives

$$k_a = aS_{ut}^b = 2.70(49^{-0.265}) = 0.963$$

For axial loading, from Eq. (6–21), the size factor $k_b = 1$, and from Eq. (6–26) the loading factor is $k_c = 0.85$. The temperature factor $k_d = 1$, since we accounted for the temperature in modifying the ultimate strength and consequently the endurance limit. For 99 percent reliability, from Table 6–5, $k_e = 0.814$. Finally, since no other conditions were given, the miscellaneous factor is $k_f = 1$. The endurance limit for the part is estimated by Eq. (6–18) as

$$\begin{aligned} S_e &= k_a k_b k_c k_d k_e k_f S'_e \\ &= 0.963(1)(0.85)(1)(0.814)(1)24.5 = 16.3 \text{ kpsi} \end{aligned}$$

Answer

Example 4–8 (continued)

For the fatigue strength at 70 000 cycles we need to construct the S-N equation. From p. 293, since $S_{ut} = 49 < 70$ kpsi, then $f = 0.9$. From Eq. (6–14)

$$a = \frac{(f S_{ut})^2}{S_e} = \frac{[0.9(49)]^2}{16.3} = 119.3 \text{ kpsi}$$

and Eq. (6–15)

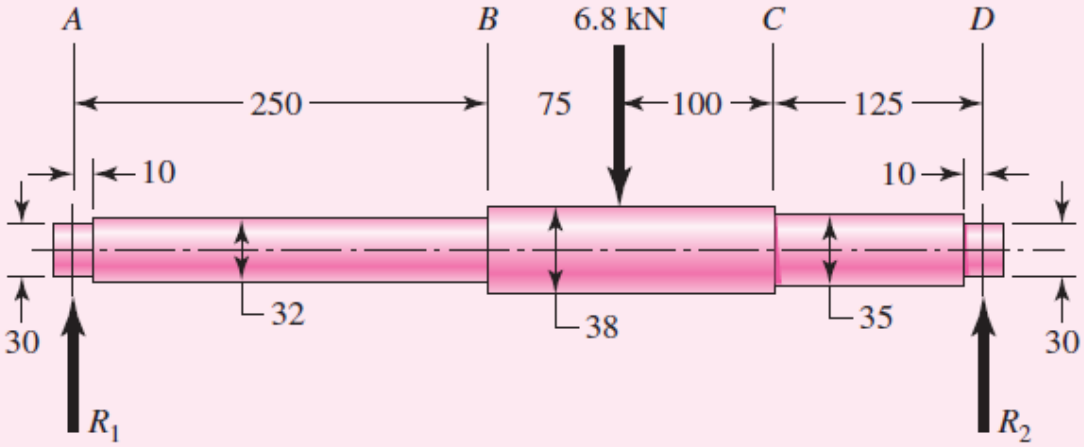
$$b = -\frac{1}{3}\log\left(\frac{f S_{ut}}{S_e}\right) = -\frac{1}{3}\log\left[\frac{0.9(49)}{16.3}\right] = -0.1441$$

Finally, for the fatigue strength at 70 000 cycles, Eq. (6–13) gives

$$S_f = a N^b = 119.3(70\,000)^{-0.1441} = 23.9 \text{ kpsi} \quad \text{Answer}$$

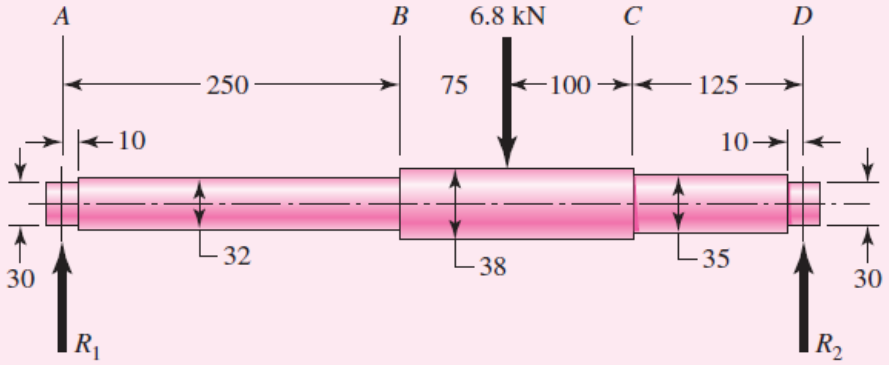
Example 4–9

Figure 6–22*a* shows a rotating shaft simply supported in ball bearings at *A* and *D* and loaded by a nonrotating force *F* of 6.8 kN. Using ASTM "minimum" strengths, estimate the life of the part.

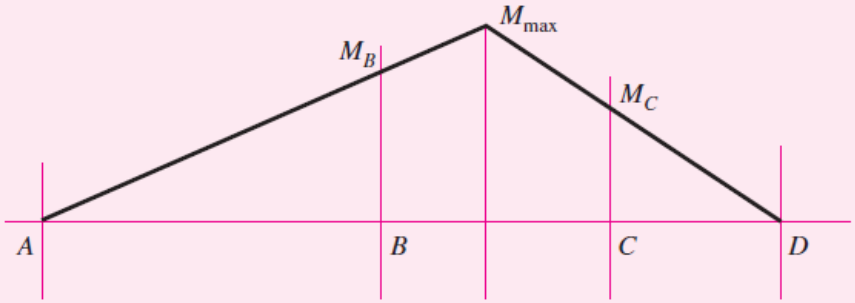


Example 4–9 (continued)

From Fig. 6–22*b* we learn that failure will probably occur at *B* rather than at *C* or at the point of maximum moment. Point *B* has a smaller cross section, a higher bending moment, and a higher stress-concentration factor than *C*, and the location of maximum moment has a larger size and no stress-concentration factor.



(*a*)



(*b*)

Example 4–9 (continued)

We shall solve the problem by first estimating the strength at point B, since the strength will be different elsewhere, and comparing this strength with the stress at the same point.

From Table A–20 we find $S_{ut} = 690$ MPa and $S_y = 580$ MPa. The endurance limit S'_e is estimated as

$$S'_e = 0.5(690) = 345 \text{ MPa}$$

From Eq. (6–19) and Table 6–2,

$$k_a = 4.51(690)^{-0.265} = 0.798$$

From Eq. (6–20),

$$k_b = (32/7.62)^{-0.107} = 0.858$$

Since $k_c = k_d = k_e = k_f = 1$,

$$S_e = 0.798(0.858)345 = 236 \text{ MPa}$$

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Example 4–9 (continued)

To find the geometric stress-concentration factor K_t we enter Fig. A–15–9 with $D/d = 38/32 = 1.1875$ and $r/d = 3/32 = 0.093\,75$ and read $K_t \doteq 1.65$. Substituting $S_{ut} = 690/6.89 = 100$ kpsi into Eq. (6–35a) yields $\sqrt{a} = 0.0622\sqrt{\text{in}} = 0.313\sqrt{\text{mm}}$. Substituting this into Eq. (6–33) gives

$$K_f = 1 + \frac{K_t - 1}{1 + \sqrt{a/r}} = 1 + \frac{1.65 - 1}{1 + 0.313/\sqrt{3}} = 1.55$$

The next step is to estimate the bending stress at point B. The bending moment is

$$M_B = R_1 x = \frac{225F}{550}250 = \frac{225(6.8)}{550}250 = 695.5\ \text{N} \cdot \text{m}$$

Just to the left of B the section modulus is $I/c = \pi d^3/32 = \pi 32^3/32 = 3.217\ (10^3)\ \text{mm}^3$. The reversing bending stress is, assuming infinite life,

$$\sigma_{\text{rev}} = K_f \frac{M_B}{I/c} = 1.55\frac{695.5}{3.217}(10)^{-6} = 335.1(10^6)\ \text{Pa} = 335.1\ \text{MPa}$$

This stress is greater than S_e and less than S_y. This means we have both finite life and no yielding on the first cycle.

Example 4–9 (continued)

For finite life, we will need to use Eq. (6–16). The ultimate strength, $S_{ut} = 690$ MPa = 100 kpsi. From Fig. 6–18, $f = 0.844$. From Eq. (6–14)

$$a = \frac{(f\ S_{ut})^2}{S_e} = \frac{[0.844(690)]^2}{236} = 1437\ \text{MPa}$$

and from Eq. (6–15)

$$b = -\frac{1}{3}\log\left(\frac{f\ S_{ut}}{S_e}\right) = -\frac{1}{3}\log\left[\frac{0.844(690)}{236}\right] = -0.1308$$

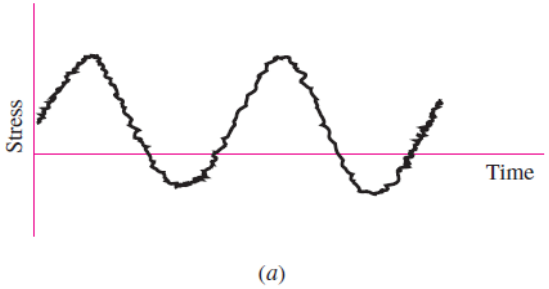
From Eq. (6–16),

$$N = \left(\frac{\sigma_{\text{rev}}}{a}\right)^{1/b} = \left(\frac{335.1}{1437}\right)^{-1/0.1308} = 68(10^3)\ \text{cycles}$$ Answer

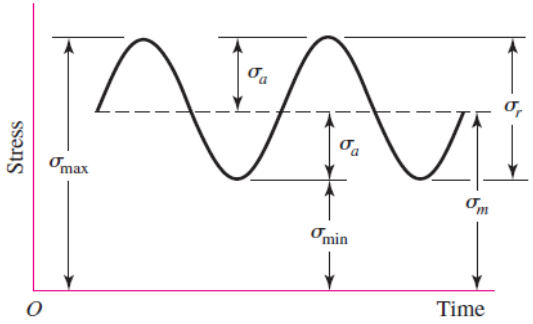
Characterizing Fluctuating Stresses

- The *S*-*N* diagram is applicable for *completely reversed* stresses
- Other fluctuating stresses exist
- Sinusoidal loading patterns are common, but not necessary
- Vary the s_m and s_a to learn about the fatigue resistance under fluctuating loading
- Three common methods of plotting results follow.

Characterizing Fluctuating Stresses

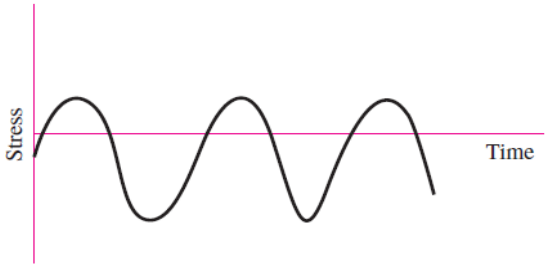


(a)

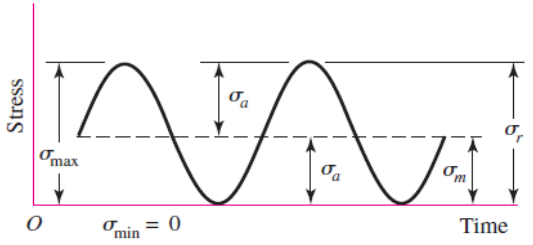


(d)

General Fluctuating

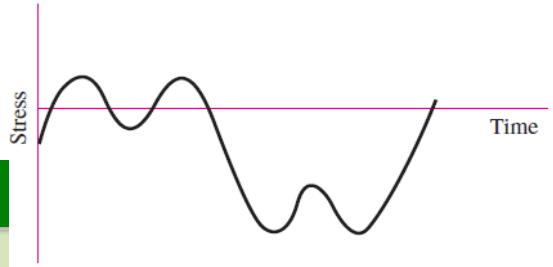


(b)

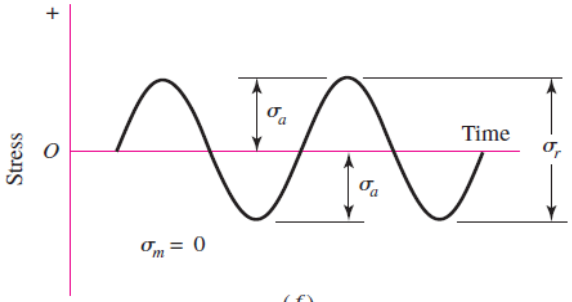


(e)

Repeated



(c)

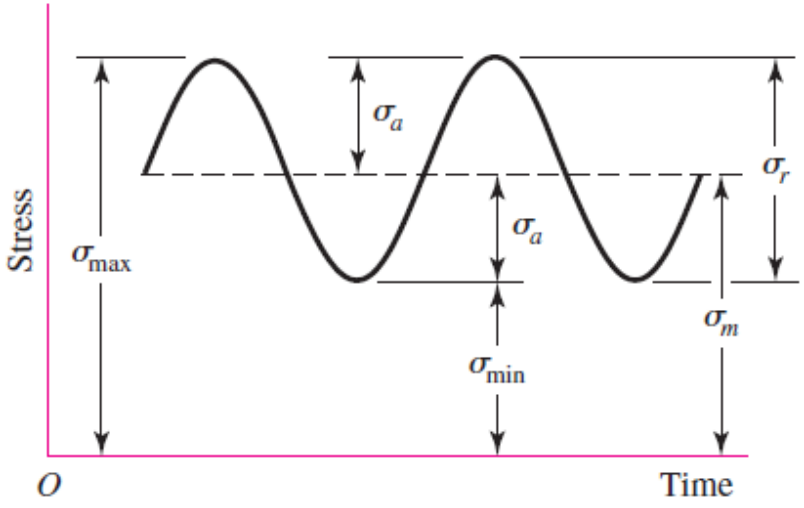


(f)

Completely Reversed

Characterizing Fluctuating Stresses

- Fluctuating stresses can often be characterized simply by the minimum and maximum stresses, $\sigma_{\min}$ and $\sigma_{\max}$
- Define σ_m as *midrange* steady component of stress (sometimes called *mean* stress) and σ_a as amplitude of *alternating* component of stress



$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

$$\sigma_a = \left| \frac{\sigma_{\max} - \sigma_{\min}}{2} \right|$$

- Other useful definitions
  - ✓ *Stress ratio*

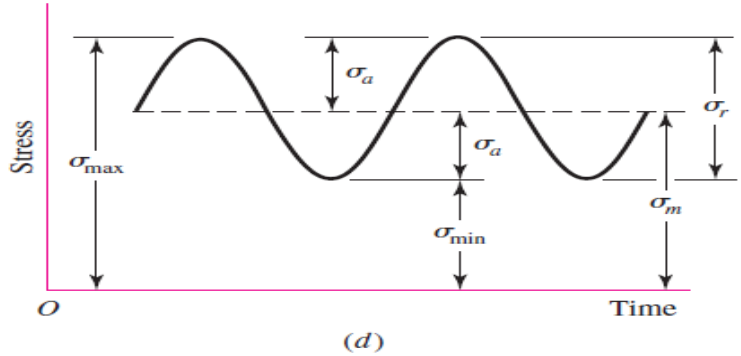
$$R = \frac{\sigma_{\min}}{\sigma_{\max}}$$

  - ✓ *Amplitude ratio*

$$A = \frac{\sigma_a}{\sigma_m}$$

Characterizing Fluctuating Stresses

General Fluctuating



(d)

$$P_m = \tfrac{1}{2}(P_{max} + P_{min})$$

$$P_a = \tfrac{1}{2}(P_{max} - P_{min})$$

$$M_m = P_m L \qquad \sigma_m = M_m/S$$

$$M_a = P_a L \qquad \sigma_a = M_a/S$$

$$\mathbf{F \sim (F_{max}, F_{min})}$$

$$F_m = \tfrac{1}{2}(F_{max} + F_{min}) \qquad T_m = \tfrac{1}{2}(T_{max} + T_{min})$$

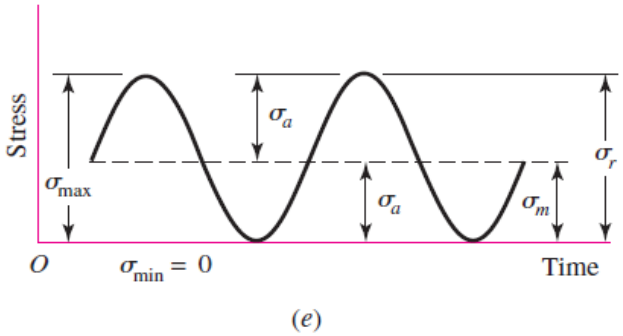
$$F_a = \tfrac{1}{2}(F_{max} - F_{min}) \qquad T_a = \tfrac{1}{2}(T_{max} - T_{min})$$

$$\sigma_m = F_m/A \qquad \tau_m = T_m/Z_p$$

$$\sigma_a = F_a/A \qquad \tau_a = T_a/Z_p$$

Characterizing Fluctuating Stresses

Repeated



(e)

$P_m = \frac{1}{2}(P_{max})$

$P_a = \frac{1}{2}(P_{max})$

$M_m = P_m L \quad \sigma_m = M_m/Z$

$M_a = P_a L \qquad \sigma_a = M_a/Z$

F ~ (F_{max}, 0)

$F_m = \frac{1}{2}(F_{max} + F_{min}) = \frac{1}{2} F_{max}$

$T_m = \frac{1}{2}(T_{max} + T_{min}) = \frac{1}{2}(T_{max})$

$F_a = \frac{1}{2}(F_{max} - F_{min}) = \frac{1}{2} F_{max}$

$T_a = \frac{1}{2}(T_{max} - T_{min}) = \frac{1}{2}(T_{max})$

$\sigma_m = F_m/A$

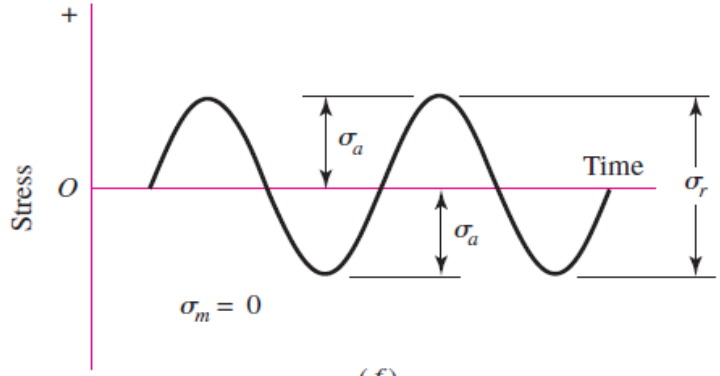
$\tau_m = T_m/Z_p$

$\sigma_a = F_a/A$

$\tau_a = T_a/Z_p$

Characterizing Fluctuating Stresses

Completely Reversed



$$F \sim (F_{max}, -F_{max})^{(f)}$$

$$P_m = \tfrac{1}{2}(P_{max} + P_{min}) = 0$$

$$P_a = \tfrac{1}{2}(P_{max} - P_{min}) = P_{max}$$

$$M_m = 0 \qquad \sigma_m = 0$$

$$M_a = P_a L \qquad \sigma_a = M_a / Z$$

$$F_m = \tfrac{1}{2}(F_{max} + F_{min}) = 0 \qquad T_m = \tfrac{1}{2}(T_{max} + T_{min}) = 0$$

$$F_a = \tfrac{1}{2}(F_{max} - F_{min}) = F_{max} \qquad T_a = \tfrac{1}{2}(T_{max} - T_{min}) = T_{max}$$

$$\sigma_m = F_m / A = 0 \qquad \tau_m = 0$$

$$\sigma_a = F_a / A = F_m / A \qquad \tau_a = T_a / Z_p$$

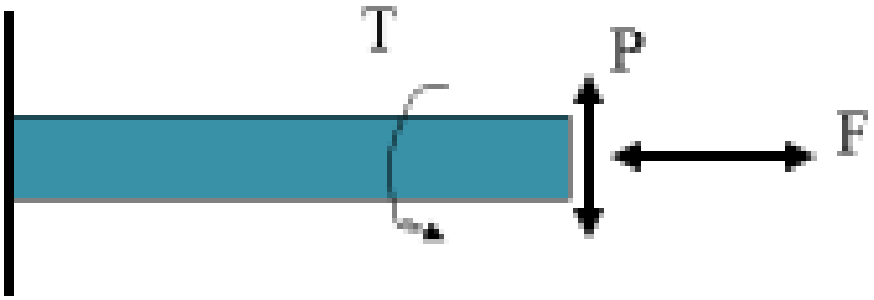
Combined Biaxial Fluctuating Stresses

1. Find: (P_m, P_a) and (F_m, F_a)

   Calculate: (M_m, M_a)

   $\sigma_m = F_m/A + M_m/Z$

   $\sigma_a = F_a/A + M_a/Z$



2. Find: (T_m, T_a)

   Calculate: (τ_m, τ_a)

3. Calculate equivalent von-Mises Normal Stresses:

$\sigma_m' = (\sigma_m^2 + 3\tau_m^2)^{½}$ $\sigma_a' = (\sigma_a^2 + 3\tau_a^2)^{½}$

Application of K_f for Fluctuating Stresses

- For fluctuating loads at points with stress concentration, the best approach is to design to avoid all localized plastic strain.
- In this case, K_f should be applied to both alternating and midrange stress components.
- When localized strain does occur, some methods (e.g. *nominal mean stress* method and *residual stress* method) recommend only applying K_f to the alternating stress.
- The *Dowling method* recommends applying K_f to the alternating stress and K_{fm} to the mid-range stress, where K_{fm} is

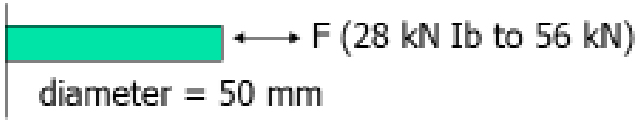
$$K_{fm} = K_f \qquad K_f|\sigma_{\max,o}| < S_y$$

$$K_{fm} = \frac{S_y - K_f\sigma_{ao}}{|\sigma_{mo}|} \qquad K_f|\sigma_{\max,o}| > S_y$$

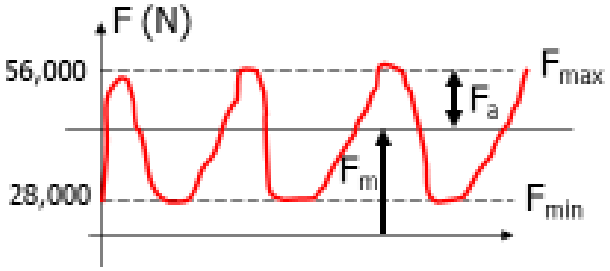
$$K_{fm} = 0 \qquad K_f|\sigma_{\max,o} - \sigma_{\min,o}| > 2S_y$$

Example 4-10

For the situation shown, find the mean and the alternating stresses in the member:



Solution



$F_m = \frac{1}{2}(F_{max} + F_{min})$ $\quad F_a = \frac{1}{2}(F_{max} - F_{min})$

$F_m = \frac{1}{2}(56{,}000 + 28{,}000) = 42{,}000$ N

$F_a = \frac{1}{2}(56{,}000 - 28{,}000) = 14{,}000$ N

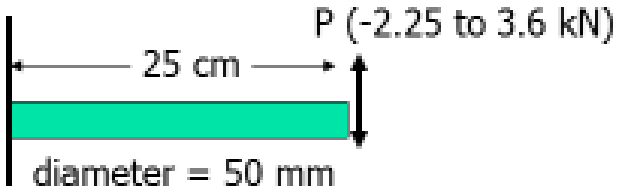
Direct Axial Load

$\sigma_m = 4F_m/\pi d^2 = 4\,(42{,}000)/\pi(50)^2 = 21$ N/mm^2 (MPa)

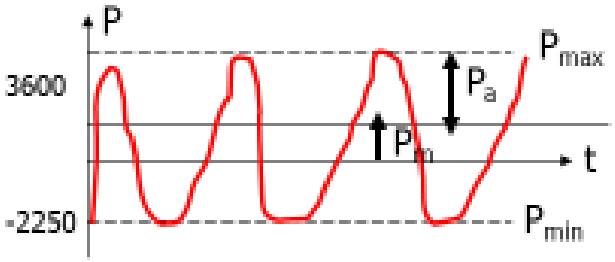
$\sigma_a = 4F_a/\pi d^2 = 4\,(14{,}000)/\pi(50)^2 = 7$ N/mm^2 (MPa)

Example 4-11

For the situation shown, find the mean and the alternating stresses in the member:



Solution



$P_m = ½(3600 + (-2250)) = 675$ N

$P_a = ½(3600 – (-2250)) = 2925$ N

$M_m = 250P_m$ $\qquad$ $M_a = 10P_a$

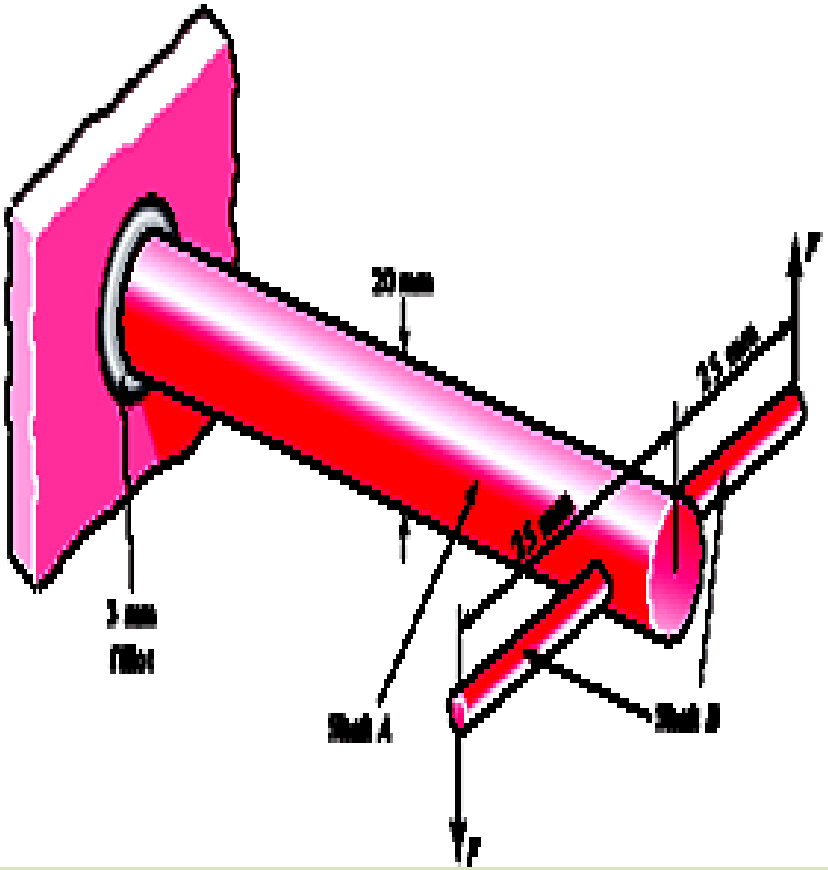
Bending Stress

$\sigma_m = 32M_m/\pi d^3 = 32\,(250P_m)/\pi(50)^3 = 14$ N/mm^2 (MPa)

$\sigma_a = 32M_a/\pi d^3 = 32\,(250P_a)/\pi(50)^3 = 60$ N/mm^2 (MPa)

Example 4-12

The force F is 2 kN (repeated). Find the mean and alternating stresses in the Member A.



$F_m = F_a = \frac{1}{2}(F_{max} - F_{min})$

$F_m = \frac{1}{2}(2 + 0) = 1$ kN

$F_a = \frac{1}{2}(2 - 0) = 1$ kN

$T_m = 50F_m = 50 \times 10^3$ N-mm

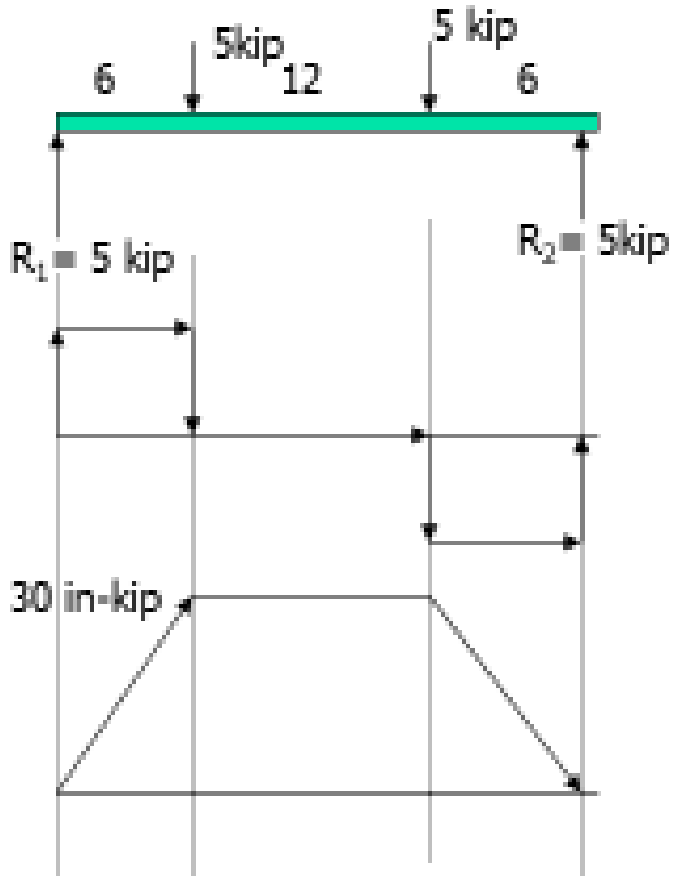
$T_a = 50F_a = 50 \times 10^3$ N-mm

$\tau_m = 16\, T_m / \pi d^3 = 32$ MPa

$\tau_a = 16\, T_a / \pi d^3 = 32$ MPa

Example 4-13

The rotating shaft is under bending loads as shown. Find the mean and alternating stresses in the shaft.



From the BMD, M = 30 x 10^3 in-Ib

Rotating shaft → Reversed Bending

$\sigma_m = 0$

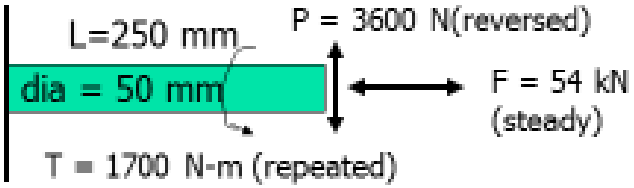
$\sigma_a = 32M_a/\pi d^3$

$= 32 \times 30 \times 10^3/\pi d^3$ psi

$= 306/d^3$ ksi

Example 4-14

Find the equivalent von-Mises mean and alternate stresses.



1. Find: $P_m = 0$ $P_a = 3600$ N
   $F_m = 54{,}000$ N $F_a = 0$

Calculate: $M_m = P_m L = 0$
$M_a = P_a L = 900 \times 10^3$ N-mm

Calculate: $\sigma_m = F_m/A + M_m/Z$
$= 4F_m/\pi d^2 + 0 = 28$ N/mm^2

Calculate $\sigma_a = F_a/A + M_a/Z = 0 + 32M_a/\pi d^3 = 73$ N/mm^2

Example 4-14 (continued)

2. Find: $T_m = 850 \times 10^3$ N-mm

$T_a = 850 \times 10^3$ N-mm

Calculate: $\tau_m = T_m/Z = 16T_m/\pi d^3 = 35$ MPa

$\tau_a = T_a/Z = 16T_a/\pi d^3 = 35$ Mpa

3. Calculate equivalent von-Mises

Normal Stresses:

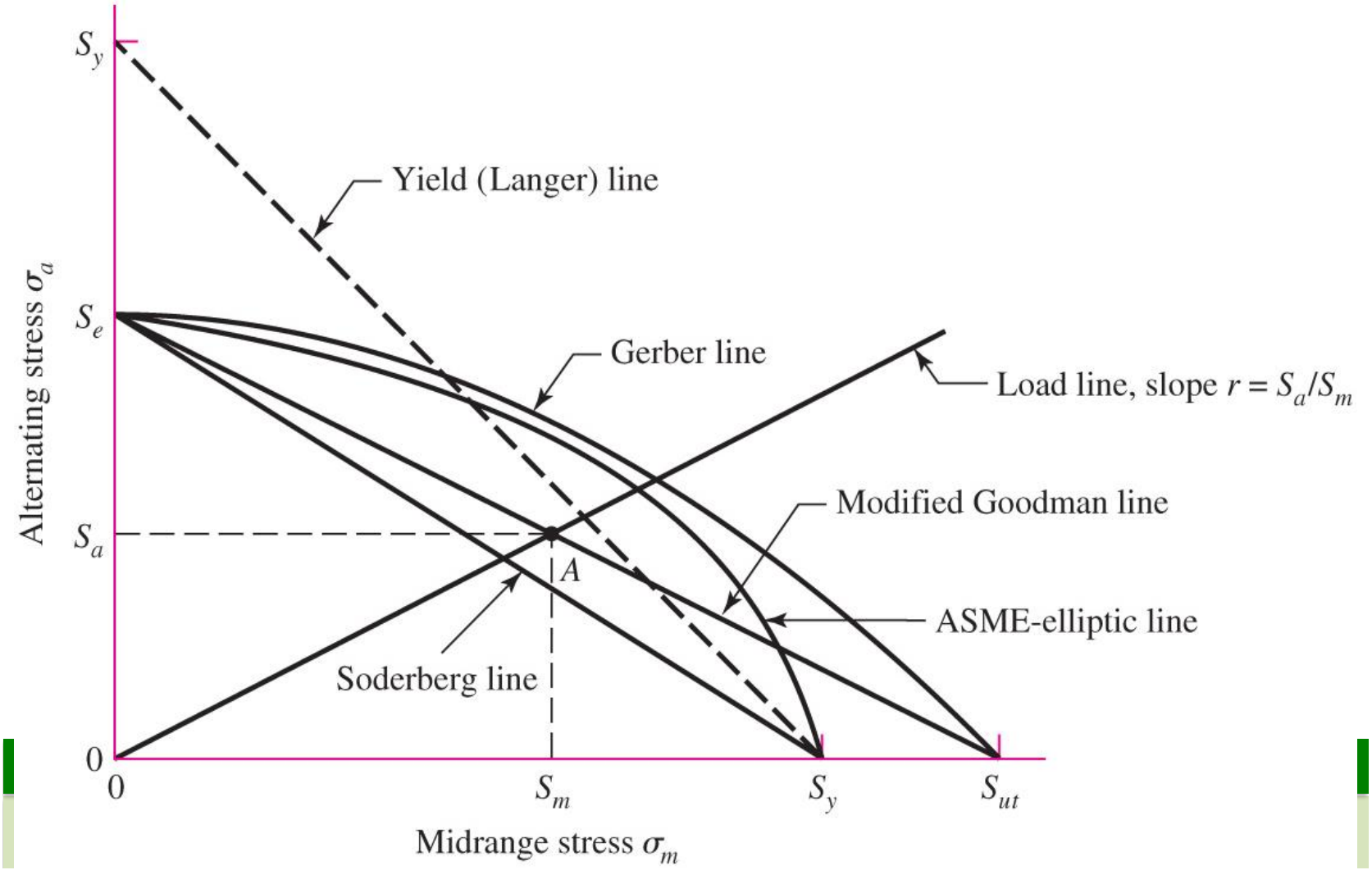
$\sigma_m' = (\sigma_m^2 + 3\tau_m^2)^{1/2} = 67$ MPa

$\sigma_a' = (\sigma_a^2 + 3\tau_a^2)^{1/2} = 95$ MPa

Failure Criteria

- Five commonly used failure criteria are
  1. Gerber passes through the data
  2. ASME-elliptic passes through data and incorporates rough yielding check
  3. Modified Goodman is linear, so simple to use for design. It is more conservative than Gerber.
  4. Soderberg provides a very conservative single check of both fatigue and yielding
  5. Langer line represents standard yield check.
- It is equivalent to comparing maximum stress to yield strength.

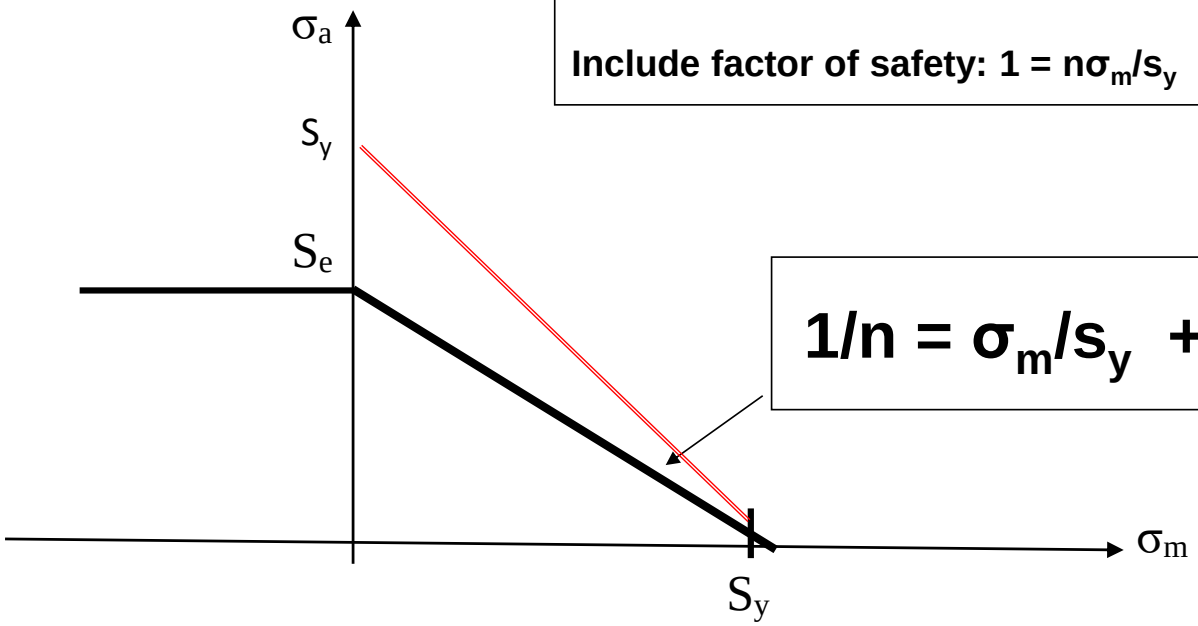
Failure Criteria



Soderberg Criterion

Equation of the Line: $1 = \sigma_m/s_y + \sigma_a/s_e$

Include factor of safety: $1 = n\sigma_m/s_y + n\sigma_a/s_e$

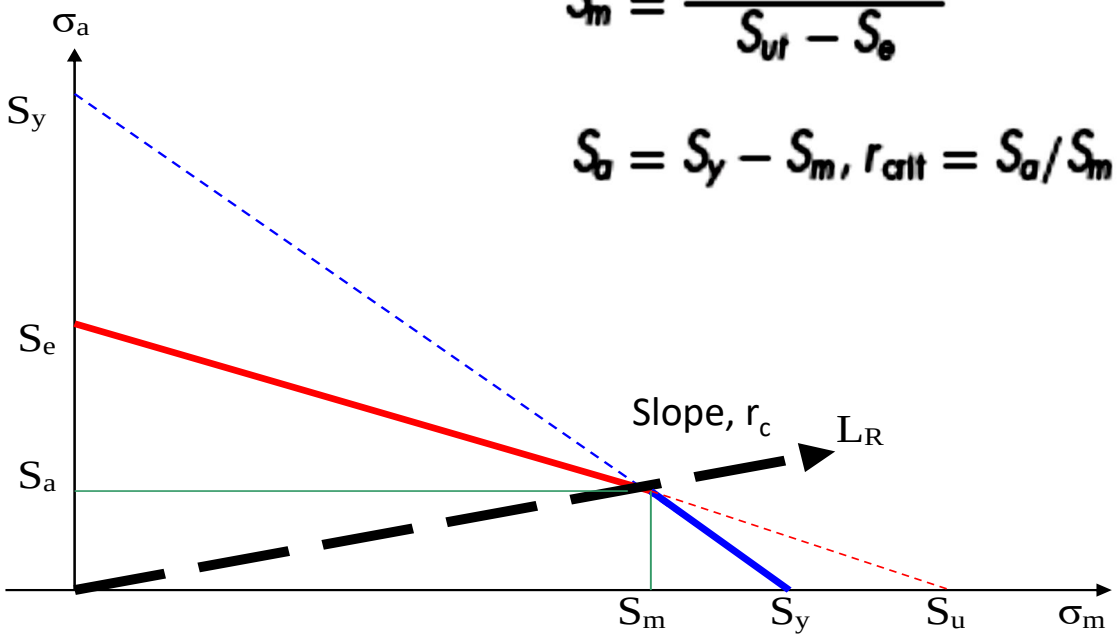


$$1/n = \sigma_m/s_y + \sigma_a/s_e$$

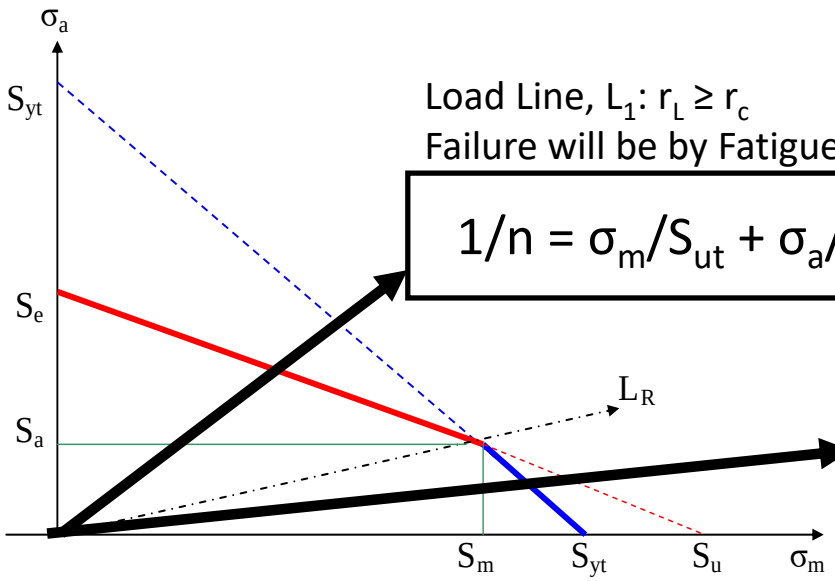
Modified Goodman Criterion

$$S_m = \frac{(S_y - S_e)\, S_{ut}}{S_{ut} - S_e}$$

$$S_a = S_y - S_m,\ r_{calt} = S_a / S_m$$



Modified Goodman Criterion



Load Line, L_1: $r_L \geq r_c$
Failure will be by Fatigue

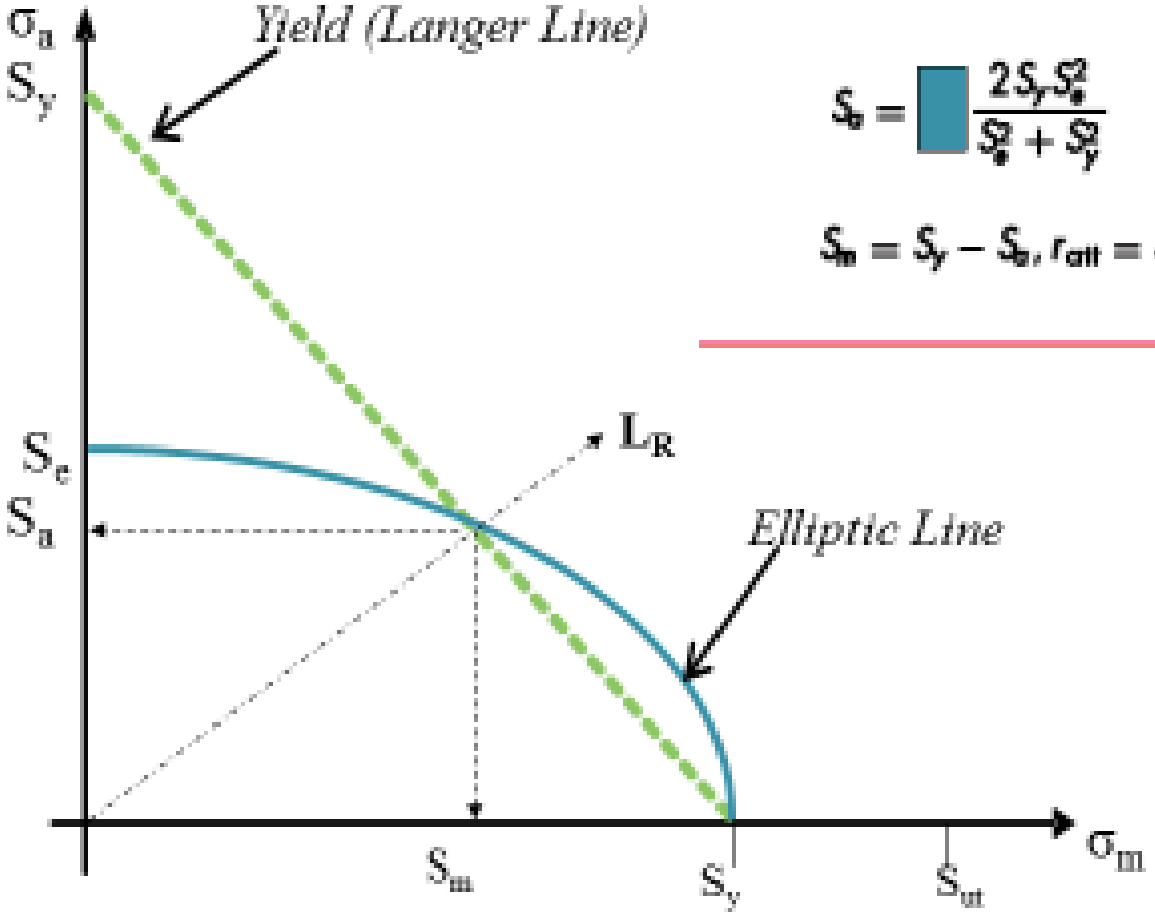
$$1/n = \sigma_m/S_{ut} + \sigma_a/S_e$$

Load Line, L_2: $r_L < r_c$
Failure will be by Yield

$$1/n = \sigma_m/S_{yt} + \sigma_a/S_{yt}$$

NOTE: Slope of load Line, $r_L = \sigma_a / \sigma_m$

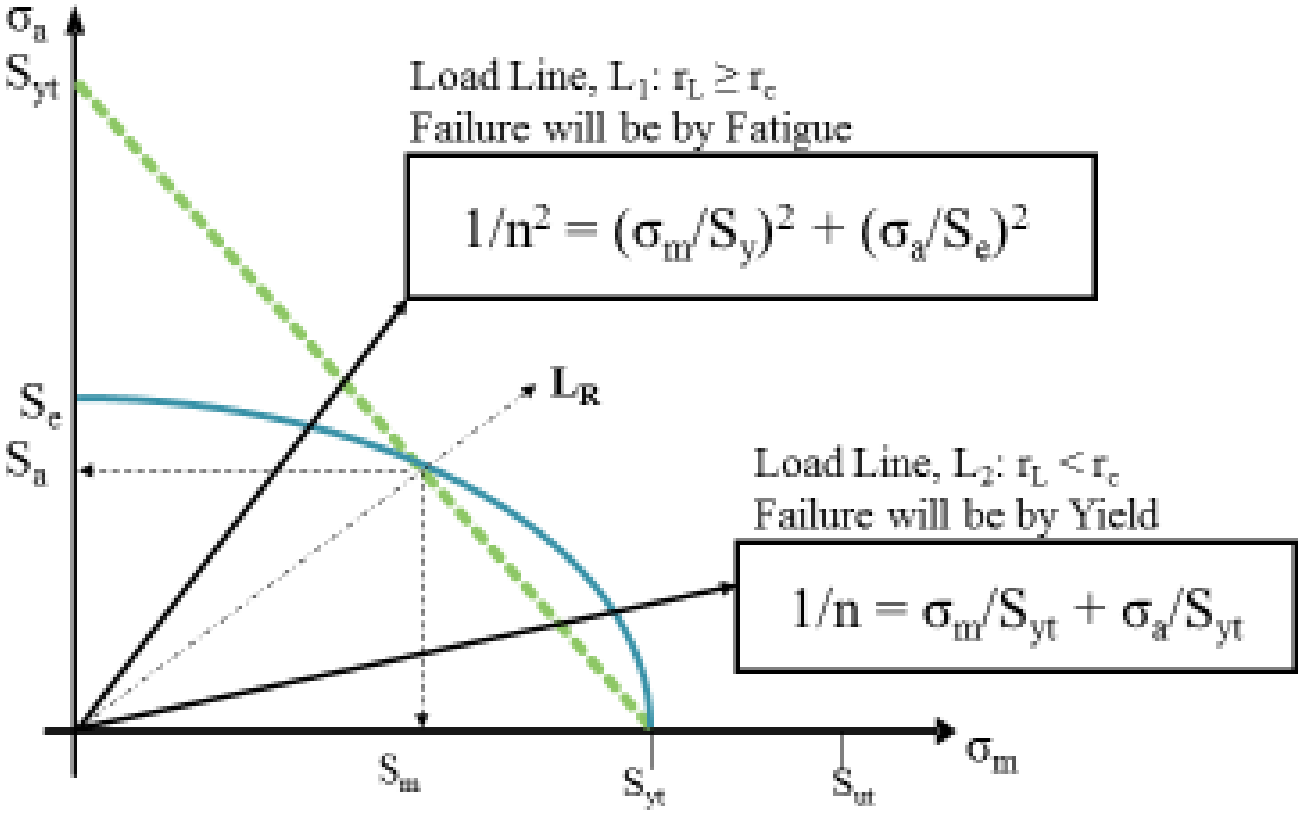
Elliptic Criterion



$$S_e = \sqrt{\frac{2S_y^2 S_e^2}{S_e^2 + S_y^2}}$$

$$S_m = S_y - S_e,\ r_{att} = S_e / S_m$$

Elliptic Criterion



Equations for the Failure Criteria

- Intersecting a constant slope load line with each failure criteria produces design equations

- *n* is the design factor or factor of safety for infinite fatigue life

Soderberg $$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n} \qquad (6\text{–}45)$$

mod-Goodman $$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n} \qquad (6\text{–}46)$$

Gerber $$\frac{n\sigma_a}{S_e} + \left(\frac{n\sigma_m}{S_{ut}}\right)^2 = 1 \qquad (6\text{–}47)$$

ASME-elliptic $$\left(\frac{n\sigma_a}{S_e}\right)^2 + \left(\frac{n\sigma_m}{S_y}\right)^2 = 1 \qquad (6\text{–}48)$$

Modification of Design Equations for Definite Life (N)

Soderberg/Fatigue

$$1/n = \sigma_m/s_y + \sigma_a/\mathbf{s_f}$$

Goodman/Fatigue

$$1/n = \sigma_m/S_{ut} + \sigma_a/\mathbf{S_f}$$

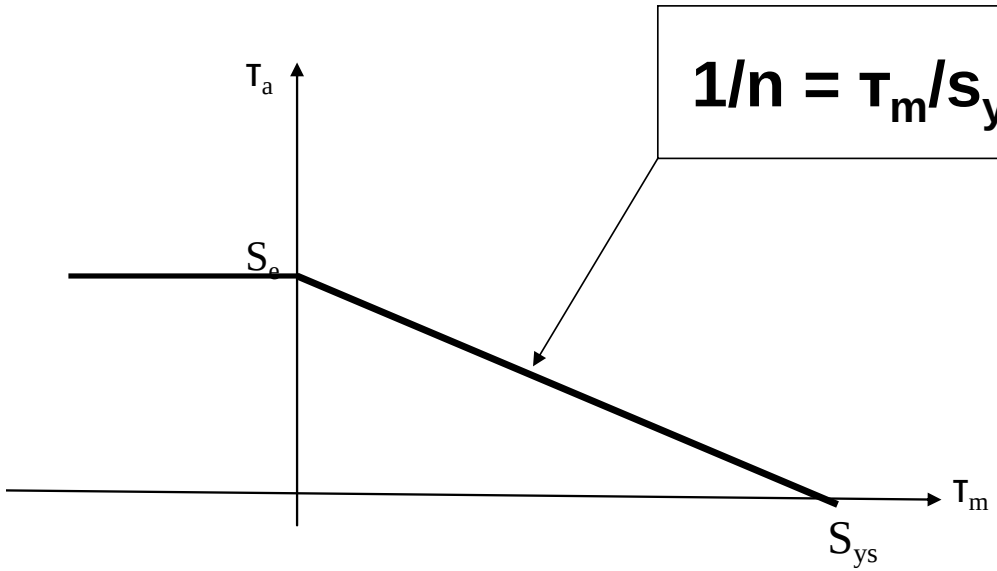
ASME/Fatigue

$$1/n^2 = (\sigma_m/S_y)^2 + (\sigma_a/\mathbf{S_f})^2$$

NOTE: $S_f = aN^b$

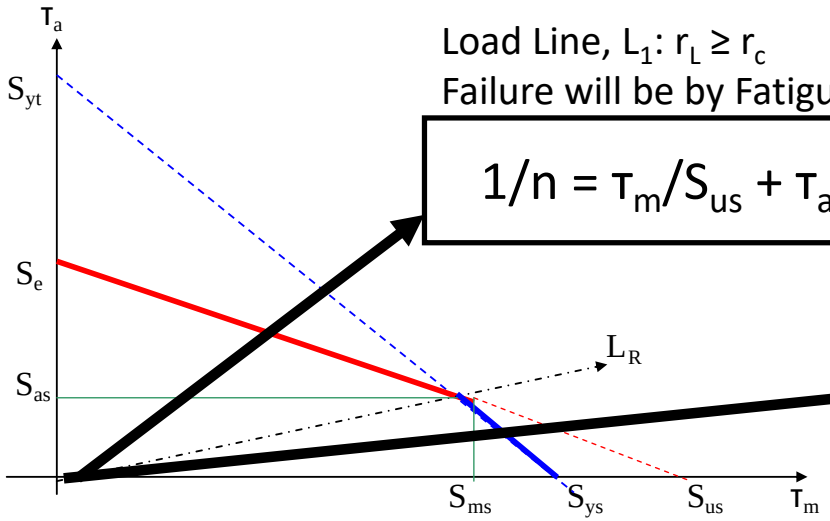
Modification of Soderberg -Shear Stress

Axes → τ_a versus τ_m



$S_{ys} = 0.577 S_y$

Modification of Goodman - Shear Stress



Load Line, L_1: $r_L \geq r_c$
Failure will be by Fatigue

$$1/n = \tau_m/S_{us} + \tau_a/S_e$$

Load Line, L_2: $r_L < r_c$
Failure will be by Yield

$$1/n = \tau_m/S_{ys} + \tau_a/S_{ys}$$

NOTE: Slope of load Line, $r_L = \tau_a / \tau_m$
Use S_{us} and S_{ys} to find S_{as} and S_{ms}
Slope of reference line, $r_c = S_{as} / S_{ms}$

$$S_{ys} = 0.577S_y$$
$$S_{us} = 0.8S_{ut}$$

Notched Parts – Corrected Design Equations

Soderberg/Fatigue →

$$1/n = \sigma_m/s_y + K_f\sigma_a/s_e$$

NOTE:
$S_f = aN^b$
Replaces S_e

Goodman/Fatigue →

$$1/n = \sigma_m/S_{ut} + K_f\,\sigma_a/S_e$$

ASME/Fatigue →

$$1/n^2 = (\sigma_m/S_y)^2 + (K_f\,\sigma_a/S_e)^2$$

Langer/Yield →

$$1/n = \sigma_m/S_y + K_f\,\sigma_a/S_y$$

Summarizing Tables for Failure Criteria

- Tables summarize the pertinent equations for Modified Goodman, Gerber, ASME-elliptic, and Langer failure criteria
- The first row gives fatigue criterion
- The second row gives yield criterion
- The third row gives the intersection of static and fatigue criteria
- The fourth row gives the equation for fatigue factor of safety
- The first column gives the intersecting equations
- The second column gives the coordinates of the intersection

Compiled by NKAY STUDY

Summarizing Tables for Failure Criteria

1 UNS No.	2 SAE and/or AISI No.	3 Process-ing	4 Tensile Strength, MPa (kpsi)	5 Yield Strength, MPa (kpsi)	6 Elongation in 2 in, %	7 Reduction in Area, %	8 Brinell Hardness
G10060	1006	HR	300 (43)	170 (24)	30	55	86
		CD	330 (48)	280 (41)	20	45	95
G10100	1010	HR	320 (47)	180 (26)	28	50	95
		CD	370 (53)	300 (44)	20	40	105
G10150	1015	HR	340 (50)	190 (27.5)	28	50	101
		CD	390 (56)	320 (47)	18	40	111
G10180	1018	HR	400 (58)	220 (32)	25	50	116
		CD	440 (64)	370 (54)	15	40	126
G10200	1020	HR	380 (55)	210 (30)	25	50	111
		CD	470 (68)	390 (57)	15	40	131
G10300	1030	HR	470 (68)	260 (37.5)	20	42	137
		CD	520 (76)	440 (64)	12	35	149
G10350	1035	HR	500 (72)	270 (39.5)	18	40	143
		CD	550 (80)	460 (67)	12	35	163
G10400	1040	HR	520 (76)	290 (42)	18	40	149
		CD	590 (85)	490 (71)	12	35	170
G10450	1045	HR	570 (82)	310 (45)	16	40	163
		CD	630 (91)	530 (77)	12	35	179
G10500	1050	HR	620 (90)	340 (49.5)	15	35	179
		CD	690 (100)	580 (84)	10	30	197
G10600	1060	HR	680 (98)	370 (54)	12	30	201
G10800	1080	HR	770 (112)	420 (61.5)	10	25	229
G10950	1095	HR	830 (120)	460 (66)	10	25	248

Summarizing Table for Modified Goodman

Intersecting Equations	Intersection Coordinates
$\frac{S_a}{S_e} + \frac{S_m}{S_{ut}} = 1$ Load line $r = \frac{S_a}{S_m}$	$S_a = \frac{r S_e S_{ut}}{r S_{ut} + S_e}$ $S_m = \frac{S_a}{r}$
$\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$ Load line $r = \frac{S_a}{S_m}$	$S_a = \frac{r S_y}{1 + r}$ $S_m = \frac{S_y}{1 + r}$
$\frac{S_a}{S_e} + \frac{S_m}{S_{ut}} = 1$ $\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$	$S_m = \frac{(S_y - S_e) S_{ut}}{S_{ut} - S_e}$ $S_a = S_y - S_m, r_{\text{crit}} = S_a / S_m$

Fatigue factor of safety

$$n_f = \frac{1}{\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}}}$$

Summarizing Table for Gerber

Intersecting Equations	Intersection Coordinates
$\frac{S_a}{S_e} + \left(\frac{S_m}{S_{ut}}\right)^2 = 1$	$S_a = \frac{r^2 S_{ut}^2}{2S_e}\left[-1 + \sqrt{1 + \left(\frac{2S_e}{r S_{ut}}\right)^2}\right]$
Load line $r = \frac{S_a}{S_m}$	$S_m = \frac{S_a}{r}$
$\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$	$S_a = \frac{r S_y}{1 + r}$
Load line $r = \frac{S_a}{S_m}$	$S_m = \frac{S_y}{1 + r}$
$\frac{S_a}{S_e} + \left(\frac{S_m}{S_{ut}}\right)^2 = 1$	$S_m = \frac{S_{ut}^2}{2S_e}\left[1 - \sqrt{1 + \left(\frac{2S_e}{S_{ut}}\right)^2\left(1 - \frac{S_y}{S_e}\right)}\right]$
$\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$	$S_a = S_y - S_m, r_{\text{crit}} = S_a / S_m$
Fatigue factor of safety	
$n_f = \frac{1}{2}\left(\frac{S_{ut}}{\sigma_m}\right)^2 \frac{\sigma_a}{S_e}\left[-1 + \sqrt{1 + \left(\frac{2\sigma_m S_e}{S_{ut}\sigma_a}\right)^2}\right]$	$\sigma_m > 0$

Summarizing Table for ASME-Elliptic

Intersecting Equations	Intersection Coordinates
$\left(\frac{S_a}{S_e}\right)^2 + \left(\frac{S_m}{S_y}\right)^2 = 1$	$S_a = \sqrt{\frac{r^2 S_e^2 S_y^2}{S_e^2 + r^2 S_y^2}}$
Load line $r = S_a/S_m$	$S_m = \frac{S_a}{r}$
$\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$	$S_a = \frac{r S_y}{1 + r}$
Load line $r = S_a/S_m$	$S_m = \frac{S_y}{1 + r}$
$\left(\frac{S_a}{S_e}\right)^2 + \left(\frac{S_m}{S_y}\right)^2 = 1$	$S_a = 0, \quad \frac{2 S_y S_e^2}{S_e^2 + S_y^2}$
$\frac{S_a}{S_y} + \frac{S_m}{S_y} = 1$	$S_m = S_y - S_a, \; r_{\text{crit}} = S_a/S_m$

Fatigue factor of safety

$$n_f = \sqrt{\frac{1}{(\sigma_a/S_e)^2 + (\sigma_m/S_y)^2}}$$

Example 4-15

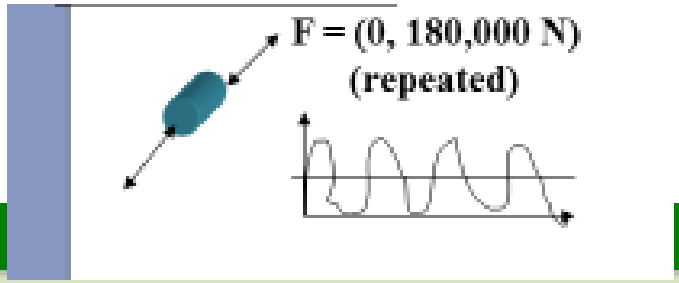
A connecting rod made of AISI 8650 steel, oil quenched and tempered is subjected to a repeated axial load of 180 kN. The yield strength of the material is 900 MPa, and the tensile strength is 1100 MPa. The following correction factors are applied to the design: Load factor = 0.7; size factor = 0.85; finish factor = 0.8. Determine the required safe diameter of the rod for indefinite life. Use:

a. Soderberg criterion
b. Modified Goodman criterion
c. Elliptic Criterion

$F_m = F_{max}/2 = 90{,}000$ N

Load & Stress Analysis

$F_a = F_m = 90{,}000$ N



$\sigma_m = 4F_m/\pi d^2 = 115 \times 10^3/d^2$

$\sigma_a = \sigma_m = 115 \times 10^3/d^2$ MPa

Slope of Load Line, $r_L = \sigma_a/\sigma_m = 1$

Example 4-15 (continued)

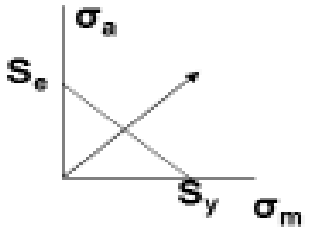
Material Properties

$S_y = 900$ MPa $S_{ut} = 1100$ MPa

$S_e = K_a K_b K_c K_d K_e (0.5 S_{ut}) = (0.8)(0.85)(0.7)(0.5 \times 1100) = 262$ MPa

a. Soderberg Criterion

$$1/n = \sigma_m / s_y + \sigma_a / s_e$$



Design Equation:

$$1/n = 115 \times 10^3/900d^2 + 115 \times 10^3/262d^2$$

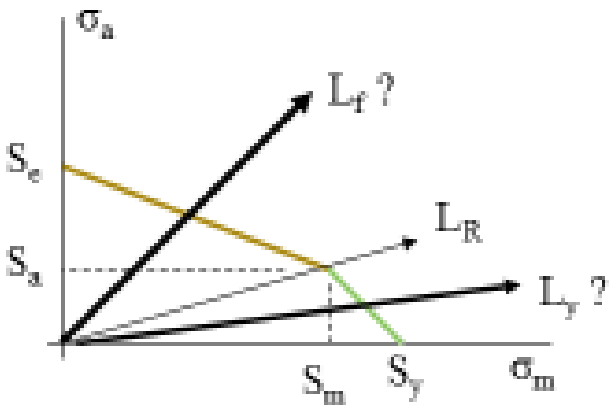
Specify n = 2

Solve d = 33.67 mm.

Use d = 34 mm

Example 4-15 (continued)

b. Modified Goodman



$S_m = (S_y - S_e)S_{ut} / (S_{ut} - S_e)$
$= 837$ MPa

$S_a = S_y - S_m$
$= 63$ MPa

$r_c = S_a / S_m = 0.075$

Note $r_L = 1.0 > r_c$
Load Line is L_f

Thus Failure will be by fatigue:

$$1/n = \sigma_m/s_{ut} + \sigma_a/s_e$$

Design Equation:

$1/n = 115 \times 10^3/1100d^2 + 115 \times 10^3/262d^2$

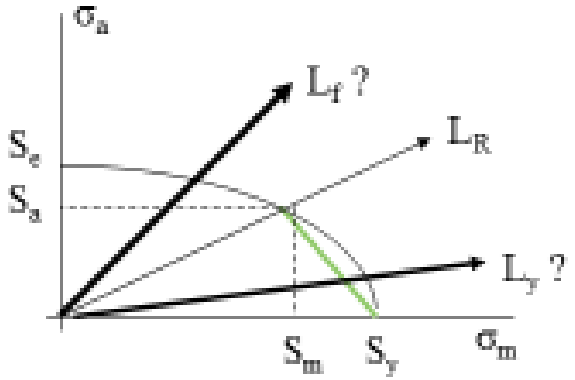
Specify n =2

Solve d = 32.97 mm

Use d = 33 mm

Example 4-15 (continued)

c. Elliptic Model



$$S_e = 0.\ \frac{2S_yS_e^2}{S_e^2 + S_y^2}$$

$$S_m = S_y - S_a,\ r_{alt} = S_a/S_m$$

S_a = 141 MPa
S_m = 759 MPa

$r_c = S_a / S_m = 0.186$
Note $r_L = 1.0 > r_c$
Failure Line = L_f

Thus Failure will be by fatigue:

$$1/n^2 = (\sigma_m/s_y)^2 + (\sigma_a/s_e)^2$$

Design Equation:

$$1/n^2 = (115 \times 10^3/900d^2)^2 + (115 \times 10^3/262d^2)^2$$

Specify n = 2
Solve d = 30.24 mm
Use d = 31 mm

Example 4-16

A 2 inch steel bar undergoes cyclic loading such that σ_{max} = 420 MPa and σ_{min} = 140 MPa. For the material, S_{ut} = 560 MPa and S_y = 450 MPa, fully corrected endurance limit is 280 MPa and f = 0.9. Using Modified Goodman Criterion, find:

a. The design factor for infinite life

b. The design factor for 10^5 life

σ_{max} = 420 MPa

σ_{min} = 140 Mpa

$\sigma_m = (\sigma_{max} + \sigma_{min})/2$

$= 280$ MPa

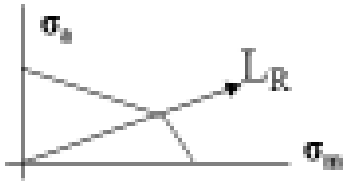
$\sigma_a = (\sigma_{max} - \sigma_{min})/2$

$= 140$ MPa

Load Slope, $r_L = \sigma_a / \sigma_m$

$= 0.5$

a. Modified Goodman (indefinite life or 10^6 ~):



Calculate S_m = 340 MPa

S_a = 110 MPa

Slope $r_c = 0.3 < r_L$

Failure by Fatigue

$1/n = \sigma_m/S_{ut} + \sigma_a/S_e$

$= 280/560 + 140/280$

Solve, n = 1 (verge of failure)

Example 4-16 (continued)

Calculation of S_{10}^5

$S_{10}^3 = fS_{ut} = 0.9\, S_{ut} = 504$ MPa

$S_f = aN^b$

$a = (S_{10}^3)^2 / S_e = 504^2/280 = 907$ MPa

$b = (-1/3) \log [S_{10}^3 / S_e] = -0.085$

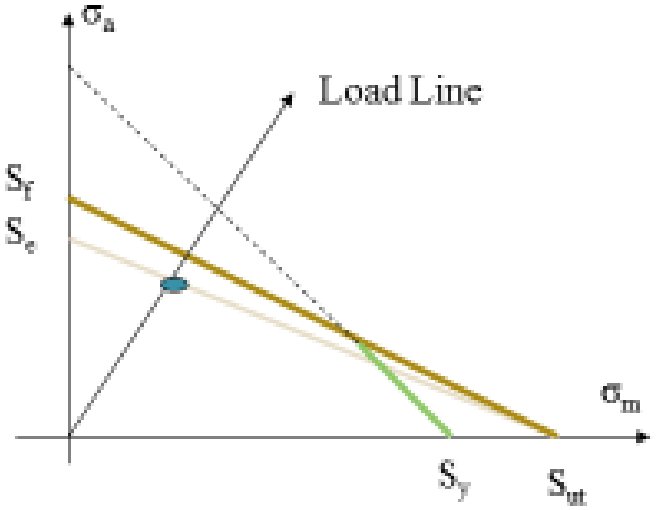
$S_f = 907(10^5)^{-0.085} = 340$ MPa

Modified Goodman:

$1/n = \sigma_m/S_{ut} + \sigma_a/S_f$

$= 280/560 + 140/340$

Solve n = 1.1 (barely safe)



Example 4-17

A commercially ground shaft made from steel transmits a fluctuating torque of T = 1000 ± 250 N-m. The material has an ultimate tensile Strength of 1.2 GPa and a yield strength of 1.0 GPa. Find the factor of safety at a section where the diameter is 30 mm. Use ASME Elliptical Theory.

Stress Analysis

Mean Torque, $T_m = 1000$ N-m

Alternating Torque, $T_a = 250$ N-m

Mean Shear stress, $\tau_m = 16T_m/\pi d^3$
$= 187$ MPa

Alt. Shear stress, $\tau_a = 16T_a/\pi d^3$
$= 47$ MPa

Sloe of Load Line, $r_L = \tau_a/\tau_m = 0.25$

Material Properties

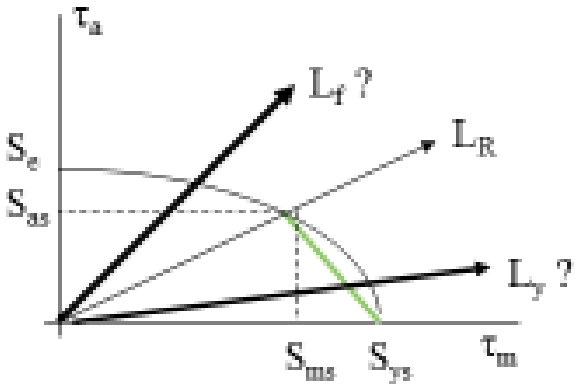
$S_y = 1000$ MPa $S_{ut} = 1200$ MPa

$S_{ys} = 0.577\ S_y = 577$ MPa

$S_{us} = 0.8\ S_{ut} = 960$ MPa

$S_e = K_a K_b K_c K_d K_e (0.5 S_{ut})$
$= (0.865)(0.86)(0.58)(1)\ (1)(0.5x1200)$
$= 258$ MPa

Example 4-17 (continued)



$$S_{as} = 2S_{ys}S_e^2/(S_e^2 + S_{ys}^2)$$

$$S_{ms} = S_{ys} - S_{as}$$

$S_{as} = 192$ MPa
$S_{ms} = 385$ MPa

$r_c = S_{as} / S_{ms} = 0.5$

Note $r_L = 0.25 < r_c = 0.5$
Load Line = L_y

Thus Failure will be by YIELD:

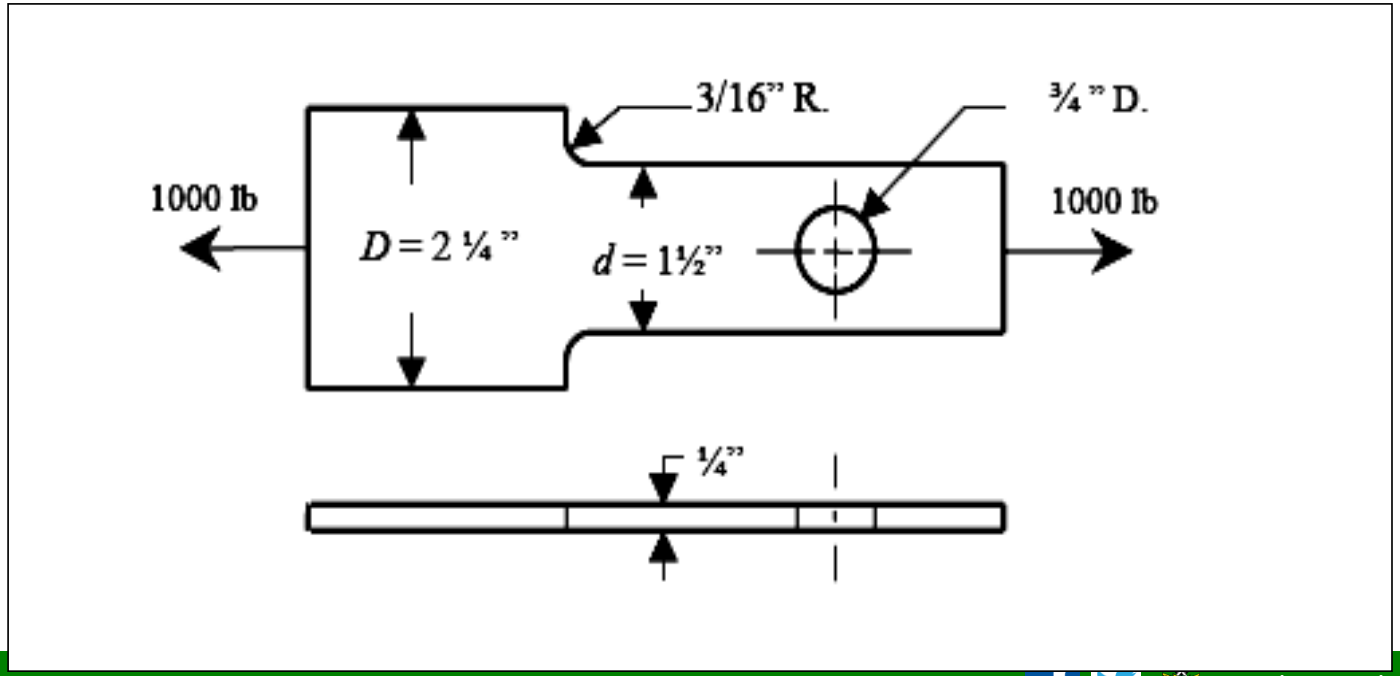
$$1/n = (\tau_m/s_{ys}) + (\tau_a/s_{YS})$$

$$1/n = (187/577) + (47/577)$$

Solve $n = 2.46$

Example 4–18

The bar is made of steel with an ultimate tensile strength of 80 ksi and yield strength of 50 ksi. Endurance limit is 19 ksi for the design. The 1000-Ib axial load is completely reversed. Find the factor of safety at the shoulder and the hole against fatigue failure.



Example 4–18 (continued)

Shoulder

Fatigue Stress Concentration Factor

D/d = 2.25/1.5 = 1.5

r/d = 0.1875/1.5 = 0.125

K_t = 1.9 (from chart)

Notch sensitivity factor, q = 0.86 (chart)

$K_f = 1 + q(K_t - 1) = 1.77$

Stress Calculation

Axial Stress, $\sigma = F/A$

Completely reversed, $\sigma_m = 0$

$\sigma = \sigma_a = F/A = F/dt = 1000/(1.5 \times 0.25)$

$= 2{,}667\ lb/in^2 = 2.667\ ksi$

Modified Goodman

$1/n = \sigma_m/S_{ut} + K_f\,\sigma_a/\mathbf{S_e}$

$\mathbf{= 0 + 1.77x2.667/19}$

Solve **n = 4.02**

Example 4–18 (continued)

Hole

Fatigue Stress Concentration Factor

w = 1.5 d = ¾ in t= 0.25 in

d/w = 0.5

K_t = 2.19 (from chart)

Notch sensitivity factor, q = 0.86 (chart)

$K_f = 1 + q(K_t - 1) = 2.023$

Stress Calculation

Axial Stress, $\sigma = F/A$

Completely reversed, $\sigma_m = 0$

$\sigma = \sigma_a = F/A = F/(w-d)t = 1000/(1.5-0.75)x0.25$

$= 5{,}333$ Ib/in^2 = 5.333 kpi

Modified Goodman

$1/n = \sigma_m/S_{ut} + K_f \sigma_a/S_e$

$= 0 + 2.023x5.333/19$

Solve **n = 1.76**

Note:

n based on shoulder = 4.02

n based on hole = 1.76

Factor of safety for the part is n = 1.76 (based on the hole)

Example 4–19



Material: AISI 1050 CD Steel; $S_{ut} = 690$ MPa, $S_y = 580$ MPa

Find factor of safety against (a) yield and (ii) against fatigue for infinite life

---

Stress Analysis (Repeated Load)

$F_m = F_a = 72000/2 = 36{,}000$ N

$\sigma_m = \sigma_a = 4F_m/\pi d^2 = 32$ MPa

Material Properties

$S_e = K_a K_b K_c K_d K_e (0.5 S_{ut})$

$= (0.798)(1)(0.85)(1)\ (1)(0.5 \times 690)$

$= 234$ MPa

Example 4-19 (continued)

Factor of Safety Against **Yield**:
Langer Yield

$1/n = \sigma_m/S_y + K_f \sigma_a/S_y$

$1/n = 32/580 + 1.85x32/580$

Solve n = 6.4

Factor of Safety Against **Fatigue**:
Modified Goodman

$1/n = \sigma_m/S_{ut} + K_f \sigma_a/S_e$
$1/n = 32/690 + 1.85x32/234$

Solve n = 3.3

NOTE:
Factor of safety against yield = 6.4
Factor of safety against fatigue = 3.3

Therefore,
Factor of safety for the part = 3.3

Example 4–20

A 1.5-in-diameter bar has been machined from an AISI 1050 cold-drawn bar. This part is to withstand a fluctuating tensile load varying from 0 to 16 kip. Because of the ends, and the fillet radius, a fatigue stress-concentration factor K_f is 1.85 for 10^6 or larger life. Find S_a and S_m and the factor of safety guarding against fatigue and first-cycle yielding, using (*a*) the Gerber fatigue line and (*b*) the ASME-elliptic fatigue line.

Solution

We begin with some preliminaries. From Table A–20, $S_{ut} = 100$ kpsi and $S_y = 84$ kpsi. Note that $F_a = F_m = 8$ kip. The Marin factors are, deterministically,

$k_a = 2.70(100)^{-0.265} = 0.797$: Eq. (6–19), Table 6–2, p. 296

$k_b = 1$ (axial loading, see k_c)

$k_c = 0.85$: Eq. (6–26), p. 298

$k_d = k_e = k_f = 1$

$S_e = 0.797(1)0.850(1)(1)(1)0.5(100) = 33.9$ kpsi: Eqs. (6–8), (6–18), p. 290, p. 295

Example 4–20 (continued)

The nominal axial stress components σ_{ao} and σ_{mo} are

$$\sigma_{ao} = \frac{4F_a}{\pi d^2} = \frac{4(8)}{\pi 1.5^2} = 4.53 \text{ kpsi} \qquad \sigma_{mo} = \frac{4F_m}{\pi d^2} = \frac{4(8)}{\pi 1.5^2} = 4.53 \text{ kpsi}$$

Applying K_f to both components σ_{ao} and σ_{mo} constitutes a prescription of no notch yielding:

$$\sigma_a = K_f \sigma_{ao} = 1.85(4.53) = 8.38 \text{ kpsi} = \sigma_m$$

(*a*) Let us calculate the factors of safety first. From the bottom panel from Table 6–7 the factor of safety for fatigue is

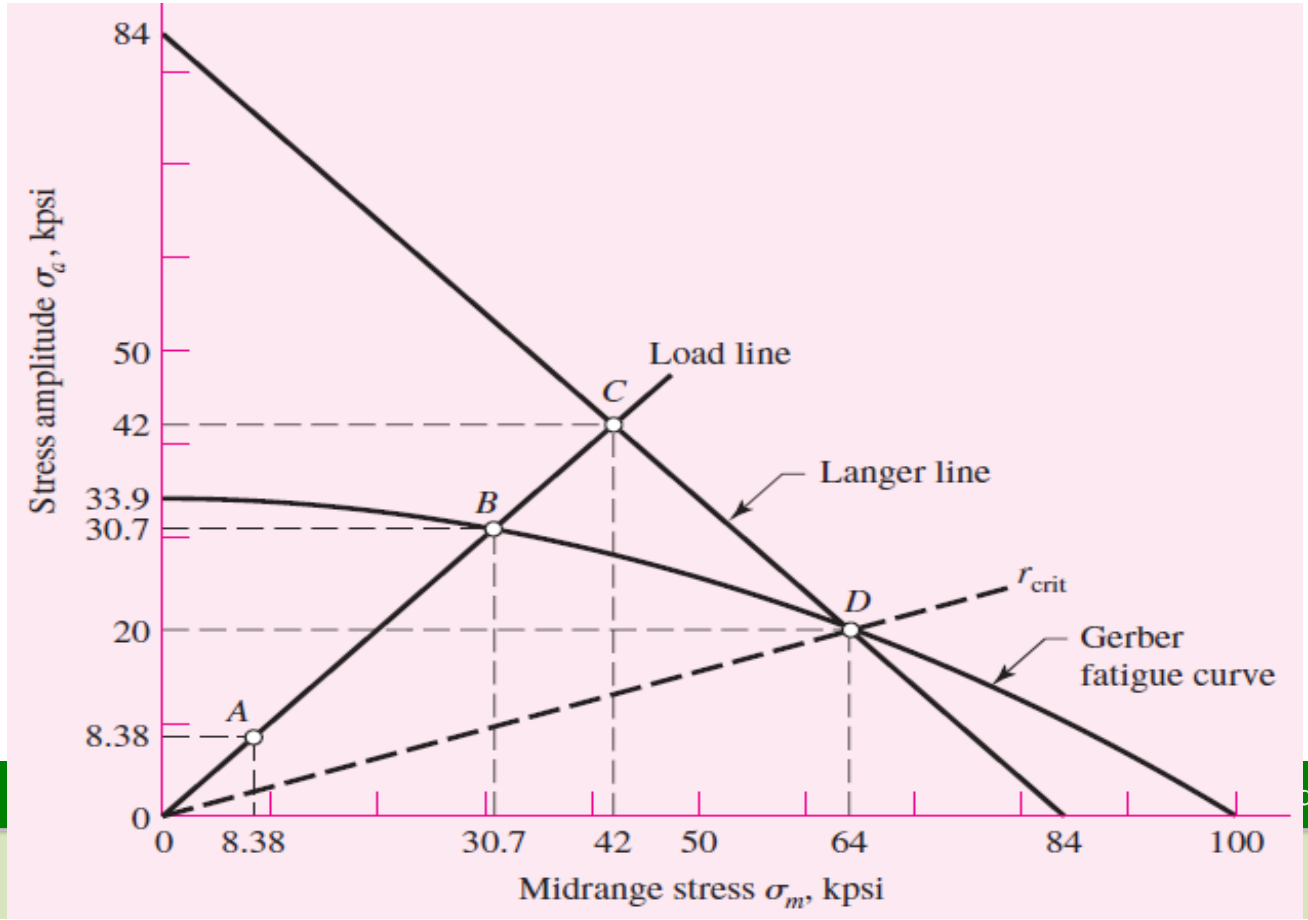
$$n_f = \frac{1}{2}\left(\frac{100}{8.38}\right)^2 \left(\frac{8.38}{33.9}\right) \left\{-1 + \sqrt{1 + \left[\frac{2(8.38)33.9}{100(8.38)}\right]^2}\right\} = 3.66 \quad \text{Answer}$$

From Eq. (6–49) the factor of safety guarding against first-cycle yield is

$$n_y = \frac{S_y}{\sigma_a + \sigma_m} = \frac{84}{8.38 + 8.38} = 5.01 \quad \text{Answer}$$

Example 4–20 (continued)

Thus, we see that fatigue will occur first and the factor of safety is 3.68. This can be seen in Fig. 6–28 where the load line intersects the Gerber fatigue curve first at point B. If the plots are created to true scale it would be seen that $n_f = OB/OA$.



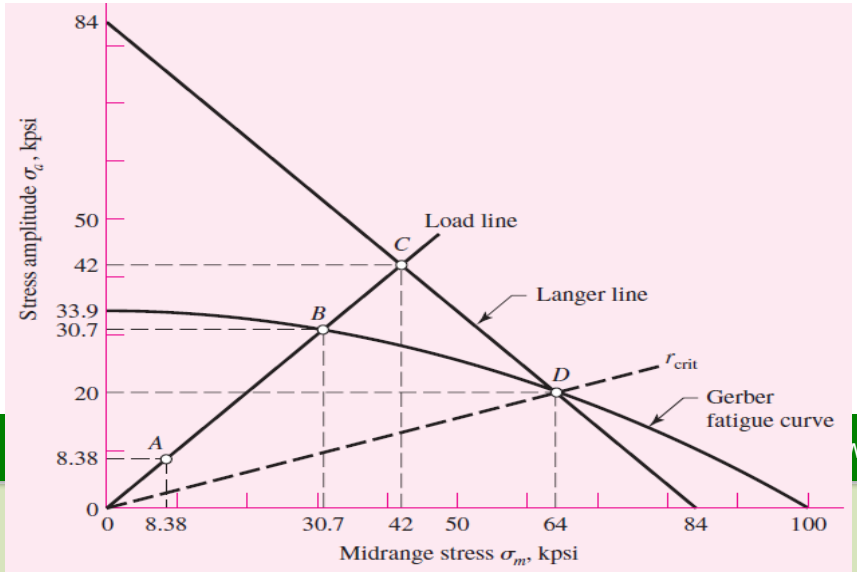
Example 4–20 (continued)

From the first panel of Table 6–7, $r = \sigma_a/\sigma_m = 1$,

$$S_a = \frac{(1)^2 100^2}{2(33.9)} \left\{ -1 + \sqrt{1 + \left[\frac{2(33.9)}{(1)100} \right]^2} \right\} = 30.7 \text{ kpsi}$$ Answer

$$S_m = \frac{S_a}{r} = \frac{30.7}{1} = 30.7 \text{ kpsi}$$ Answer

As a check on the previous result, $n_f = OB/OA = S_a/\sigma_a = S_m/\sigma_m = 30.7/8.38 = 3.66$ and we see total agreement.



Example 4–20 (continued)

We could have detected that fatigue failure would occur first without drawing Fig. 6–28 by calculating r_{crit}. From the third row third column panel of Table 6–7, the intersection point between fatigue and first-cycle yield is

$$S_m = \frac{100^2}{2(33.9)}\left[1 - \sqrt{1 + \left(\frac{2(33.9)}{100}\right)^2 \left(1 - \frac{84}{33.9}\right)}\right] = 64.0 \text{ kpsi}$$

$$S_a = S_y - S_m = 84 - 64 = 20 \text{ kpsi}$$

The critical slope is thus

$$r_{\text{crit}} = \frac{S_a}{S_m} = \frac{20}{64} = 0.312$$

which is less than the actual load line of $r = 1$. This indicates that fatigue occurs before first-cycle-yield.

Example 4–20 (continued)

(*b*) Repeating the same procedure for the ASME-elliptic line, for fatigue

$$n_f = \sqrt{\frac{1}{(8.38/33.9)^2 + (8.38/84)^2}} = 3.75$$ Answer

Again, this is less than $n_y = 5.01$ and fatigue is predicted to occur first. From the first row second column panel of Table 6–8, with $r = 1$, we obtain the coordinates S_a and S_m of point B in Fig. 6–29 as

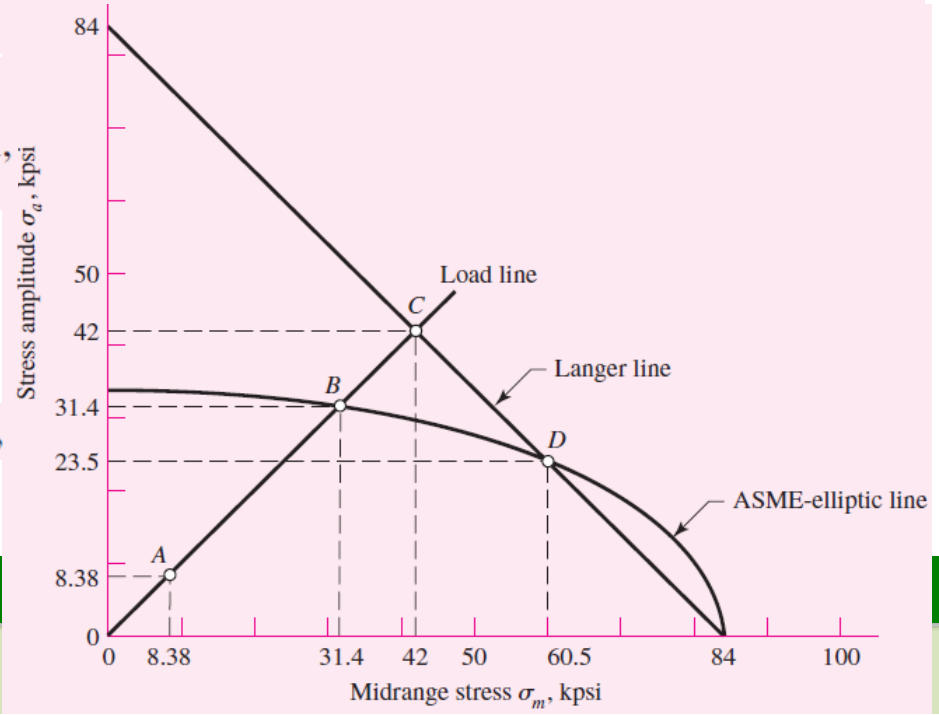
$$S_a = \sqrt{\frac{(1)^2 33.9^2 (84)^2}{33.9^2 + (1)^2 84^2}} = 31.4 \text{ kpsi},$$

$$S_m = \frac{S_a}{r} = \frac{31.4}{1} = 31.4 \text{ kpsi}$$

Answer

To verify the fatigue factor of safety,

$$n_f = S_a/\sigma_a = 31.4/8.38 = 3.75.$$



Example 4–20 (continued)

As before, let us calculate r_{crit}. From the third row second column panel of Table 6–8,

$$S_a = \frac{2(84)33.9^2}{33.9^2 + 84^2} = 23.5 \text{ kpsi}, \quad S_m = S_y - S_a = 84 - 23.5 = 60.5 \text{ kpsi}$$

$$r_{\text{crit}} = \frac{S_a}{S_m} = \frac{23.5}{60.5} = 0.388$$

which again is less than $r = 1$, verifying that fatigue occurs first with $n_f = 3.75$.

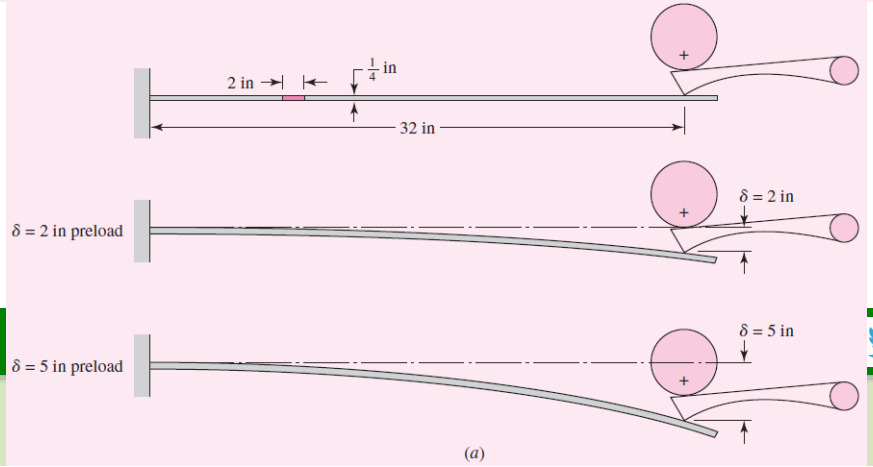
The Gerber and the ASME-elliptic fatigue failure criteria are very close to each other and are used interchangeably. The ANSI/ASME Standard B106.1M–1985 uses ASME-elliptic for shafting.

Example 4–21

A flat-leaf spring is used to retain an oscillating flat-faced follower in contact with a plate cam. The follower range of motion is 2 in and fixed, so the alternating component of force, bending moment, and stress is fixed, too. The spring is preloaded to adjust to various cam speeds. The preload must be increased to prevent follower float or jump. For lower speeds the preload should be decreased to obtain longer life of cam and follower surfaces. The spring is a steel cantilever 32 in long, 2 in wide, and $\frac{1}{4}$ in thick, as seen in Fig. 6–30*a*. The spring strengths are $S_{ut} = 150$ kpsi, $S_y = 127$ kpsi, and $S_e = 28$ kpsi fully corrected. The total cam motion is 2 in. The designer wishes to preload the spring by deflecting it 2 in for low speed and 5 in for high speed.

(*a*) Plot the Gerber-Langer failure lines with the load line.

(*b*) What are the strength factors of safety corresponding to 2 in and 5 in preload?



Example 4–21 (continued)

We begin with preliminaries. The second area moment of the cantilever cross section is

$$I = \frac{bh^3}{12} = \frac{2(0.25)^3}{12} = 0.00260 \text{ in}^4$$

Since, from Table A–9, beam 1, force F and deflection y in a cantilever are related by $F = 3EIy/l^3$, then stress σ and deflection y are related by

$$\sigma = \frac{Mc}{I} = \frac{32Fc}{I} = \frac{32(3EIy)}{l^3}\frac{c}{I} = \frac{96Ecy}{l^3} = Ky$$

where $K = \dfrac{96Ec}{l^3} = \dfrac{96(30 \cdot 10^6)0.125}{32^3} = 10.99(10^3)$ psi/in $= 10.99$ kpsi/in

Now the minimums and maximums of y and σ can be defined by

$$y_{\min} = \delta \qquad y_{\max} = 2 + \delta$$

$$\sigma_{\min} = K\delta \qquad \sigma_{\max} = K(2 + \delta)$$

Example 4–21 (continued)

The stress components are thus

$$\sigma_a = \frac{K(2+\delta) - K\delta}{2} = K = 10.99 \text{ kpsi}$$

$$\sigma_m = \frac{K(2+\delta) + K\delta}{2} = K(1+\delta) = 10.99(1+\delta)$$

For $\delta = 0$, $\sigma_a = \sigma_m = 10.99 = 11$ kpsi

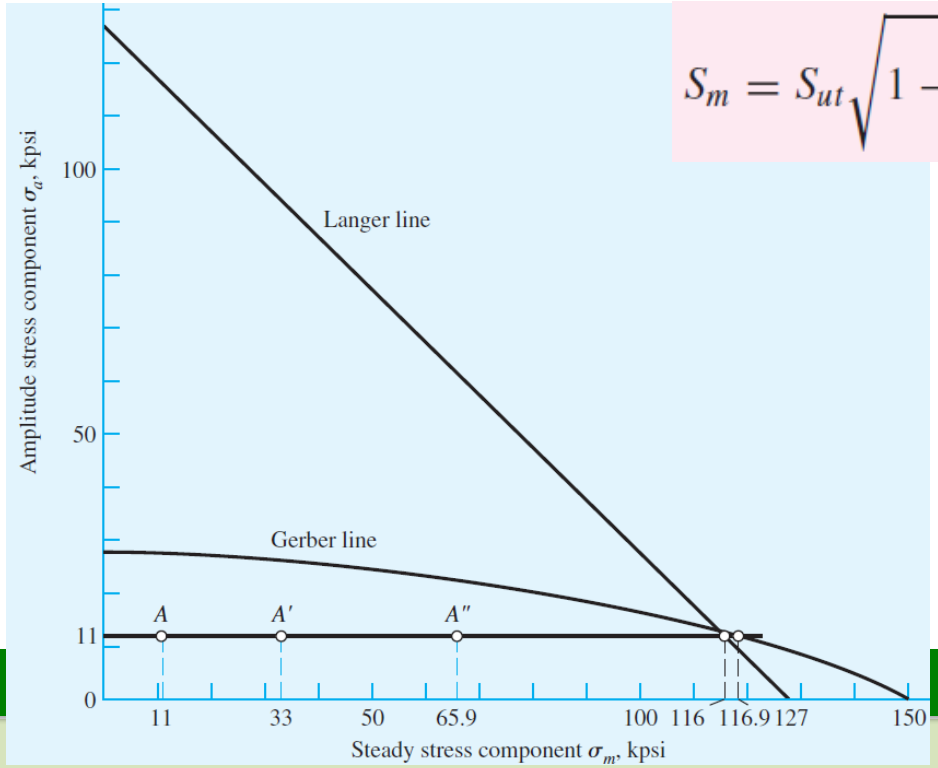
For $\delta = 2$ in, $\sigma_a = 11$ kpsi, $\sigma_m = 10.99(1+2) = 33$ kpsi

For $\delta = 5$ in, $\sigma_a = 11$ kpsi, $\sigma_m = 10.99(1+5) = 65.9$ kpsi

Example 4–21 (continued)

(*a*) A plot of the Gerber and Langer criteria is shown in Fig. 6–30*b*. The three preload deflections of 0, 2, and 5 in are shown as points A, A', and A''. Note that since σ_a is constant at 11 kpsi, the load line is horizontal and does not contain the origin. The intersection between the Gerber line and the load line is found from solving Eq. (6–42) for S_m and substituting 11 kpsi for S_a:

$$S_m = S_{ut}\sqrt{1 - \frac{S_a}{S_e}} = 150\sqrt{1 - \frac{11}{28}} = 116.9 \text{ kpsi}$$



Example 4–21 (continued)

The intersection of the Langer line and the load line is found from solving Eq. (6–44) for S_m and substituting 11 kpsi for S_a:

$$S_m = S_y - S_a = 127 - 11 = 116 \text{ kpsi}$$

The threats from fatigue and first-cycle yielding are approximately equal.

(*b*) For $\delta = 2$ in,

$$n_f = \frac{S_m}{\sigma_m} = \frac{116.9}{33} = 3.54 \qquad n_y = \frac{116}{33} = 3.52$$ Answer

and for $\delta = 5$ in,

$$n_f = \frac{116.9}{65.9} = 1.77 \qquad n_y = \frac{116}{65.9} = 1.76$$ Answer

Example 4–22

A steel bar undergoes cyclic loading such that $\sigma_{max} = 60$ kpsi and $\sigma_{min} = -20$ kpsi. For the material, $S_{ut} = 80$ kpsi, $S_y = 65$ kpsi, a fully corrected endurance limit of $S_e = 40$ kpsi, and $f = 0.9$. Estimate the number of cycles to a fatigue failure using:

(*a*) Modified Goodman criterion.

(*b*) Gerber criterion.

Solution

From the given stresses,

$$\sigma_a = \frac{60 - (-20)}{2} = 40 \text{ kpsi} \qquad \sigma_m = \frac{60 + (-20)}{2} = 20 \text{ kpsi}$$

(*a*) For the modified Goodman criterion, Eq. (6–46), the fatigue factor of safety based on infinite life is

$$n_f = \frac{1}{\dfrac{\sigma_a}{S_e} + \dfrac{\sigma_m}{S_{ut}}} = \frac{1}{\dfrac{40}{40} + \dfrac{20}{80}} = 0.8$$

Example 4–22 (continued)

This indicates a finite life is predicted. The *S-N* diagram is only applicable for completely reversed stresses. To estimate the finite life for a fluctuating stress, we will obtain an equivalent completely reversed stress that is expected to be as damaging as the fluctuating stress. A commonly used approach is to assume that since the modified Goodman line represents all stress situations with a constant life of 10^6 cycles, other constant-life lines can be generated by passing a line through $(S_{ut}, 0)$ and a fluctuating stress point (σ_m, σ_a). The point where this line intersects the σ_a axis represents a completely reversed stress (since at this point $\sigma_m = 0$), which predicts the same life as the fluctuating stress.

This completely reversed stress can be obtained by replacing S_e with σ_{rev} in Eq. (6–46) for the modified Goodman line resulting in

$$\sigma_{\text{rev}} = \frac{\sigma_a}{1 - \dfrac{\sigma_m}{S_{ut}}} = \frac{40}{1 - \dfrac{20}{80}} = 53.3 \text{ kpsi}$$

Example 4–22 (continued)

From the material properties, Eqs. (6–14) to (6–16), p. 293, give

$$a = \frac{(f S_{ut})^2}{S_e} = \frac{[0.9(80)]^2}{40} = 129.6 \text{ kpsi}$$

$$b = -\frac{1}{3}\log\left(\frac{f S_{ut}}{S_e}\right) = -\frac{1}{3}\log\left[\frac{0.9(80)}{40}\right] = -0.0851 \qquad (1)$$

$$N = \left(\frac{\sigma_{\text{rev}}}{a}\right)^{1/b} = \left(\frac{\sigma_{\text{rev}}}{129.6}\right)^{-1/0.0851}$$

Substituting σ_{rev} into Eq. (1) yields

$$N = \left(\frac{53.3}{129.6}\right)^{-1/0.0851} = 3.4(10^4) \text{ cycles}$$ Answer

Example 4–22 (continued)

(*b*) For Gerber, similar to part (*a*), from Eq. (6–47),

$$\sigma_{\text{rev}} = \frac{\sigma_a}{1 - \left(\frac{\sigma_m}{S_{ut}}\right)^2} = \frac{40}{1 - \left(\frac{20}{80}\right)^2} = 42.7 \text{ kpsi}$$

Again, from Eq. (1),

$$N = \left(\frac{42.7}{129.6}\right)^{-1/0.0851} \doteq 4.6(10^5) \text{ cycles}$$ Answer

Comparing the answers, we see a large difference in the results. Again, the modified Goodman criterion is conservative as compared to Gerber for which the moderate difference in S_f is then magnified by a logarithmic S, N relationship.

Fatigue Criteria for Brittle Materials

- For many brittle materials, the first quadrant fatigue failure criteria follows a concave upward Smith-Dolan locus,

$$\frac{S_a}{S_e} = \frac{1 - S_m/S_{ut}}{1 + S_m/S_{ut}}$$

- Or as a design equation,

$$\frac{n\sigma_a}{S_e} = \frac{1 - n\sigma_m/S_{ut}}{1 + n\sigma_m/S_{ut}}$$

- For a radial load line of slope r, the intersection point is

$$S_a = \frac{rS_{ut} + S_e}{2}\left[-1 + \sqrt{1 + \frac{4rS_{ut}S_e}{(rS_{ut} + S_e)^2}}\right]$$

- In the second quadrant,

$$S_a = S_e + \left(\frac{S_e}{S_{ut}} - 1\right) S_m \qquad -S_{ut} \le S_m \le 0 \quad \text{(for cast iron)}$$

Fatigue Criteria for Brittle Materials

- Table gives properties of gray cast iron, including endurance limit
- The endurance limit already includes k_a and k_b
- The average k_c for axial and torsional is 0.9

Example 4–23

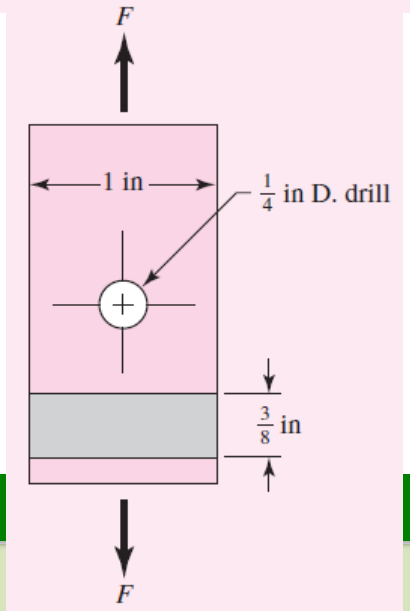
A grade 30 gray cast iron is subjected to a load F applied to a 1 by $\frac{3}{8}$-in cross-section link with a $\frac{1}{4}$-in-diameter hole drilled in the center as depicted in Fig. 6–31*a*. The surfaces are machined. In the neighborhood of the hole, what is the factor of safety guarding against failure under the following conditions:

(*a*) The load $F = 1000$ lbf tensile, steady.

(*b*) The load is 1000 lbf repeatedly applied.

(*c*) The load fluctuates between -1000 lbf and 300 lbf without column action.

Use the Smith-Dolan fatigue locus.



Example 4–23 (continued)

Some preparatory work is needed. From Table A–24, $S_{ut} = 31$ kpsi, $S_{uc} = 109$ kpsi, $k_a k_b S'_e = 14$ kpsi. Since k_c for axial loading is 0.9, then $S_e = (k_a k_b S'_e)k_c = 14(0.9) = 12.6$ kpsi. From Table A–15–1, $A = t(w - d) = 0.375(1 - 0.25) = 0.281$ in^2, $d/w = 0.25/1 = 0.25$, and $K_t = 2.45$. The notch sensitivity for cast iron is 0.20 (see p. 304), so

$$K_f = 1 + q(K_t - 1) = 1 + 0.20(2.45 - 1) = 1.29$$

(*a*) Since the load is steady, $\sigma_a = 0$, the load is static. Based on the discussion of cast iron in Sec. 5–2, K_t, and consequently K_f, need not be applied. Thus, $\sigma_m = F_m/A = 1000(10^{-3})/0.281 = 3.56$ kpsi, and

$$n = \frac{S_{ut}}{\sigma_m} = \frac{31.0}{3.56} = 8.71 \qquad \text{Answer}$$

Example 4–23 (continued)

(*b*)

$$F_a = F_m = \frac{F}{2} = \frac{1000}{2} = 500 \text{ lbf}$$

$$\sigma_a = \sigma_m = \frac{K_f F_a}{A} = \frac{1.29(500)}{0.281}(10^{-3}) = 2.30 \text{ kpsi}$$

$$r = \frac{\sigma_a}{\sigma_m} = 1$$

From Eq. (6–52),

$$S_a = \frac{(1)31 + 12.6}{2}\left[-1 + \sqrt{1 + \frac{4(1)31(12.6)}{[(1)31 + 12.6]^2}}\right] = 7.63 \text{ kpsi}$$

$$n = \frac{S_a}{\sigma_a} = \frac{7.63}{2.30} = 3.32 \quad \text{Answer}$$

Example 4–23 (continued)

(c) $F_a = \frac{1}{2}|300 - (-1000)| = 650 \text{ lbf}$ $\quad \sigma_a = \frac{1.29(650)}{0.281}(10^{-3}) = 2.98 \text{ kpsi}$

$F_m = \frac{1}{2}[300 + (-1000)] = -350 \text{ lbf}$ $\quad \sigma_m = \frac{1.29(-350)}{0.281}(10^{-3}) = -1.61 \text{ kpsi}$

$$r = \frac{\sigma_a}{\sigma_m} = \frac{3.0}{-1.61} = -1.86$$

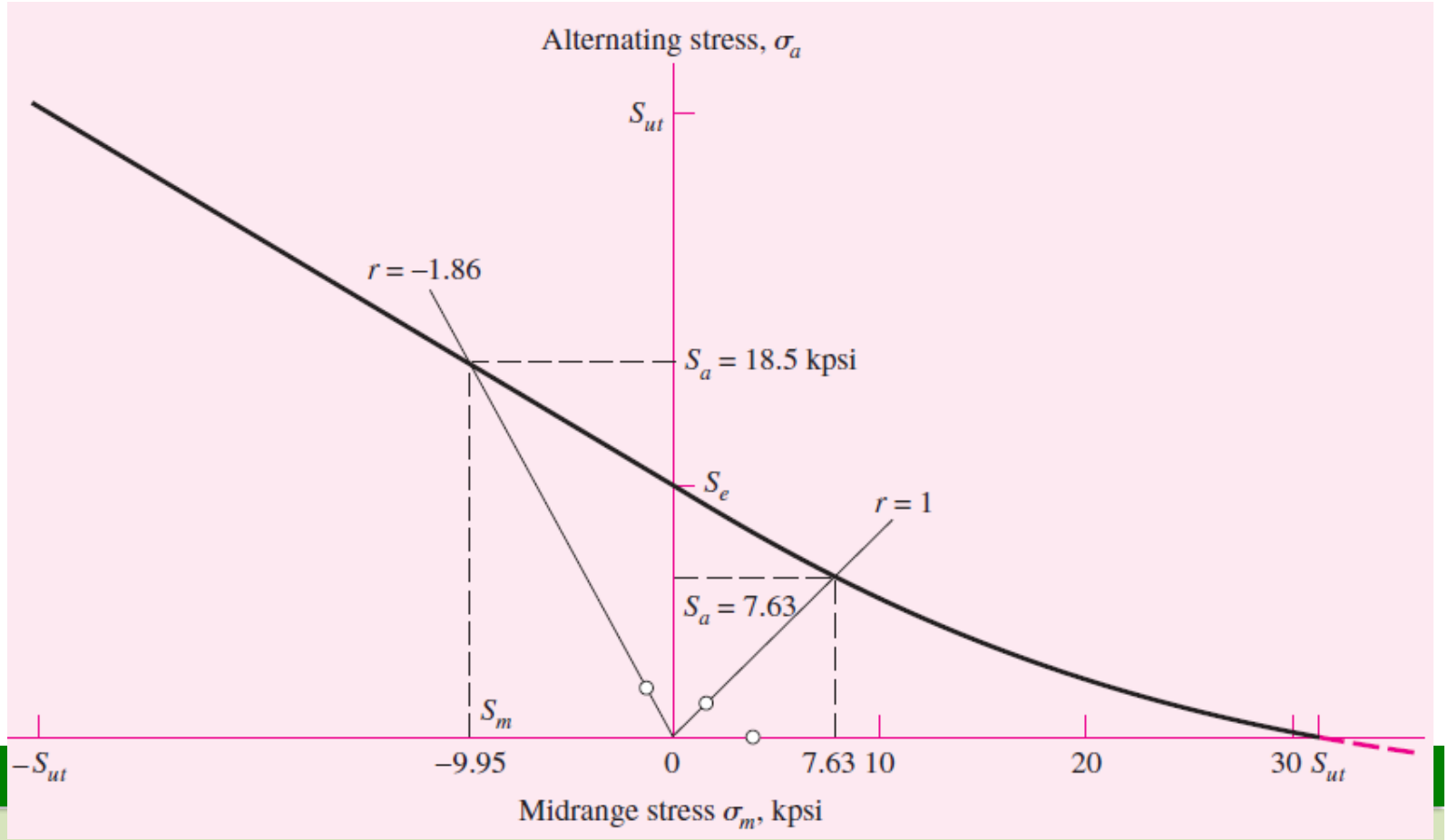
From Eq. (6–53), $S_a = S_e + (S_e/S_{ut} - 1)S_m$ and $S_m = S_a/r$. It follows that

$$S_a = \frac{S_e}{1 - \frac{1}{r}\left(\frac{S_e}{S_{ut}} - 1\right)} = \frac{12.6}{1 - \frac{1}{-1.86}\left(\frac{12.6}{31} - 1\right)} = 18.5 \text{ kpsi}$$

$$n = \frac{S_a}{\sigma_a} = \frac{18.5}{2.98} = 6.20$$ Answer

Example 4–23 (continued)

Figure 6–31*b* shows the portion of the designer's fatigue diagram that was constructed.



THANK YOU